A Procedural Content Generation Framework for Light-Based Puzzles: Exploring Algorithmic Control of Difficulty and Rule Interaction

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Abstract

Procedural Content Generation (PCG) is widely used in game development to automate the creation of game content. However, generating puzzle levels that remain solvable while also supporting difficulty analysis remains a challenge. Existing puzzle generation approaches often rely on validation after generation, while difficulty is commonly estimated using simple structural measures such as path length or Total Objects.

This study proposes Constraint-Guided Reverse Path Generation (CGRPG), a procedural generation approach for grid-based light puzzles that use reflection and refraction mechanics. The approach constructs a solution path from the receiver towards the emitter while applying placement constraints during generation. This study also investigates Mechanic Interaction Density (MID), a weighted interaction-based metric whose coefficients were selected through pilot calibration experiments, as a measure for examining whether mechanic interactions can help describe puzzle complexity and solver effort.

The framework and metric were evaluated using a dataset of 950 generated puzzles. A custom solver agent was used to assess puzzle solvability and to collect solver-performance measures, including Solve Time and Search Nodes. Additional analyses were conducted using structural baseline measures, statistical comparison testing, and incremental contribution analysis to examine whether MID provided information beyond traditional object-count descriptors.

MID showed weak positive relationships with solver-effort measures. Comparison with structural baseline measures produced mixed results: MID showed the strongest association with Solve Time, whereas object-count measures were more strongly associated with Search Nodes. These findings suggest that MID is more appropriately interpreted as an exploratory interaction-based descriptor that may complement existing structural measures rather than replace them. Overall, the study contributes a procedural generation framework for solvable light-based puzzles, an interaction-based metric for analyzing mechanic relationships, and an empirical comparison between interaction-based and structural approaches to puzzle analysis.

Abbreviations, Symbols and Notation

MID - Mechanic Interaction Density
Cl - Component Density
Ch - Rule Heterogeneity
Ci - Component Complexity
α - Weight assigned to Cl
β - Weight assigned to Ch
γ - Weight assigned to Ci
PCG - Procedural Content Generation
CGRPG - Constraint-Guided Reverse Path Generation

Chapter 1 Introduction

Procedural Content Generation (PCG) is widely used in game development to automate content creation. It can reduce development effort, increase replay ability, and generate large amounts of content. Recent research has also explored combining PCG with artificial intelligence techniques to improve content quality and flexibility (Farrokhi Maleki and Zhao, 2024).

Puzzle generation remains a challenging area of PCG because generated puzzles must be both valid and solvable. Many existing systems use generate-and-test approaches, where puzzles are generated and then validated. Although effective, invalid puzzles often need to be discarded or repaired, increasing computational cost (del Solar-Zavala and Barriga, 2026; McConnell and Zhao, 2025).

This research focuses on light-based puzzles that use reflection and refraction mechanics. Puzzle elements are placed on a two-dimensional grid, and solvability depends on the arrangement of emitters, receivers, mirrors, and refractors. This creates a procedural generation challenge because generated puzzles must remain both solvable and meaningful.

In addition to solvability, controlling puzzle difficulty is another important challenge. Existing methods often estimate difficulty using measures such as path length, Total Objects, or solution depth. However, these measures may not fully capture interactions between multiple puzzle mechanics (Kristensen and Burelli, 2024; Lazaridis and Fragulis, 2024).

To address these challenges, this research proposes a procedural generation framework for grid-based light puzzles. The framework uses a constraint-guided reverse path generation approach that constructs solution paths from the receiver towards the emitter during generation. It also investigates Mechanic Interaction Density (MID), an interaction-based metric that combines component density, rule heterogeneity, and component complexity using calibrated weighting coefficients derived from pilot experiments. The metric is evaluated as a potential indicator of puzzle difficulty.

The study is exploratory in nature and does not attempt to establish a universal model of puzzle difficulty. Instead, it investigates whether mechanic-interaction information can contribute useful analytical insight when evaluating generated puzzles. The research therefore focuses on examining relationships between puzzle structure, mechanic interactions, and solver behaviour within a controlled puzzle domain.

1.1 Research Motivation

Procedural puzzle generation presents a unique challenge within procedural content generation because generated content must satisfy multiple requirements simultaneously. In addition to producing diverse content, generated puzzles must remain valid, solvable, and capable of providing meaningful problem-solving experiences. Failure to satisfy any of these requirements can reduce the usefulness of generated content regardless of the generation method employed.

Light-based puzzles provide an interesting domain for investigating these challenges because puzzle solutions emerge through interactions between multiple mechanics rather than from individual puzzle objects alone. Reflection and refraction mechanics can create a wide variety of possible solution structures, making it difficult to maintain both solvability and controllability during generation.

These characteristics motivate the investigation of generation approaches that incorporate solution information directly into the construction process. They also motivate the exploration of alternative approaches to puzzle analysis that consider mechanic interactions alongside traditional structural puzzle descriptors.

1.2 Research Objectives

The primary objective of this research is to investigate procedural generation methods for grid-based light puzzles while examining the usefulness of mechanic-interaction information for puzzle analysis.

To achieve this objective, the study aims to:

1.Develop a procedural generation framework capable of constructing solvable light-based puzzles.

2.Implement a solver agent for evaluating generated puzzles.

3.Develop the Mechanic Interaction Density (MID) metric to describe interactions between puzzle mechanics.

4.Evaluate relationships between MID and solver-performance measures.

5.Compare MID with structural baseline measures to assess the additional information provided by mechanic-interaction descriptors.

1.3 Research Questions

To support these aims, this research addresses the following questions:

- RQ1 (Solvability): Can Constraint-Guided Reverse Path Generation (CGRPG) generate solvable light-based puzzles?
- RQ2 (Difficulty Modelling): To what extent can a calibrated MID metric help describe puzzle complexity and solver effort?

1.4 Research Contributions

This dissertation makes three main contributions.

1.Constraint-Guided Reverse Path Generation (CGRPG)

The first contribution is the development of Constraint-Guided Reverse Path Generation (CGRPG), a procedural puzzle-generation approach that constructs solution paths during generation rather than relying exclusively on post-generation validation. The framework investigates whether solvability can be incorporated directly into the generation process while maintaining diverse puzzle configurations.

2. Mechanic Interaction Density (MID)

The second contribution is the introduction of Mechanic Interaction Density (MID), an interaction-based metric designed to describe puzzle characteristics through mechanic interactions. MID combines component density, rule heterogeneity, and component complexity into a single descriptor and is intended as an exploratory analytical measure rather than a direct measure of puzzle difficulty.

3. Empirical Evaluation of Interaction-Based Puzzle Analysis

The third contribution is an empirical comparison between interaction-based and structural puzzle descriptors. By comparing MID with commonly used object-count measures and analyzing relationships with solver-performance metrics, the study investigates whether interaction-related information provides additional insight beyond traditional structural approaches.

Collectively, these contributions support a broader investigation of how procedural generation and puzzle analysis can be integrated within a single framework. By combining puzzle generation, solver-based evaluation, and interaction-based analysis, the study provides a foundation for exploring relationships between puzzle structure, mechanic interactions, and solver behaviour within light-based puzzle environments.

1.5 Dissertation Structure

This dissertation is organised into six chapters.

Chapter 1 introduces the research background, research problem, objectives, and research questions. The chapter also outlines the motivation for investigating procedural generation and puzzle-complexity modelling within light-based puzzle environments.

Chapter 2 reviews existing literature related to procedural content generation, puzzle generation, and puzzle-difficulty evaluation. Relevant studies are analysed to identify research gaps that motivate the development of CGRPG and MID.

Chapter 3 describes the proposed methodology, including the CGRPG framework, solver agent, MID metric, and evaluation design used throughout the study.

Chapter 4 presents the experimental results and statistical analyses. The findings are reported in relation to both research questions.

Chapter 5 discusses the results in the context of previous research, identifies strengths and limitations of the proposed approach, and interprets the significance of the findings.

Chapter 6 concludes the dissertation by summarising the main contributions of the study and outlining directions for future research.

Chapter 2 Literature Review

2.1 Procedural Content Generation

Procedural Content Generation (PCG) refers to the automated creation of game content using computational methods. PCG can generate levels, environments, mechanics, and other game elements while reducing development effort and increasing content variety (Shaker, Togelius and Nelson, 2016; Farrokhi Maleki and Zhao, 2024).

PCG approaches include constructive methods, search-based generation, machine learning techniques, and hybrid systems. Each approach offers different advantages and limitations between generation speed, controllability, and content diversity. Recent research has increasingly combined traditional PCG methods with artificial intelligence techniques to improve generation quality and adaptability (Farrokhi Maleki and Zhao, 2024).

Despite these advances, generating high-quality content remains challenging. Different PCG approaches often involve trade-offs between generation speed, controllability, content diversity, and computational requirements (Shaker, Togelius and Nelson, 2016; Farrokhi Maleki and Zhao, 2024).

Modern PCG systems are expected not only to generate valid content but also to control design properties such as difficulty, diversity, and player experience. These requirements are particularly important in puzzle generation, where content must remain both valid and solvable (Lazaridis and Fragulis, 2024; del Solar-Zavala and Barriga, 2026).

As a result, puzzle generation remains an active area of PCG research. The next section reviews procedural puzzle generation and examines the challenges of maintaining puzzle validity, solvability, and difficulty control.

2.2 Procedural Puzzle Generation

Procedural puzzle generation focuses on automatically creating puzzles that players can solve through reasoning and problem-solving. Unlike many other game assets, generated puzzles must remain both valid and solvable to provide a meaningful gameplay experience (del Solar-Zavala and Barriga, 2026).

A major challenge in puzzle generation is balancing diversity and correctness. Random generation can increase content variety, but it may also produce invalid or unsolvable puzzles. As a result, many puzzle generators use constraints or validation procedures to improve reliability (del Solar-Zavala and Barriga, 2026).

Two widely used approaches are constraint-based generation and generate-and-test generation. Constraint-based methods guide the construction process using predefined rules, while generate-and-test methods generate candidate puzzles before applying validation procedures. Although effective, validation-based approaches can increase computational cost when many generated puzzles fail verification (del Solar-Zavala and Barriga, 2026; McConnell and Zhao, 2025).

Difficulty is another important consideration in puzzle generation. An effective puzzle generation system should not only produce solvable puzzles but also provide control over puzzle difficulty. This becomes particularly important when puzzle complexity is influenced by interactions between multiple gameplay mechanics (Kristensen and Burelli, 2024).

These challenges highlight the need for puzzle generation approaches that maintain solvability while supporting difficulty control. The following section examines solvability in greater detail and reviews existing methods for ensuring that generated puzzles remain playable.

2.3 Solvability in Puzzle Generation

Solvability is a fundamental requirement in procedural puzzle generation. A generated puzzle must contain at least one valid solution; otherwise, it becomes unusable regardless of its structure or complexity. For this reason, maintaining solvability remains a central challenge in puzzle generation research (del Solar-Zavala and Barriga, 2026).

Existing approaches commonly use constraint-based generation or generate-and-test pipelines to ensure puzzle correctness. Constraint-based methods apply predefined rules during generation, whereas generate-and-test methods to validate puzzles after construction. Although effective, validation-based approaches may increase computational cost when many generated puzzles fail verification (del Solar-Zavala and Barriga, 2026; McConnell and Zhao, 2025).

Many recent systems still rely on post-generation validation. For example, SPaRC applies additional verification procedures to ensure that generated puzzle instances satisfy predefined requirements (Kaesberg et al., 2025). While these methods can improve reliability, they introduce an additional processing stage that may reduce generation efficiency.

Although post-generation validation can improve reliability, it does not prevent invalid puzzle configurations from being generated in the first place. Consequently, computational resources may still be spent creating puzzles that are ultimately discarded during verification. This limitation highlights a trade-off between generation flexibility and generation efficiency, which remains a common challenge in procedural puzzle generation research.

From a practical perspective, the choice between validation-based and generation-based solvability control depends on the objectives of the generation system. Validation-based approaches offer flexibility because generation and verification are performed independently, allowing a wider range of puzzle structures to be produced. However, computational resources may still be spent generating puzzles that are later discarded during verification. In contrast, generation-based approaches integrate solvability directly into the construction process, potentially reducing verification costs while imposing additional restrictions on puzzle structure. Consequently, procedural puzzle generation often involves a trade-off between computational efficiency, solvability control, and content diversity.

These limitations highlight the need for approaches that incorporate solvability directly into the generation process. By constructing valid solution paths during generation, it may be possible to reduce reliance on post-generation validation while maintaining puzzle correctness. This challenge motivates the first research question of this dissertation.

2.4 Difficulty Modelling and Player Engagement

Difficulty is an important factor in player experience. In procedural content generation, difficulty must be represented in a form that can be measured and controlled automatically. Existing studies commonly estimate difficulty using gameplay data, solver performance, or player analytics (Kristensen and Burelli, 2024).

Many approaches measure difficulty using structural features such as solution length, branching factor, search depth, or the number of puzzle elements. While these measures provide useful information about puzzle structure, they do not always capture how different mechanics interact during gameplay (Kristensen and Burelli, 2024; Lazaridis and Fragulis, 2024).

Recent research has highlighted the importance of controllable puzzle generation and interaction-based difficulty evaluation. For example, SPaRC includes an interaction score that considers the presence of multiple rule types within a puzzle, suggesting that puzzle difficulty may be influenced by relationships between mechanics rather than by structure alone (Kaesberg et al., 2025).

To explore this challenge, this study proposes Mechanic Interaction Density (MID) as a metric for describing puzzle characteristics based on mechanic interactions. Rather than assuming that MID fully explains puzzle difficulty, the study evaluates whether MID can help represent differences in puzzle complexity. This challenge motivates the second research question of the dissertation.

Many existing difficulty models rely on structural features because they are simple to calculate and compare across large puzzle datasets. Common examples include TotalObjects, solution length, and branching factor. These measures provide useful information about puzzle composition but may not fully represent how different mechanics interact during problem solving (Van Kreveld, Löffler and Mutser, 2015).

Recent studies have therefore explored broader perspectives on puzzle evaluation, suggesting that difficulty may emerge from multiple interacting factors rather than any single characteristic alone (Kristensen and Burelli, 2024). This observation motivates the investigation of mechanic-interaction descriptors in the present study and provides the foundation for the proposed MID metric.

2.5 Comparison of Existing Difficulty Modelling Approaches

Existing puzzle-difficulty models can generally be grouped into three categories: structural approaches, solver-based approaches, and interaction-based approaches. Each category attempts to describe puzzle complexity from a different perspective and therefore exhibits different strengths and limitations.

Structural approaches focus on measurable properties of a puzzle, such as Total Objects, branching factor, or solution length. These measures are relatively easy to calculate and are often used because they provide simple and interpretable descriptions of puzzle composition (Van Kreveld, Löffler and Mutser, 2015; Lazaridis and Fragulis, 2024). However, structural descriptors primarily represent the arrangement of puzzle elements and may not fully capture how those elements interact during problem solving.

Solver-based approaches evaluate puzzles through the behaviour of an automated solving system. Metrics such as solving time, search depth, or explored states are commonly used to estimate puzzle complexity because they directly reflect the effort required to find a solution (Kristensen and Burelli, 2024). These measures provide objective and repeatable evaluations, although the resulting difficulty estimates may depend on the characteristics of the solver itself rather than the puzzle alone.

Interaction-based approaches attempt to capture relationships between puzzle mechanics instead of focusing solely on puzzle structure or solver performance. Recent research suggests that interactions between mechanics can influence puzzle complexity and controllability, particularly in procedurally generated environments (Kaesberg et al., 2025). However, interaction-based descriptors remain relatively underexplored compared with traditional structural measures.

Table 1 summarises the strengths and limitations of these approaches. As shown, no individual method fully captures every aspect of puzzle complexity. This observation suggests that puzzle difficulty may be more appropriately viewed as a multidimensional concept requiring multiple complementary perspectives. The present study adopts this viewpoint by evaluating an interaction-based descriptor alongside established structural measures rather than treating them as competing alternatives.

Approach	Example Measures	Strengths	Limitations
Structural	Total Objects, solution length	Simple, interpretable, efficient	May overlook mechanic interactions
Solver-based	Solve time, search nodes	Objective and repeatable	Depends on solver characteristics
Interaction-based	MID, interaction metrics	Captures relationships between mechanics	Less established and requires validation

Table 1 Comparison of Puzzle Difficulty Modelling Approaches

Although these approaches are often presented as separate categories, they are not necessarily mutually exclusive. Structural measures, solver-based evaluations, and interaction-based descriptors each capture different characteristics of a puzzle. As a result, the strengths of one approach may compensate for limitations in another.

For example, structural descriptors provide efficient large-scale analysis but may overlook interaction effects, whereas solver-based evaluations can capture search effort without directly explaining why that effort occurs. Interaction-based approaches attempt to address this limitation by describing relationships between puzzle components, although such approaches require additional validation.

These observations suggest that puzzle difficulty is unlikely to be fully explained by any single evaluation method. Consequently, difficulty modelling may benefit from combining multiple complementary perspectives when analysing puzzle complexity.

2.6 Light-Based Puzzle Systems

Light-based puzzles require players to manipulate light paths to achieve a specific objective. Common mechanics include reflection and refraction, which alter the direction or behavior of light within a puzzle. These mechanics can create a variety of solution structures and reasoning challenges.

The interaction between reflection and refraction mechanics can create a wide range of possible solution structures. Simple puzzles may rely on a single mechanic, whereas more complex puzzles may require multiple mechanic transitions before a valid path can be established. Consequently, light-based puzzle environments provide an appropriate setting for investigating how interactions between mechanics influence puzzle structure and solver behaviour.

From an analytical perspective, light-based puzzles are particularly suitable for studying mechanic interactions because solution paths can be represented explicitly through object relationships. This characteristic makes it possible to examine not only whether a puzzle is solvable but also how different mechanics contribute to the construction of valid solutions.

From a procedural generation perspective, light-based puzzles must remain both solvable and meaningful. The placement of optical elements directly affects solution paths and puzzle difficulty. As a result, maintaining solvability while supporting mechanic interactions becomes an important challenge in puzzle generation.

This research focuses on grid-based light puzzles that combine reflection and refraction mechanics. These characteristics provide an appropriate domain for investigating puzzle solvability and interaction-based difficulty modelling. The next section reviews reverse generation approaches that can support these objectives.

2.7 Goal-Directed and Reverse Generation Approaches

Goal-directed generation starts from a predefined goal and works backwards during content construction. In puzzle generation, this approach allows solution requirements to be considered during generation rather than after construction has finished.

One form of goal-directed generation is reverse generation, where construction begins from a goal state and progresses towards an initial state. Because solution paths are incorporated during generation, reverse approaches can provide greater control over puzzle structure and solvability.

While reverse generation can improve control over solvability, it may also introduce limitations. Because generation begins from predefined goals, overly restrictive constraints can reduce the diversity of generated content or favour particular solution structures. As a result, reverse-generation systems must balance solvability guarantees with the ability to produce varied and interesting puzzle configurations.

Another consideration is that reverse generation systems often depend on assumptions about the desired solution structure before generation begins. As a result, the quality and diversity of generated puzzles may be influenced by how those assumptions are defined. Generators that rely heavily on predefined solution characteristics may achieve reliable solvability while producing a narrower range of puzzle configurations.

These trade-offs highlight an important design consideration within procedural puzzle generation. Methods that emphasise solvability may not always maximise diversity, while methods that prioritise diversity may increase the likelihood of invalid puzzle configurations. Consequently, generation approaches often require a balance between reliability and variety rather than optimising for a single objective.

The approach used in this dissertation follows the principle of reverse generation. The generator starts from the receiver and constructs a solution path towards the emitter while applying placement constraints throughout the process. This allows solvability to be considered during generation rather than through extensive post-generation validation.

These principles form the basis of the proposed Constraint-Guided Reverse Path Generation (CGRPG) approach, which is described in detail in Chapter 3.

2.8 Challenges in Puzzle Difficulty Evaluation

Evaluating puzzle difficulty remains a challenging problem because different evaluation methods often focus on different aspects of gameplay. Some studies rely on structural properties, while others examine solver performance or player behaviour (Van Kreveld, Löffler and Mutser, 2015; Kristensen and Burelli, 2024).

Structural measures provide a simple and efficient way to analyse puzzle complexity, but they may overlook interactions between puzzle mechanics. Solver-based evaluations offer objective and repeatable measurements, although they do not necessarily reflect how human players perceive difficulty. Consequently, puzzle difficulty is often viewed as a combination of structural, behavioural, and interaction-related characteristics rather than a single measurable property (Kristensen and Burelli, 2024).

These limitations suggest that puzzle difficulty is best viewed as a multidimensional concept rather than a single measurable property. This challenge motivates the investigation of approaches that combine structural, behavioural, and interaction-related information when analysing puzzle complexity.

2.9 Research Gaps

The literature review identified two challenges that remain insufficiently addressed in procedural puzzle generation. The first concerns maintaining puzzle solvability during generation without relying heavily on post-generation validation. Although existing approaches can produce valid puzzles, many still depend on additional verification or repair procedures after generation (Kaesberg et al., 2025; McConnell and Zhao, 2025).

The second challenge concerns difficulty modelling. Existing methods frequently use structural measures such as solution length or Total Objects, but fewer studies have examined how interactions between multiple mechanics contribute to puzzle difficulty (Kristensen and Burelli, 2024; Kaesberg et al., 2025).

To address these challenges, this study proposes Constraint-Guided Reverse Path Generation (CGRPG) for generating solvable light-based puzzles and investigates Mechanic Interaction Density (MID) as a metric for describing puzzle difficulty. These challenges form the basis of the two research questions presented in Chapter 1.

2.10 Summary

This chapter reviewed literature related to procedural content generation, procedural puzzle generation, puzzle solvability, and difficulty modelling. The review showed that maintaining solvability and controlling difficulty remain important challenges in procedural puzzle generation.

Existing approaches often rely on validation after generation and commonly estimate difficulty using structural measures.

Two research gaps were identified. The first concerns generating solvable puzzles during construction, while the second concerns modelling difficulty through interactions between puzzle mechanics. To address these challenges, this study proposes Constraint-Guided Reverse Path Generation (CGRPG) and investigates Mechanic Interaction Density (MID). The next chapter describes the design and implementation of the proposed framework.

Chapter 3 Methodology

3.1 Framework Overview

This chapter describes the proposed framework for generating light-based puzzles. The framework was developed in Unity and focuses on two objectives: generating solvable puzzles and investigating puzzle difficulty.

The framework consists of two main components. The first is a puzzle generator that constructs solution paths during generation. The second is a difficulty modelling component based on Mechanic Interaction Density (MID), which is used to analyze generated puzzles.

Generated puzzles are represented using a two-dimensional grid containing emitters, receivers, mirrors, and refractors. Puzzle solvability depends on constructing a valid path between the emitter and receiver.

The following sections describe the puzzle representation, the Constraint-Guided Reverse Path Generation (CGRPG) approach, the MID metric, and the evaluation method used in this study.

3.2 Game Overview and Spatial Representation

3.2.1 Grid Representation

The puzzle environment is represented using a two-dimensional grid. Each grid cell can contain a puzzle object or remain empty. This representation is used to manage object placement and support the generation process.

The grid contains several puzzle elements, including emitters, receivers, mirrors, refractors, walls, and doors. Each element occupies a single grid cell and follows predefined gameplay rules.

Using a grid-based representation simplifies object placement, constraint checking, and path construction. It also provides a consistent structure for puzzle generation and evaluation.

3.2.2 Puzzle Mechanics

The objective of the puzzle is to guide light from an emitter to a receiver. Players must arrange puzzle elements so that a valid light path connects the two objects.

The puzzle uses two main mechanics: reflection and refraction. Mirrors change the direction of the light path, while refractors modify the path according to predefined rules. Different combinations of these mechanics create different solution paths.

Puzzle complexity depends on the arrangement of puzzle elements and the sequence of mechanics encountered along the solution path. These mechanics form the basis of the generated puzzles used throughout this study.

3.2.3 Design Inspiration

The puzzle design was inspired by rule-based puzzle games such as *The Witness*. Similar to many grid-based puzzle games, players must reason about the relationships between puzzle elements to find a valid solution. However, the focus of this study is procedural puzzle generation rather than player interaction design.

Light-based puzzle systems are particularly suitable for procedural generation because puzzle solutions emerge from interactions between multiple mechanics. Unlike puzzles that depend primarily on spatial navigation or object collection, light-based puzzles require players to reason about relationships between components and how those relationships affect solution paths.

Reflection and refraction mechanics provide a useful foundation for procedural puzzle generation because they allow multiple valid interaction structures to exist within the same puzzle environment. Small changes in object placement can produce substantially different solution paths, making such systems well suited for investigating procedural generation and puzzle-complexity modelling.

These characteristics influenced the design of the puzzle environment used in this study. Rather than focusing on narrative progression or player progression systems, the environment was designed to emphasise mechanic interactions and solution-path construction.

3.3 Constraint-Guided Reverse Path Generation (CGRPG)

Constraint-Guided Reverse Path Generation (CGRPG) is the generation approach proposed in this study. The method constructs a solution path from the receiver towards the emitter while applying placement constraints during generation. By creating the solution path first, the framework aims to generate solvable puzzles without relying heavily on post-generation validation.

The reverse generation strategy was selected because puzzle solvability is a primary objective of this study. By starting from the receiver and constructing the solution path before completing the puzzle layout, the framework maintains a valid route between the receiver and emitter throughout generation. This reduces reliance on generating a complete puzzle first and verifying solvability afterwards.

The term *Constraint-Guided Reverse Path Generation* reflects the main characteristics of the approach:

- Constraint-Guided – puzzle elements are placed according to predefined constraints.
- Reverse – generation starts from the receiver and progresses towards the emitter.
- Path Generation – the solution path is constructed before the remaining puzzle elements are added.

A detailed description of the generation process is provided in Section 3.3.1.

3.3.1 Step-by-Step Generation Process

The CGRPG approach generates puzzles by constructing a solution path from the receiver towards the emitter. The process consists of five main stages.

Step 1: Receiver Placement

Generation begins by placing a receiver, which represents the goal state of the puzzle.

Step 2: Reverse Path Construction

The generator builds a solution path backwards from the receiver while remaining within the grid boundaries.

Step 3: Constraint Checking

Placement constraints are applied throughout generation to prevent invalid object placement and maintain a valid solution path.

Step 4: Mechanic Placement

Mirrors and refractors are placed along the path when required. Rule transitions between mechanics are also recorded for later MID calculation.

Step 5: Emitter and Puzzle Completion

When the required path length is reached, an emitter is placed at the end of the path. Optional decoy objects may then be added to increase puzzle complexity.

3.3.2 Algorithm Flowchart

Figure 1 illustrates the overall CGRPG process. The generator places a receiver, constructs a solution path in reverse while applying constraints, places puzzle mechanics, and completes the puzzle by adding an emitter and optional decoy objects.

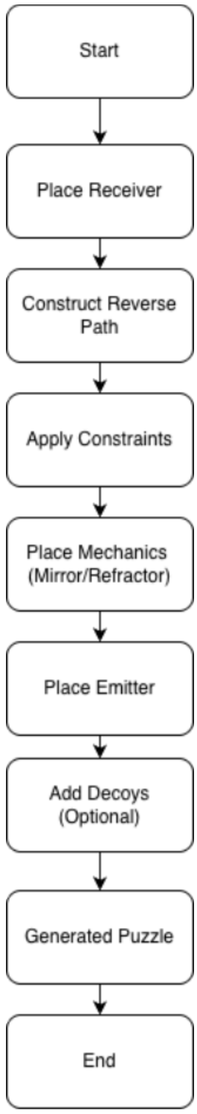

Figure 1 Flowchart of the Constraint-Guided Reverse Path Generation (CGRPG) Process.

3.3.3 Example Puzzle Environment

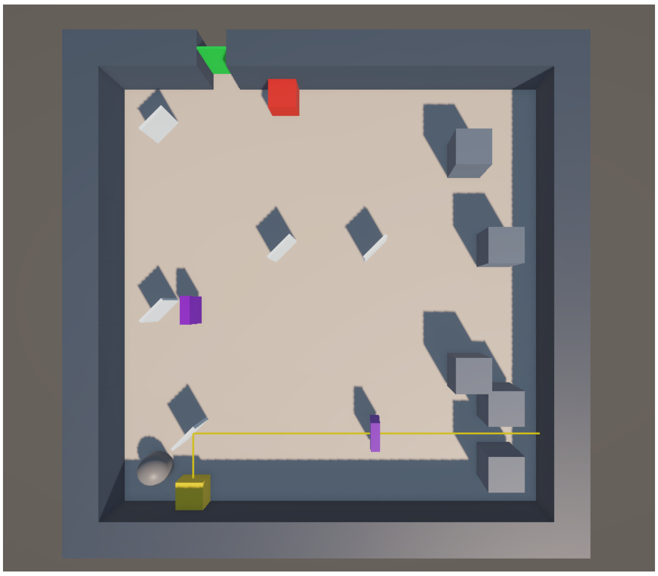

Figure 2 Example puzzle environment used in the proposed framework.

Figure 2 illustrates an example puzzle environment generated by the proposed framework. The environment contains an emitter (yellow cube), a receiver (red cube), mirrors (transparent rectangular blocks), refractors (purple rectangular blocks), obstacle pillars (grey columns), a door mechanism (green door), and a solver agent represented by the capsule model.

The objective of the puzzle is to direct light from the emitter to the receiver through interactions with mirrors and refractors. A puzzle is considered successfully solved when light reaches the receiver, triggering the door mechanism and allowing progression. Different arrangements of puzzle elements can produce different solution paths and interaction structures.

Obstacle pillars act as static environmental constraints within the puzzle space. Unlike mirrors and refractors, obstacle pillars do not directly participate in light interactions. Instead, they restrict movement, object placement, and available spatial configurations. As a result, obstacle pillars influence puzzle layouts indirectly by shaping the environment in which mechanic interactions occur.

The puzzle environment combines both interactive and non-interactive elements. Interactive components influence the behaviour of light paths directly, whereas environmental objects affect the spatial structure of the puzzle. Together, these elements create diverse puzzle configurations that can vary in solution structure, interaction patterns, and solver requirements.

This example demonstrates the core mechanics investigated in the study. Although the puzzle environment is constructed from a relatively small set of object types, different combinations and arrangements can produce substantially different interaction structures. These interaction structures provide the conceptual basis for the Mechanic Interaction Density (MID) metric described in the following section.

3.3.4 Puzzle Object Definitions

The proposed puzzle environment is constructed from a set of interactive objects that collectively form puzzle configurations. Each object fulfils a specific role within the puzzle and contributes to the interaction structures analysed throughout this study.

1.Emitter

The emitter is the source of the light beam within a puzzle. Every generated puzzle contains at least one emitter that produces light in a predefined direction. The objective of the puzzle is ultimately based on successfully directing light from the emitter to the intended receiver through interactions with other puzzle components.

2.Receiver

The receiver represents the primary puzzle objective. A puzzle is considered successfully solved when the receiver receives light from the emitter through a valid interaction path. The receiver therefore functions as the endpoint of the puzzle solution and provides a clear success condition for both the solver agent and the generation framework.

3.Mirror

Mirrors alter the direction of incoming light through reflection. By changing the trajectory of a light beam, mirrors enable puzzles to create indirect solution paths that cannot be achieved through direct emitter-to-receiver connections. Mirrors therefore increase the range of possible interaction structures that can occur within a puzzle.

4.Refractor

Refractors modify light behaviour differently from mirrors by redirecting or transforming light paths according to predefined interaction rules. As a result, refractors introduce additional mechanic diversity and can increase the number of interaction possibilities available within a puzzle configuration.

5.Door

Doors function as progression elements that change state based on puzzle conditions. In the prototype environment, a door may open when the receiver successfully receives light from the emitter. Doors therefore provide a visible representation of puzzle completion and reinforce the relationship between mechanic interactions and puzzle progression.

Together, these objects form the interaction structures analysed by MID. Rather than examining Total Objects alone, the proposed metric focuses on how puzzle components interact with one another to produce valid solution paths and puzzle behaviours.

3.3.5 Puzzle Mechanics Interaction Examples

Puzzle behaviour emerges through interactions between puzzle objects rather than from the presence of individual components alone. Consequently, understanding how mechanics interact is important for interpreting both generated puzzle complexity and the MID metric proposed in this study.

A simple interaction example consists of an emitter, a mirror, and a receiver. In this configuration, light emitted from the emitter is reflected by the mirror before reaching the receiver. Although only a small number of objects are involved, successful puzzle completion depends on the interaction between multiple components rather than on any individual object acting independently.

More complex interactions can occur when additional mechanics are introduced. For example, a puzzle may require light to pass through a refractor before being redirected by one or more mirrors toward the receiver. In this situation, solving the puzzle depends on coordinating multiple interaction steps that occur across different mechanic types.

Interaction complexity may also arise from alternative solution possibilities. Different object arrangements can create multiple potential interaction pathways, some of which may successfully direct light to the receiver while others fail. As a result, puzzles with similar numbers of objects may exhibit substantially different interaction structures and search requirements.

These examples illustrate the distinction between object-based and interaction-based descriptions of a puzzle. Object-based measures primarily describe what components are present, whereas interaction-based approaches examine how those components influence one another during puzzle solving. This distinction provides the conceptual motivation for MID, which was designed to describe interaction-related characteristics within generated puzzles.

Interaction structures may also vary in length. Some puzzles require only a small number of mechanic interactions before reaching a valid solution, whereas others require longer chains involving multiple transitions between mirrors and refractors. As interaction chains become longer, the number of potential solution states may increase, creating additional search requirements for both players and automated solvers.

From an analytical perspective, these interaction chains provide a useful way of examining differences between puzzles. Two puzzles may contain similar numbers of objects while exhibiting substantially different interaction structures depending on how those objects are connected throughout the solution path.

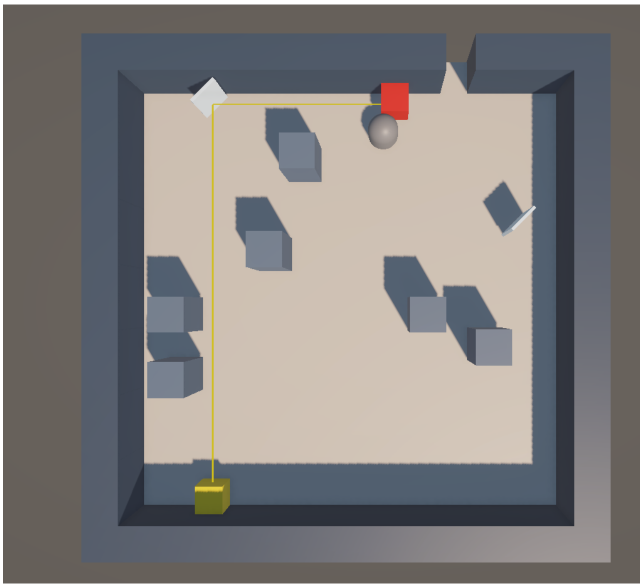

Figure 3 Emitter -> Mirrors -> Receiver

Figure 3 illustrates a simple interaction structure consisting of an emitter, a mirror, and a receiver. In this configuration, the mirror serves as an intermediary object that redirects the light path towards the receiver. Although only a small number of objects are involved, successful completion still depends on the correct interaction between multiple components rather than on direct object connectivity alone.

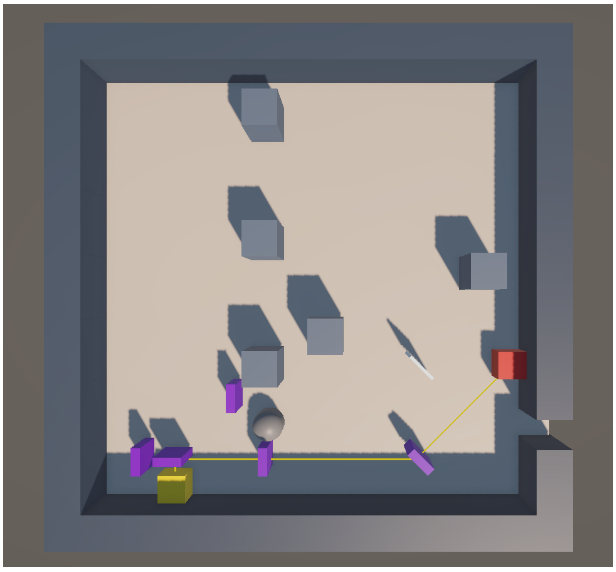

Figure 4 Emitter -> Refactors -> Receiver

Figure 4 demonstrates a configuration involving a refractor. Compared with the previous example, the refractor introduces an additional mechanic that alters the behaviour of the light path. Consequently, the interaction structure becomes more diverse, requiring the puzzle solution to account for multiple mechanic types rather than a single interaction rule.

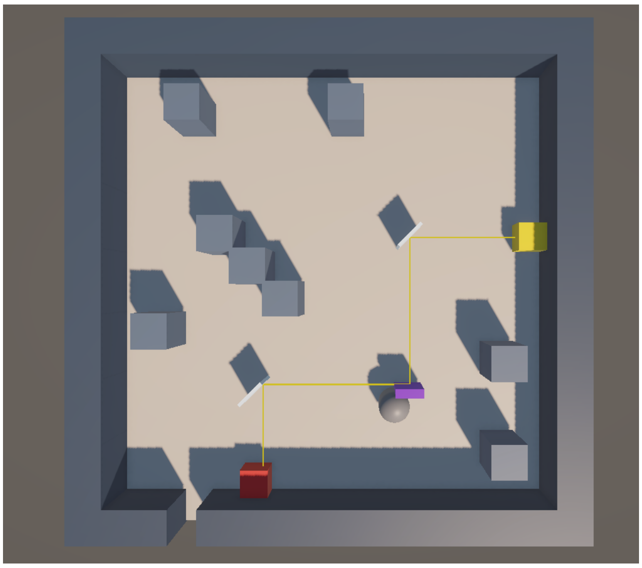

Figure 5 Emitter ->Mirrors -> Refactors -> Receiver

Figure 5 illustrates a more complex interaction structure that combines mirrors and refractors within the same solution path. Such configurations may contain multiple interaction stages and transitions between mechanic types. As a result, puzzles with similar object counts can still exhibit substantially different interaction structures depending on how puzzle mechanics are connected throughout the solution path.

3.3.6 Generation Constraints

The generation process applies several constraints to maintain puzzle validity and reduce the likelihood of producing trivial or invalid configurations.

First, all puzzle objects must remain within grid boundaries throughout generation.

Second, solution-path construction must preserve a continuous route between the receiver and the emitter.

Third, object placement must avoid invalid overlaps and conflicting mechanic configurations.

Finally, optional decoy objects may be introduced only when they do not invalidate the constructed solution path.

These constraints were intended to guide generation toward valid puzzle configurations while preserving variation in puzzle structure and interaction patterns.

3.4 Difficulty Modelling: Mechanic Interaction Density (MID)

This study investigates whether interactions between puzzle mechanics can help describe puzzle difficulty. To support this investigation, a metric called Mechanic Interaction Density (MID) is proposed. MID focuses on mechanic interactions rather than relying only on simple measures such as path length or Total Objects.

The purpose of MID is not to define puzzle difficulty directly. Instead, it is used to examine whether mechanic interactions can help explain differences between generated puzzles.

3.4.1 Motivation for MID

Difficulty modelling is a common challenge in procedural puzzle generation. Previous studies have measured difficulty using structural features such as solution length, branching factors, or puzzle entropy (Chen, White, and Sturtevant, 2023). These measures provide useful information about puzzle structure, but they do not always consider how multiple mechanics interact within a puzzle.

The MID metric is proposed as an experimental measure for examining mechanic interactions. The goal is not to replace existing difficulty measures but to investigate whether mechanic interactions contribute useful information when describing puzzle complexity.

3.4.2 Mathematical Formulation of the MID

The MID score is calculated as a weighted combination of three puzzle features:

$MID = \alpha Cl + \beta Ch + \gamma Ci$

Where:

- Cl = Component Density.
- Ch = Rule Heterogeneity.
- Ci = Component Complexity.
- α, β, γ = weighting coefficients
- $\alpha + \beta + \gamma = 1$

The selected weights are used for evaluation and do not represent fixed difficulty values.

3.4.3 Feature Metric Definitions

Component Density (Cl)

Component Density measures how many objects are placed within a puzzle. Component Density is calculated from the initial puzzle state at generation time. Since mirrors and refractors can be relocated during solving, the measured density does not necessarily remain constant throughout puzzle interaction.

$$Cl = \frac{Ntotalobjects}{\text{GridSize}}$$

Higher values indicate a greater number of puzzle objects.

Rule Heterogeneity (Ch)

Rule Heterogeneity measures how often the solution path changes between mechanic types.

$$Ch = \frac{Rule\ Transitions}{Nsolutionobjects - 1}$$

Higher values indicate more mechanic transitions.
Component Complexity (Ci)
Component Complexity measures the proportion of refractors within the solution path.

$$Ci = \frac{Nrefractors}{Nsolutionobjects}$$

Higher values indicate greater use of refractors.

3.4.4 Rationale for MID Components

The MID metric was designed to capture mechanic-interaction characteristics that may not be fully represented by traditional structural measures. Three components were selected: Component Density (Cl), Rule Heterogeneity (Ch), and Component Complexity (Ci). Each component describes a different aspect of puzzle interactions and collectively forms the overall MID score.

Component Density (Cl) measures the concentration of interactive puzzle elements within a generated puzzle. The rationale for including this component is that puzzles containing larger numbers of interacting objects may provide more opportunities for mechanic interactions to occur. As the number of interactive elements increases, the number of possible relationships between those elements may also increase. Consequently, Component Density was included to represent the overall interaction potential within a puzzle.

Rule Heterogeneity (Ch) measures the diversity of mechanics present in a puzzle. Puzzles that contain only one mechanic type may require repeated application of similar actions, whereas puzzles that combine multiple mechanics may require players or solvers to consider a wider range of behaviours and interactions. The purpose of this component is therefore to represent variation in mechanic types rather than the number of objects alone.

Component Complexity (Ci) measures the complexity of interaction structures created by puzzle mechanics. This component was included because some puzzles may contain relatively few objects while still requiring multiple interaction steps before a solution can be achieved. By incorporating interaction complexity, MID attempts to represent the organisational structure of mechanic relationships rather than simply their quantity.

These three components were selected because they represent complementary perspectives on puzzle interactions. Density reflects how many interactions may occur, heterogeneity describes the variety of mechanics involved, and complexity represents how interactions are organised. Combining the three components allows MID to provide a broader description of mechanic interactions than could be obtained from any individual component alone.

3.4.5 Assumptions of MID

The design of MID is based on several assumptions regarding mechanic interactions within light-based puzzles. These assumptions do not claim to directly measure puzzle difficulty. Instead, they provide a conceptual foundation for describing interaction-related characteristics that may influence solver behaviour.

The first assumption is that puzzles containing a greater concentration of interactive elements are likely to provide more opportunities for mechanic interactions to occur. As the number of interactive components increases, the number of possible relationships between those components may also increase. This assumption motivated the inclusion of Component Density (Cl).

The second assumption is that puzzles containing a wider variety of mechanics may require consideration of a broader range of interactions. Puzzles that rely on a single mechanic often involve repeated application of similar actions, whereas puzzles combining multiple mechanics may require additional coordination between puzzle elements. This assumption motivated the inclusion of Rule Heterogeneity (Ch).

The third assumption is that the organisation of interactions may contribute to puzzle complexity independently of the number of objects present. Some puzzles may contain relatively few components while still requiring multiple interaction steps before a solution can be achieved. This assumption motivated the inclusion of Component Complexity (Ci).

Together, these assumptions suggest that mechanic interactions can be described from multiple complementary perspectives. Consequently, MID combines density, heterogeneity, and complexity in an attempt to provide a broader description of interaction-related puzzle characteristics than any individual component alone.

3.4.6 Example MID Calculation

To illustrate the calculation process, this section presents a simplified example of how MID is computed for a generated puzzle.

Consider a puzzle containing an emitter, a receiver, two mirrors, and one refractor. The puzzle requires interactions between these components to direct light successfully from the emitter to the receiver. Based on the definitions introduced previously, values for Component Density (Cl), Rule Heterogeneity (Ch), and Component Complexity (Ci) are first calculated from the puzzle configuration.

Suppose the resulting values are:

Cl = 0.60

Ch = 0.50

Ci = 0.80

Using the weighting configuration adopted in this study (0.50, 0.30, and 0.20 respectively), the MID value can be computed as follows:

MID = (0.50 × Cl) + (0.30 × Ch) + (0.20 × Ci)

MID = (0.50 × 0.60) + (0.30 × 0.50) + (0.20 × 0.80)

MID = 0.30 + 0.15 + 0.16

MID = 0.61

In this example, the resulting MID score is 0.61. This value does not directly represent puzzle difficulty. Instead, it represents an interaction-based description of the puzzle configuration derived from the density, diversity, and complexity of mechanic interactions. MID values obtained from generated puzzles can subsequently be compared with solver-performance measures such as Solve Time and Search Nodes to investigate whether interaction-related characteristics are associated with differences in solver effort.

In practical terms, different MID values represent different interaction characteristics. Lower MID values generally indicate puzzles containing fewer mechanic interactions, lower mechanic diversity, or simpler interaction structures. Higher MID values indicate greater concentrations of interactions, larger numbers of mechanic transitions, or more complex interaction arrangements.

It is important to note that different MID values do not automatically imply greater or lower puzzle difficulty. Instead, MID is intended to describe interaction-related characteristics that can subsequently be compared with external measures such as Solve Time and Search Nodes.

This example highlights one of the primary motivations for MID. Traditional object-count measures may identify how many components are present within a puzzle, but they do not necessarily describe how those components interact. By incorporating interaction density, mechanic diversity, and interaction complexity, MID attempts to provide a richer description of puzzle structure that extends beyond simple object-count representations. Consequently, two puzzles may contain similar numbers of objects while exhibiting different interaction structures and solver requirements. The metric therefore serves as an analytical tool for investigating interaction-related characteristics and their relationships with solver behaviour rather than as a direct measure of puzzle difficulty.

3.5 Model Calibration and Evaluation Setup

3.5.1 Weight Selection

The MID metric combines three components: Component Density (Cl), Rule Heterogeneity (Ch), and Component Complexity (Ci). A pilot calibration process was conducted to determine an initial weighting configuration for evaluation.

The pilot study used 50 generated puzzle instances that were separate from the main evaluation dataset. Correlation analysis was performed to examine the relationships between the individual MID components and external solver-performance measures, including Solve Time and Search Nodes.

The pilot results indicated that Component Density (Cl) showed the strongest overall relationship with the selected solver-effort measures, followed by Component Heterogeneity (Ch) and Component Complexity (Ci). Based on this ordering, a greater weighting was assigned to Cl than to Ch, while Ci received the lowest weighting. Several candidate weighting configurations satisfying the constraint $\alpha + \beta + \gamma = 1$ were then evaluated. The configuration $\alpha = 0.5$, $\beta = 0.3$, and $\gamma = 0.2$ demonstrated the strongest overall alignment with the selected solver-performance measures (The correlation between MID and Solve Time, and the correlation between MID and Search Node, are at their highest values.) and was therefore adopted for the main evaluation.

Similar to previous difficulty-modelling studies that explored feature weighting through empirical observations (Van Kreveld, Löffler and Mutser, 2015), the purpose of the calibration process was to determine a reasonable starting configuration rather than to establish a validated difficulty model. Consequently, the selected coefficients should be regarded as exploratory and specific to the current experimental setting.

3.5.2 Pilot Calibration

A pilot study using 50 generated puzzles was conducted to verify that puzzle generation and MID calculations functioned correctly. The pilot results confirmed that the framework produced puzzles with different structures and MID values. These puzzles were not included in the final evaluation. The pilot dataset was used only for calibration purposes and was excluded from the final evaluation to avoid circular evaluation effects.

3.5.3 Main Evaluation Setup

The main evaluation used 950 generated puzzles. For each puzzle, the framework recorded:

- MID Score
- Solve Time (ms)
- Search Nodes

These measurements were used to examine relationships between puzzle characteristics and solver performance. Correlation analysis was performed to evaluate whether MID was associated with differences in puzzle-solving behavior.

For each generated puzzle, MID values were first calculated from the generated puzzle state. The puzzle was then evaluated using the solver agent to determine solvability and record solver-performance measurements. Finally, statistical analyses were performed using the collected measurements to investigate relationships between interaction characteristics, structural descriptors, and solver effort.

This workflow ensured that all generated puzzles were evaluated under identical conditions, allowing comparisons to be performed consistently across the full dataset.

3.5.4 Generated Puzzle Dataset

The evaluation was conducted using a dataset of 950 generated puzzles produced by the proposed Constraint-Guided Reverse Path Generation (CGRPG) framework. The dataset was created to provide a sufficiently large collection of puzzle instances for analysing puzzle solvability, solver performance, and mechanic interaction characteristics.

Each generated puzzle was constructed using combinations of the puzzle objects described previously, including emitters, receivers, mirrors, refractors, and other interactive components. Although all puzzles shared the same underlying mechanics, variations in object placement and interaction structures resulted in diverse puzzle configurations with different solution requirements and search characteristics.

The dataset was intended to capture a broad range of puzzle structures rather than a single puzzle type. Consequently, generated puzzles differed in terms of object composition, interaction patterns, and solution arrangements. This variability was important because the study aimed to examine whether interaction-based information remained useful across multiple puzzle configurations rather than within a small set of manually designed examples.

A generated dataset was selected in preference to a small collection of handcrafted puzzles for two reasons. First, a larger dataset reduces the likelihood that observed relationships are dominated by a small number of individual puzzle instances. Second, statistical analyses such as correlation testing, Steiger's Z tests, and incremental contribution analyses require sufficient variation across puzzle characteristics to produce meaningful comparisons.

The resulting dataset formed the basis of all subsequent evaluations. Each puzzle was analysed using the solver agent, structural baseline measures, and the proposed Mechanic Interaction Density (MID) metric. The collected measurements were then used to investigate relationships between puzzle characteristics and solver-performance outcomes.

The use of a procedurally generated dataset also supports reproducibility and consistency throughout the evaluation process. Because all puzzles were produced using the same generation framework and constraint system, differences observed during analysis are more likely to reflect variations in puzzle structure and interaction characteristics rather than inconsistencies in puzzle design. This consistency was important for examining relationships between MID, structural descriptors, and solver-performance measures across the full dataset.

3.5.5 Solver Agent

To evaluate generated puzzles, this study uses a custom solver agent implemented in Unity. The solver provides a consistent method for assessing puzzle solvability and measuring solver performance across a large number of generated puzzles.

The solver operates under predefined search procedures and stopping conditions. If a valid solution is not found before the search limits are reached, the puzzle is recorded as unsolved.

During evaluation, the solver records two performance measures: Solve Time and Search Nodes. Solve Time represents the time required to identify a valid solution, whereas Search Nodes represent the amount of search effort performed during the evaluation process. These measures are subsequently used to analyze relationships between generated puzzle characteristics and solver performance.

3.5.6 Solver Search Phases

The solver evaluates generated puzzles through a sequence of predefined search phases that gradually increase search effort. This phased approach allows puzzles of varying complexity to be analysed within a consistent evaluation framework.

The initial phases focus on relatively simple solution possibilities, whereas later phases perform broader exploration when a solution cannot be found. As the search progresses, the number of explored configurations increases, requiring greater computational effort to identify valid solutions.

The phased search strategy enables differences in solver effort to be observed across generated puzzles. Puzzles solved during earlier phases generally require less exploration, whereas puzzles reaching later phases tend to produce larger Search Node counts and longer Solve Times. Consequently, the resulting measurements provide useful indicators of solver effort throughout the evaluation process.

3.5.7 Solver Design Rationale

The custom solver was developed to provide a repeatable method for evaluating generated puzzles. Human-player studies can provide valuable insights into perceived difficulty, but they require participant recruitment, controlled testing conditions, and additional data collection. In contrast, a solver-based approach allows puzzle characteristics to be analysed efficiently across a large, generated dataset.

A solver-based evaluation was selected because the primary objective of this study was to investigate relationships between puzzle characteristics and solver effort. Although solver behaviour cannot fully represent human experience, it provides an objective basis for comparing generated puzzles within the scope of this research.

Consequently, the solver was used as an analytical tool for investigating puzzle complexity rather than as a direct model of human puzzle-solving behaviour.

3.5.8 Structural Baseline Comparison Design

In addition to evaluating MID against solver-performance measures, a comparison study was conducted using structural baseline measures. The purpose of this comparison was to determine whether MID provided information beyond simple structural characteristics of generated puzzles.

Two baseline measures were selected: Total Objects and Solution Objects. Total Objects represents the overall number of puzzle elements in a generated puzzle, whereas Solution Objects represents the number of objects involved in the solution path. These measures were chosen because object-count-based descriptors are commonly used as simple indicators of puzzle complexity in previous puzzle-generation and difficulty-modelling studies (Van Kreveld, Löffler and Mutser, 2015; Kristensen and Burelli, 2024).

Pearson correlation analysis was used to compare MID and the baseline measures with two solver-performance variables: Solve Time and Search Nodes. To determine whether differences between correlation coefficients were statistically meaningful, Steiger's Z test was applied.

Finally, an Incremental Contribution Analysis was performed to examine whether MID provided additional explanatory value beyond object-count measures alone. Together, these analyses were used to evaluate whether mechanic-interaction information offered a different perspective on puzzle complexity when compared with traditional structural descriptors.

The evaluation design was intended to examine puzzle complexity from multiple perspectives. Solver-performance measures were used to represent search effort, while structural descriptors were included to provide baseline comparisons. Combining these measures allowed the study to investigate whether interaction-based information offered additional insight beyond traditional object-count approaches.

3.6 Threats to Validity

Several factors may influence the validity of the results presented in this study.

First, the evaluation relies on a custom solver agent rather than human participants. Although the solver provides a consistent and repeatable evaluation process, the recorded measurements primarily reflect solver behaviour and may not fully represent human puzzle-solving experiences. Consequently, relationships identified between puzzle characteristics and solver effort should not be interpreted as direct indicators of player-perceived difficulty.

Second, the study focuses on a single puzzle domain consisting of light-based mechanics. The framework, solver, and MID metric were developed and evaluated within this specific environment. As a result, the findings may not generalise directly to puzzle systems that contain substantially different mechanics or interaction structures.

Third, the MID metric is based on a set of design assumptions regarding mechanic interactions. While these assumptions provide a conceptual basis for describing interaction-related puzzle characteristics, they have not been independently validated through human-player studies. Therefore, MID should be interpreted as an exploratory descriptor rather than a validated difficulty measure.

Finally, the weighting configuration used in MID was selected through a calibration process conducted within the scope of this study. Although sensitivity analysis suggested that the observed relationships were relatively stable across alternative weighting configurations, the selected coefficients should not be considered universally optimal.

To reduce these threats, the study employed a consistent evaluation framework, structural baseline comparisons, and statistical analyses. Nevertheless, the findings should be interpreted within the scope and limitations of the current experimental setting.

Chapter 4 Results

4.1 Overview

This chapter presents the results of the proposed framework. The analysis focuses on two objectives: evaluating whether CGRPG can generate solvable puzzles and investigating whether MID is associated with differences in puzzle-solving behaviour. Results are presented using data collected from 950 generated puzzles and analysed using the custom solver agent.

The results presented in this chapter address both research questions introduced in Chapter 1. The first part examines the ability of CGRPG to generate solvable puzzles, whereas the second part investigates relationships between MID and solver-performance measures.

Results are first presented descriptively before statistical analyses are applied. Correlation analysis, Steiger's Z testing, and incremental contribution analysis are subsequently used to examine whether MID provides information beyond traditional structural puzzle descriptors.

4.2 RQ1: Solvability Evaluation

4.2.1 Generated Puzzle Summary

This section evaluates whether the proposed Constraint-Guided Reverse Path Generation (CGRPG) approach can generate solvable light-based puzzles. A total of 950 puzzles were generated and evaluated using the custom solver agent described in Chapter 3.

Table 2 presents a summary of the generated puzzle dataset.

Metric	Value
Generated Puzzles	950
Solved Puzzles	459
Unsolved Puzzles	491
Success Rate	48.3%

Table 2 Generated Puzzle Summary

4.2.2 Solvability Results

The results show that 459 out of 950 generated puzzles were successfully solved by the solver agent, corresponding to a success rate of 48.3%. The remaining 491 puzzles could not be solved during evaluation.

These results indicate that the proposed framework was capable of generating a substantial number of solvable puzzles. However, successful puzzle generation was not achieved across the entire dataset.

The solver operated under predefined search procedures and stopping conditions. Therefore, an unsuccessful solver outcome indicates that the solver was unable to find a solution within the evaluation process, rather than proving that no valid solution exists.

4.2.3 Discussion for RQ1

The results provide partial support for RQ1. The CGRPG approach was able to generate many puzzles that could be solved by the solver agent, demonstrating that solvability can be incorporated into the generation process.

However, the overall success rate of 48.3% indicates that constructing a solution path during generation does not guarantee that every generated puzzle will be solved successfully. Factors such as puzzle layout, mechanic placement, and solver limitations may contribute to unsuccessful outcomes.

These findings suggest that further refinement of the generation process may be required to improve the proportion of solvable puzzles produced by the framework.

4.3 RQ2: MID Evaluation

4.3.1 Overview

This section evaluates whether Mechanic Interaction Density (MID) can help describe puzzle difficulty. The analysis uses MID scores together with data collected from the solver agent, including solve time, search nodes, and solve phase.

Measure	Description
MID Score	Mechanic Interaction Density
Solve Time	Time required to solve a puzzle
Search Nodes	Number of nodes explored by the solver

Table 3 Summaries the evaluation measures used in this study

4.3.2 MID Distribution

Table 4 summarises the distribution of MID scores across the generated puzzle dataset. The average MID score was 0.2085, with values ranging from 0.0039 to 0.447. Figure 2 shows that most puzzles were concentrated in the lower and the middle MID ranges, while relatively fewer puzzles appeared at the lowest or highest ranges. These results indicate that highly interactive puzzle configurations occurred less frequently than simpler interaction structures within the generate dataset. It contained a broad range of mechanic-interaction levels, providing a suitable basis for subsequent analyses.

Measure	Value
Mean MID	0.2085
Minimum MID	0.0039
Maximum MID	0.447

Table 4 MID Statistic

The MID values reported in this section were calculated using the calibrated weighting configuration described in Section 3.5.1 ($\alpha = 0.5$, $\beta = 0.3$, $\gamma = 0.2$). Figure 6 shows that most generated puzzles were concentrated within the middle MID ranges, while relatively few appeared at either end of the distribution. This variation provides a suitable basis for examining potential relationships between MID and solver-performance measures in the following sections. The observed distribution also indicates that the generation process produced puzzles with varying levels of mechanic interaction rather than concentrating solely on a narrow range of MID values. This variation is important because correlation analysis requires sufficient diversity across the independent variable to identify potential relationships with solver-performance measures. The presence of puzzles spanning both lower and higher MID ranges therefore provides a suitable basis for evaluating whether mechanic-interaction characteristics are associated with solving behaviour.

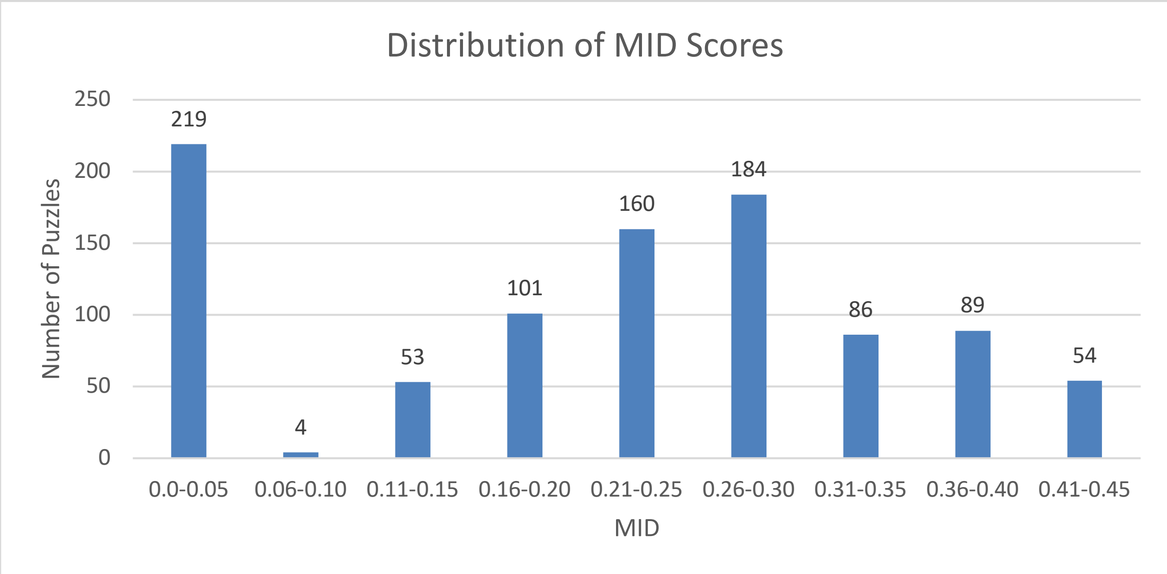

Figure 6 Distribution of MID Scores

4.3.3 Solver Outcomes

The solver records the phase at which a puzzle was solved or whether it remained unsolved. Earlier phases generally indicate less solver effort, whereas later phases require additional search or fallback procedures. Figure 7 illustrates the distribution of solver outcomes across the generated puzzle dataset.

- Trivial – solved without object movement or rotation.
- 1A – solved using rotation-based search.
- 1B – solved using relocation-based search.
- Sweep-S1 – solved using an additional rotation search.
- Sweep-S2 – solved after relocation followed by rotation search.
- Sweep-S3 – solved using an exhaustive rotation search.
- Sweep – no solution was found.
- None – no valid solver outcome was recorded.

The distribution of solver outcomes also suggests that generated puzzles exhibited different levels of search complexity. While a subset of puzzles could be solved during earlier phases with relatively limited exploration, a large proportion of puzzles required progression to later search procedures or remained unsolved. This variation indicates that the generated dataset contains puzzles with substantially different solving requirements, providing a useful basis for evaluating relationships between puzzle characteristics and solver effort.

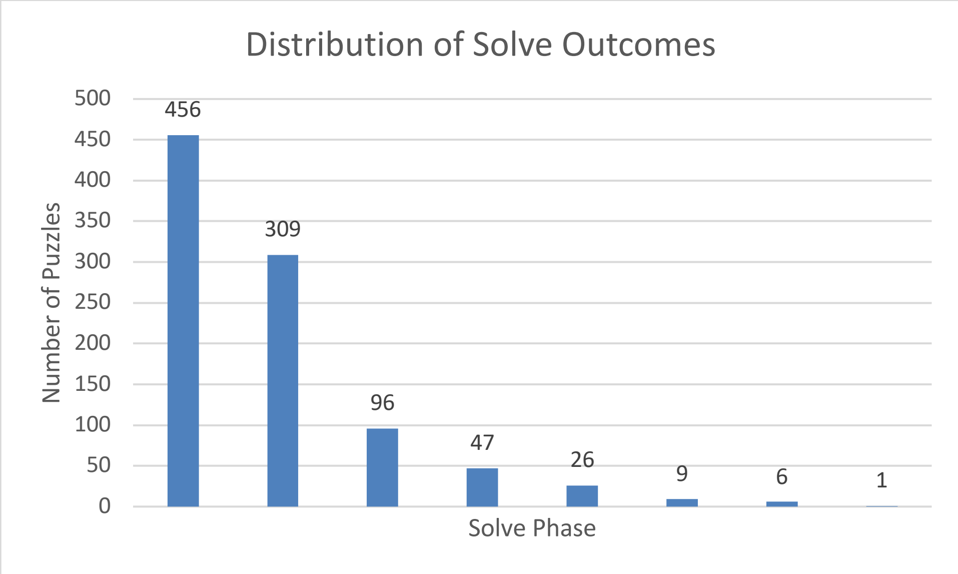

Figure 7 Distribution of Solver Outcomes

Table 5 summarizes solver outcomes for all generated puzzles. Most successful puzzles were solved during the Sweep-S1 phase (309 puzzles), followed by the 1B phase (96 puzzles). In contrast, 456 puzzles reached the Sweep phase without a successful solution. Overall, the results show substantial variation in solver effort across the generated dataset.

Solve Phase	Successful Puzzle outcomes	Failed Puzzle outcome	Total SolvePhase
1A	13	34	47
1B	96	0	96
Sweep	0	456	456
Sweep-S1	309	0	309
Sweep-S2	9	0	9
Sweep-S3	6	0	6
Trivial	26	0	26
None	0	1	1
Total	459	491	950

Table 5 Successful and Failed Puzzle Outcome

Most solved puzzles were completed during the Sweep-S1 phase (309 puzzles), followed by the 1B phase (96 puzzles). In contrast, 456 puzzles reached the Sweep phase without a successful solution.

4.3.4 Relationship Between MID and Solver Performance

To investigate RQ2, MID scores were compared with two solver-performance measures: Solve Time and Search Nodes. Solve Time represents the time required for the solver to find a solution, whereas Search Nodes indicate the amount of search effort performed during problem solving. Pearson correlation analysis was used to examine the strength of the relationship between MID and these measures.

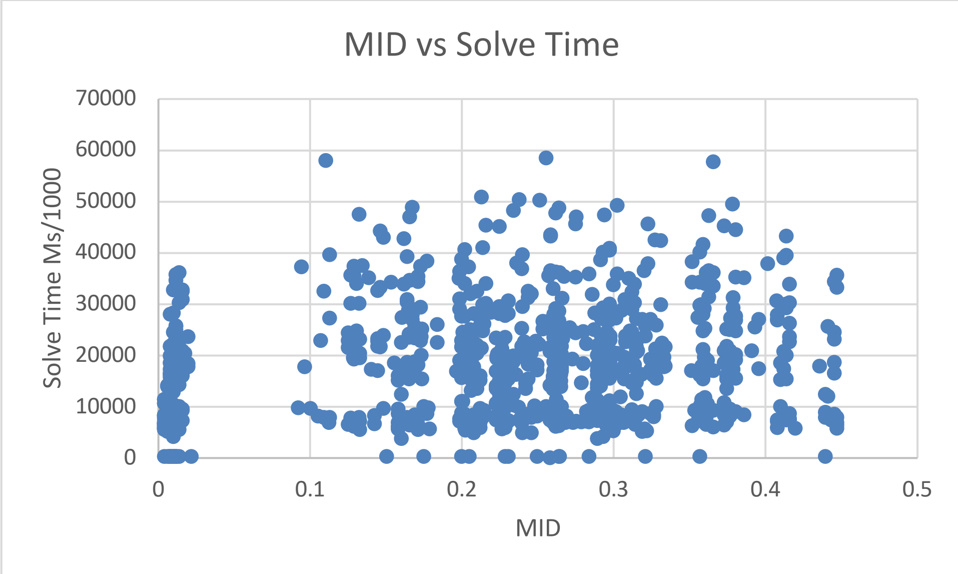

Figure 8 MID vs Solve Time

Figure 8 shows the relationship between MID and Solve Time. The results indicate considerable variation in solve time across different MID values. No clear linear relationship between MID and solve time was visually observed. Puzzle with similar MID scores often required different amounts of solving time. Solve time was recorded milliseconds and converted to seconds for presentation purposes. Correlation analysis also indicated only a weak positive relationship between MID and solve time (r=0.1670), which is consistent with the visual pattern observed in the figure.

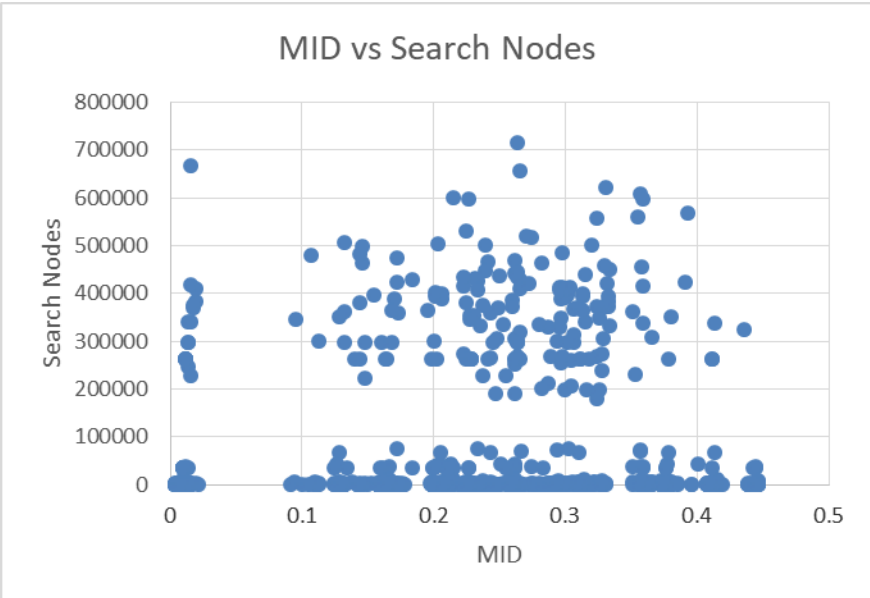

Figure 9 MID vs Search Nodes

Figure 9 presents the relationship between MID and Search Nodes. Similar to Solve Time, the relationship was positive but relatively weak (r = 0.0974). This result suggests that puzzles with higher MID values tended to require greater search effort, although MID alone does not fully explain solver behaviour.

4.3.5 Discussion for RQ2

MID was proposed as an interaction-based metric intended to describe differences in puzzle structure rather than directly predict puzzle difficulty. The analysis presented in this section examined whether variations in MID were associated with variations in solver performance.

The results showed weak positive relationships between MID and both Solve Time (r = 0.1670) and Search Nodes (r = 0.0974). Although higher MID values were generally associated with greater solver effort, the observed relationships remained relatively weak. Puzzles with similar MID scores often exhibited different levels of solving effort, suggesting that mechanic interaction alone does not fully explain solver behaviour.

These findings provide limited support for the use of MID as an indicator of puzzle difficulty. However, the existence of positive correlations suggests that mechanic interaction may contribute some information regarding puzzle complexity. To investigate this further, the following section compares MID with alternative weighting schemes, individual MID components, and commonly used structural measures.

4.3.6 Comparison with Structural Measures

To further investigate RQ2, additional analyses were conducted to evaluate the robustness of MID and compare it with commonly used structural measures. These analyses examined the influence of alternative weighting schemes, the contribution of individual MID components, and the relationship between MID and object-count-based measures.

4.3.6.1 Weight Sensitivity Analysis

Weight Set (Cl/Ch/Ci)	Correlation Solve Time	Correlation Search Nodes
Current (0.5/0.3/0.2)	0.1985	0.1549
Equal (0.33/0.33/0.33)	0.1818	0.1075
Cl (0.7/0.2/0.1)	0.1823	0.1173
Ch (0.2/0.6/0.2)	0.1734	0.1041
Ci (0.2/0.2/0.6)	0.1811	0.1027

Table 6 Weight Sensitity Analysis

Table 6 presents the weight sensitivity analysis of the MID formulation using five alternative weighting schemes for Cl, Ch, and Ci. The current weighting scheme (0.5/0.3/0.2) produced the highest correlations with both Solve Time (r=0.1985) and Search Nodes (r=0.1549). The lowest correlations were observed for the Ch-dominant scheme in relation to Solve Time (r=0.1734) and the Ci-dominant scheme in relation to Search Nodes (r=0.1027). These results indicate that assigning the greatest weight to Cl, followed by Ch and Ci, provides the strongest observed association with solving performance among the tested alternatives. Nevertheless, all correlations were weak, ranging from 0.1027 to 0.1985. Therefore, the current weighting scheme can be regarded as the most suitable empirically among the evaluated schemes, but not as statistically optimal.

An additional observation is that the differences between weighting configurations remained relatively small. Although the current weighting scheme produced the strongest observed relationships, alternative configurations generated correlation values within a similar range. This suggests that the observed association between MID and solver-performance measures was not driven exclusively by a single weighting configuration. Instead, the general relationship appears reasonably stable across different component-weighting assumptions.

4.3.6.2 Component-Level Analysis

Correlation	Solve Time	Search Node
Cl	0.1658	0.2381
Ch	0.1614	0.0983
Ci	0.1701	0.0919

Table 7 Component-Level Analysis

Table 7 presents the component-level correlation analysis between the three components (Cl, Ch, and Ci) and two solving-performance measures: Solve Time and Search Node. All correlation coefficients are positive but weak, ranging from 0.0919 to 0.2381. For Solve Time, Ci shows the highest correlation (r=0.1701), followed by Cl (r=0.1658) and Ch (r=0.1614). For Search Node, Cl has the highest correlation (r=0.2381), whereas Ch and Ci show very weak correlations (r=0.0983 and r=0.0919, respectively). These findings suggest that the individual components have only limited linear associations with solving performance, and no component demonstrates a strong relationship with either Solve Time or Search Node.

The component-level results also indicate that the three MID components may contribute different types of information. Cl exhibited the strongest relationship with Search Nodes, whereas Ci showed the strongest relationship with Solve Time. This pattern suggests that the components may capture different aspects of puzzle structure and solver behaviour rather than representing a single underlying characteristic.

4.3.6.3 Structural Baseline Comparison

Existing puzzle-difficulty studies commonly employ simple structural descriptors, such as Total Objects and solution-related characteristics, as indicators of puzzle complexity. (Van Kreveld, Löffler and Mutser, 2015; Kristensen and Burelli, 2024).

To examine whether MID captures information beyond these measures, comparisons were performed using Total Objects and Solution Objects as baseline metrics.

Measure	Solve Times	Search Nodes
MID	0.1985	0.1549
Solution Objects	0.1453	0.2770
Total Objects	0.1660	0.2376

Table 8 Structural Baseline Comparison

Table 8 compares MID with two structural baseline measures. For Solve Time, MID produced the strongest correlation (r = 0.1985), slightly exceeding Total Objects (r = 0.1660) and Solution Objects (r = 0.1453). This suggests that interaction-based information may be more closely related to solving time than object-count measures alone.

For Search Nodes, a different pattern was observed. Both Solution Objects (r = 0.2770) and Total Objects (r = 0.2376) produced stronger correlations than MID (r =0.1549). This indicates that object-count measures remained useful predictors of search effort. Overall, the results suggest that MID and structural measures may capture different aspects of puzzle complexity.

An additional observation is that the relative performance of the measures differed depending on the selected outcome variable. MID demonstrated the strongest relationship with Solve Time, whereas structural measures performed better for Search Nodes. This pattern suggests that different descriptors may be sensitive to different characteristics of puzzle-solving behaviour. Consequently, a single metric may not be sufficient to capture all dimensions of puzzle complexity.

4.3.6.4 Statistical Testing

Because MID and Total Objects were correlated with the same outcome variable (Solve Time) using the same solved-puzzle dataset, Steiger's Z test was used to determine whether the observed difference between the two dependent correlation coefficients was statistically significant.

Table 8 presents the results of the statistical comparison between MID and Total Objects correlations.

Test	Result
Steiger's Z	-0.784
p-value	0.433

Table 9 Statistical Testing

Steiger's test was conducted to compare the correlation between MID and Solve Time (r=.1985) with the correlation between Total Object and Solve Time (r=.1660), while accounting for the correlation between MID and Total Object (r=.6141). Table 9 presents he result was not statistically significant, Z=--0.784, p-value = 0.433>0.05. Therefore, the two dependent correlations did not differ significantly. This is evidence that the MID and Total Object metrics may not adequately cover the different aspects of puzzle complexity.

The absence of a statistically significant difference does not imply that MID and Total Objects are identical measures. Instead, the result indicates that the observed difference between the two correlations was insufficient to conclude that one measure was clearly superior within the evaluated dataset. Consequently, both measures may provide useful, although partially overlapping, information regarding puzzle complexity.

4.3.6.5 Incremental Contribution Analysis

Outcome	**Total Object Only $(R^2)_1$**	**Components (MID) +Interaction $(R^2)_2$**	**P-Value (MID)**
Solve Time	0.6508	0.6577	0.0025
Search Node	0.0933	0.0955	0.9664

Table 10 Incremental Contribution Analysis

Table 10 assesses whether MID provides more information than Total Objects. For the Solve Time case, adding MID (p-value=0.0025< 0.05) slightly increased the R^2 value from 0.6508 to 0.6577. Although the improvement is not significant, the results suggest that MID can retain Solve Time information that is not fully represented by Total Objects. For the Search Node case, there is insufficient evidence to conclude that adding MID increases the R^2 value (Since the p-value was $0.9664 > 0.05$, no statistically significant relationship was found between Search Node and MID.)

Although the statistical comparison did not identify a significant difference between the two dependent correlations, the result remains informative because it indicates that both measures exhibited comparable levels of association with the evaluated outcome. This finding suggests that MID and Total Objects may provide partially overlapping information while still reflecting different properties of the generated puzzles.

The incremental analysis is particularly relevant because it examines whether interaction-based information contributes beyond structural descriptors that are already available. The observed increase in explanatory power for Solve Time was relatively small, but it indicates that MID captured some variation not fully represented by object-count measures alone. This observation provides additional support for investigating interaction-based descriptors as complementary analytical tools rather than direct replacements for structural measures.

4.4 Summary

This chapter presented the results obtained from the proposed framework using a dataset of 950 generated puzzles. The solvability evaluation showed that 459 puzzles were successfully solved by the custom solver agent, corresponding to a success rate of 48.3%. These results indicate that the CGRPG framework was capable of generating a substantial number of solvable light-based puzzles, although not all generated puzzles could be solved successfully.

The chapter also evaluated Mechanic Interaction Density (MID) using solver-performance measures, including Solve Time and Search Nodes. MID showed weak positive relationships with both measures. Additional analyses demonstrated that these relationships were relatively stable across alternative weighting configurations and that individual MID components contributed differently to solver performance.

Comparison with structural baseline measures showed mixed results. MID produced the strongest correlation with Solve Time, whereas Total Objects measures remained more strongly associated with Search Nodes. Statistical testing indicated significant differences between the compared relationships, and incremental contribution analysis suggested that MID may provide additional information beyond Total Objects measures alone.

Overall, the findings provide partial support for MID as an interaction-based descriptor of puzzle complexity and suggest that mechanic-interaction information may complement traditional structural measures rather than replace them. Taken together, the results indicate that different evaluation measures captured different aspects of puzzle behaviour. Structural measures generally showed stronger associations with search effort, whereas MID demonstrated stronger relationships with Solve Time and provided a modest incremental contribution beyond object-count descriptors. These findings suggest that puzzle complexity may emerge from multiple characteristics rather than from a single measurable property. Consequently, the results support further investigation of approaches that combine structural and interaction-based information when analysing procedurally generated puzzles.

Chapter 5 Discussion

5.1 Overview

This chapter discusses the findings of the study in relation to the two research questions and the broader literature on procedural puzzle generation and difficulty modelling. The discussion focuses on two main themes. First, it examines the effectiveness of Constraint-Guided Reverse Path Generation (CGRPG) for generating solvable light-based puzzles. Second, it evaluates the usefulness of Mechanic Interaction Density (MID) as an interaction-based descriptor of puzzle complexity.

Previous studies have suggested that puzzle difficulty is influenced by multiple interacting factors rather than a single structural property, and several approaches have been proposed to estimate or model difficulty using structural features, interaction patterns, or learned feature weightings (Van Kreveld, Löffler and Mutser, 2015; Kristensen and Burelli, 2024; Kaesberg et al., 2025). Against this background, the discussion considers whether MID provides meaningful information beyond traditional structural measures and how its current limitations affect the interpretation of the results.

Rather than treating MID as a validated difficulty metric, the findings are interpreted as evidence regarding the potential contribution of mechanic-interaction information to puzzle analysis. This perspective is particularly important given the exploratory nature of the proposed metric and the limitations identified during evaluation.

5.2 Discussion of RQ1

5.2.1 Can CGRPG generate solvable puzzles?

RQ1 examined whether Constraint-Guided Reverse Path Generation (CGRPG) could generate solvable light-based puzzles. The evaluation showed that 459 out of 950 generated puzzles were successfully solved by the solver agent, corresponding to a success rate of 48.3%. These results indicate that the proposed generation approach was capable of producing a substantial number of puzzle instances that could be completed through the evaluation process.

A key characteristic of CGRPG is that it constructs a candidate solution path before generating the final puzzle layout. This differs from traditional generate-and-test approaches, where puzzles are generated first and subsequently validated for solvability. Previous research has noted that validation and repair stages can become computationally expensive when large numbers of generated puzzles fail verification (del Solar-Zavala and Barriga, 2026; McConnell and Zhao, 2025). By embedding solvability considerations directly into the generation process, CGRPG attempts to reduce reliance on post-generation verification and support the production of valid puzzle content.

The successful generation of 459 solved puzzles suggests that reverse-path construction can be used to produce light-based puzzles that are generally solvable under the evaluation framework. Although the overall success rate remained below 50%, the findings suggest that incorporating solution-path construction directly into generation represents a viable approach for maintaining solvability within the proposed framework.

The findings also suggest that solvability should not be viewed as a binary property. Rather than classifying generated puzzles simply as solvable or unsolvable, the results indicate that different puzzles may exhibit varying levels of solution accessibility. Some puzzles can be solved using relatively straightforward search procedures, whereas others require substantially greater search effort despite containing valid solution paths. This observation highlights the importance of considering not only whether a puzzle can be solved, but also how easily a solution can be discovered within the generated puzzle space.

An additional observation concerns the relationship between solvability and generation control. The results suggest that reverse-path construction can improve the likelihood of generating valid puzzles by embedding solution information within the generation process. However, the findings also indicate that solvability alone is insufficient as a measure of generation quality. The ability to generate puzzles with manageable search spaces may be equally important when evaluating the practical usefulness of procedurally generated puzzle content.

5.2.2 Why were many puzzles unsolved?

Despite the use of reverse-path construction, 491 generated puzzles were recorded as unsolved during evaluation. This finding indicates that embedding a valid solution path within the generation process does not necessarily guarantee successful solver completion. Puzzle layout, object placement, search complexity, and limitations of the solver may all contribute to unsuccessful outcomes.

Similar challenges have been reported in procedural puzzle generation research, where generated puzzles may contain valid solutions but still present large search spaces that make solution discovery difficult for automated solving systems (Kaesberg et al., 2025; del Solar-Zavala and Barriga, 2026). In the present study, object rotation and relocation increased the number of possible puzzle states, suggesting that some failures may reflect search complexity rather than generation errors. Consequently, the findings provide only partial support for RQ1 and indicate that further refinement of puzzle construction constraints and solver-guidance mechanisms may be required.

The results also highlight the trade-off between puzzle generation and puzzle verification. A generated puzzle may contain a valid solution while still presenting a large search space that makes solution discovery difficult for the solver. This distinction is important because successful puzzle generation does not necessarily imply efficient puzzle solving. Consequently, evaluation methods should consider both solvability and search effort when assessing generated puzzle quality.

From a procedural generation perspective, these findings suggest that generation quality should not be assessed solely in terms of successful puzzle creation. Characteristics such as search complexity, solution accessibility, and interaction structure may also influence how generated puzzles are experienced and evaluated.

These findings have broader implications for procedural puzzle generation research. In practice, a generation method that produces theoretically solvable puzzles may still generate content that is difficult to verify or analyse due to search complexity. Consequently, future procedural generation systems may benefit from considering search-space characteristics during generation rather than treating solvability and search effort as entirely separate properties.

5.3 Discussion of RQ2

5.3.1 MID as an Exploratory Metric

RQ2 investigated whether Mechanic Interaction Density (MID) could help describe puzzle difficulty. MID was motivated by the idea that puzzle complexity may depend not only on structural characteristics, such as Total Objects and solution length, but also on interactions between mechanics. This view is consistent with recent difficulty-modelling research, which suggests that puzzle complexity emerges from multiple interacting factors rather than a single structural property (Kristensen and Burelli, 2024; Kaesberg et al., 2025).

The evaluation showed weak positive relationships between MID and both Solve Time and Search Nodes. MID produced the strongest association with Solve Time, whereas object-count-based measures showed stronger relationships with Search Nodes. These findings suggest that mechanic interaction information may be related to some aspects of solver effort, although the overall relationships remained relatively weak.

Nevertheless, the results do not invalidate the underlying concept of mechanic interaction. Instead, they suggest that mechanic interaction represents one aspect of puzzle complexity that should be considered alongside structural and behavioural factors. Consequently, MID is more appropriately interpreted as an exploratory interaction-based descriptor that may complement existing approaches to puzzle analysis.

5.3.2 Weighting Assumptions and Component Contributions

The MID formulation combines Component Density (Cl), Rule Heterogeneity (Ch), and Component Complexity (Ci). The final MID formulation used a calibrated weighting configuration of 0.5, 0.3, and 0.2 for Component Density, Rule Heterogeneity, and Component Complexity, respectively. Determining feature importance is a recognised challenge in difficulty modelling, particularly when limited empirical evidence is available (Kristensen and Burelli, 2024).

Component-level analysis showed that the observed relationships between MID components and solver effort did not fully align with the assumed weighting hierarchy. However, sensitivity analysis demonstrated that the overall MID–solver relationship remained relatively stable across alternative weighting configurations. These findings suggest that the current weighting configuration should be interpreted as exploratory rather than empirically validated.

Unlike the data-driven approach of Van Kreveld, Löffler and Mutser (2015), where feature weights were derived from human difficulty ratings, the MID weighting scheme was informed by calibration analysis but has not been independently validated. Consequently, the current coefficients should be viewed as provisional rather than definitive.

5.3.3 Object Mobility and MID Limitations

A significant limitation of MID is that it is calculated from the puzzle state at generation time. Although the metric captures structural characteristics of the generated layout, it does not account for dynamic changes that occur during puzzle solving. Mirrors and refractors can be moved or reoriented, creating additional puzzle configurations that are not represented within the current MID formulation.

As a result, the MID value remains fixed even though the effective interaction structure of the puzzle may change substantially during solver execution. This distinction is important because solver effort is influenced not only by the initial puzzle configuration but also by the number of alternative states that become reachable through object manipulation. Consequently, a puzzle may exhibit relatively modest MID values at generation time while still producing a large dynamic search space during solving.

Previous research has highlighted the importance of state-space growth and interaction dynamics in puzzle analysis. Kristensen and Burelli (2024) argue that puzzle complexity emerges from multiple interacting factors and may not be fully represented by static structural descriptors alone. Similar observations have been reported in procedural puzzle generation research, where movable puzzle elements can substantially increase the number of possible configurations that must be considered during search.

This limitation may provide one explanation for the observed differences between Solve Time and Search Nodes. While MID showed its strongest relationship with Solve Time, the metric exhibited much weaker associations with Search Nodes. One possible interpretation is that Search Nodes are influenced more heavily by dynamic state expansion, whereas MID primarily reflects generation-time interaction structures.

The findings therefore suggest that future interaction-based metrics should move beyond static puzzle descriptions. Rather than measuring only generation-time interactions, future formulations could incorporate information about object mobility, state transitions, reachable configurations, or interaction probabilities observed during puzzle solving.

For example, movable mirrors and refractors can create large numbers of additional puzzle states that are not represented within the current MID formulation. A mobility-aware extension of MID could therefore account not only for the interactions present in the generated puzzle layout, but also for the range of interaction opportunities that become available during solving. Such an extension may better represent dynamic puzzle complexity and improve the ability of interaction-based descriptors to explain solver behavior, particularly for measures related to search-space exploration.

More generally, future versions of MID may benefit from incorporating probabilistic interaction information. Rather than treating all interactions as equally important, interaction weights could be adjusted according to how frequently particular interaction patterns occur during solving. This approach may provide a more realistic representation of mechanic influence and enable MID to capture aspects of puzzle behavior that emerge only during gameplay.

5.3.4 MID versus Structural Measures

Many difficulty-modelling approaches rely on structural measures because they are simple to calculate and often provide useful estimates of puzzle complexity. (Van Kreveld, Löffler and Mutser, 2015; Kristensen and Burelli, 2024; Lazaridis and Fragulis, 2024).

However, the results of the present study showed a more mixed pattern. MID produced the strongest correlation with Solve Time, whereas Total Object measures remained more strongly associated with Search Nodes.

One possible explanation for the observed differences is that MID and Total Object measures capture different characteristics of a puzzle. Total Object measures primarily describe puzzle size, whereas MID attempts to describe the density and complexity of interactions between puzzle components. As a result, the measures may be sensitive to different aspects of solver behaviour.

The findings therefore suggest that structural and interaction-based measures should not necessarily be viewed as competing alternatives. Instead, they may provide complementary perspectives on puzzle complexity. Structural descriptors capture the overall composition of a puzzle, while interaction-based descriptors attempt to represent relationships that emerge between mechanics during problem solving.

5.3.5 Additional Value of Interaction Information

Although MID did not consistently outperform the structural baselines across all evaluation measures, the incremental contribution analysis suggests that interaction-based information was not entirely redundant with object-count measures. The inclusion of MID produced a small increase in explanatory power, indicating that mechanic interactions may capture aspects of puzzle behaviour that are not fully represented by structural descriptors alone.

Therefore, the primary contribution of MID lies not in establishing a superior difficulty metric, but in demonstrating the potential value of incorporating mechanic-interaction information alongside traditional structural measures. Consistent with previous research suggesting that puzzle complexity emerges from multiple interacting factors (Kristensen and Burelli, 2024; Kaesberg et al., 2025), the findings indicate that structural and interaction-based descriptors may provide complementary perspectives on puzzle complexity rather than functioning as competing alternatives.

5.3.6 Solver-Based Evaluation and Difficulty Ground Truth

The interpretation of MID is also affected by the use of a solver-based evaluation framework. The study relied on a custom solver agent and solver-effort measures, including Solve Time and Search Nodes, rather than human difficulty ratings. Consequently, the observed relationships describe algorithmic search behavior rather than player-perceived difficulty.

This distinction is important because human players may employ different reasoning strategies, prior knowledge, visual perception processes, and learning behaviors that are not represented within the solver. As a result, a puzzle that appears difficult for an automated solver may not necessarily be perceived as equally difficult by human players.

The limitation is particularly relevant when compared with previous difficulty-modelling studies that use human difficulty ratings as ground truth (Van Kreveld, Löffler and Mutser, 2015). Unlike these approaches, the present study evaluates MID using solver-performance measures only. Therefore, the findings should be interpreted as evidence regarding relationships between MID and solver effort rather than evidence that MID predicts human difficulty directly.

Future work should investigate the relationship between MID and human difficulty assessments through controlled user studies. Such validation would provide a stronger basis for determining whether interaction-based descriptors capture aspects of puzzle complexity that are meaningful from a player perspective.

5.3.7 Scope and Generalizability

The findings indicate that MID captures some aspects of puzzle structure and mechanic interaction. However, the observed relationships with solver-effort measures remained relatively weak and the current evidence is insufficient to support MID as a validated or comprehensive difficulty model.

In addition, the study focuses exclusively on grid-based light puzzles implemented within a single Unity environment. The generated puzzles rely primarily on reflection and refraction mechanics, and the evaluation is based on a specific solver implementation. Consequently, the findings may not generalize directly to other puzzle genres or procedural generation systems.

The domain-specific nature of the puzzle environment may be particularly important. Interaction structures that emerge in light-based puzzles are strongly influenced by the behavior of mirrors and refractors. Other puzzle genres may exhibit different forms of interaction complexity that are not adequately represented by the current MID formulation.

Therefore, MID should be interpreted as an exploratory interaction-based analytical framework developed within the specific context of light-based puzzles. Further validation across multiple puzzle domains, alternative mechanic sets, and human-player evaluations would be required before broader conclusions regarding its general applicability can be drawn.

5.4 Chapter Summary

This chapter discussed the findings of the study in relation to the two research questions and the broader literature on procedural puzzle generation and difficulty modelling.

For RQ1, the findings demonstrated that CGRPG can generate solvable light-based puzzles through reverse-path construction, although successful generation does not guarantee solver completion.

For RQ2, the findings suggest that MID provides useful interaction-based information and captures aspects of puzzle complexity that are not fully represented by object-count measures alone. While MID produced the strongest relationship with Solve Time, structural measures remained more strongly associated with Search Nodes. These findings indicate that interaction-based and structural descriptors may provide complementary perspectives on puzzle complexity. Consequently, MID should be interpreted as an exploratory descriptor rather than a validated difficulty metric.

These findings provide the foundation for the conclusions and future research directions presented in Chapter 6.

Chapter 6 Conclusion and Future Work

6.1 Overview

This chapter concludes the dissertation by summarising the main findings and contributions of the study. The chapter revisits the two research questions, outlines the key contributions of the proposed framework, and identifies directions for future research.

6.2 Research Contributions

This study made three primary contributions to the field of procedural puzzle generation and puzzle analysis.

First, the study proposed Constraint-Guided Reverse Path Generation (CGRPG), a procedural generation framework for light-based puzzles. Unlike traditional generate-and-test approaches, CGRPG incorporates solution-path construction directly into the generation process. The evaluation demonstrated that the framework was capable of generating a substantial number of solvable puzzle instances, indicating that reverse-path construction can support solvable puzzle generation within the proposed environment. From a procedural-generation perspective, this contribution demonstrates an alternative approach to handling solvability during puzzle construction. Rather than treating solvability as a validation problem that must be checked after generation, the framework incorporates solution-path information directly into content creation. This design choice contributes to ongoing discussions concerning the balance between generation efficiency, controllability, and puzzle correctness within procedural puzzle-generation research.

Second, the study introduced Mechanic Interaction Density (MID), an interaction-based metric designed to describe relationships between puzzle mechanics. The metric combines Component Density, Rule Heterogeneity, and Component Complexity to provide a structured representation of mechanic interaction within generated puzzles. The evaluation showed that MID produced the strongest association with Solve Time, while object-count measures remained more strongly associated with Search Nodes. These findings suggest that interaction-based and structural measures may provide complementary perspectives on puzzle complexity. The significance of this contribution lies in its focus on mechanic interactions rather than puzzle structure alone. Although MID is not presented as a validated difficulty measure, it provides a structured way of describing interaction-related characteristics within generated puzzles. This perspective extends existing approaches that primarily rely on total objects, solution lengths, or other structural descriptors and demonstrates how mechanic relationships may be incorporated into puzzle analysis. In addition, MID contributes a structured framework for investigating puzzle mechanics beyond simple object-count approaches. By representing interaction-related characteristics through multiple components, the metric provides a basis for exploring how puzzle elements combine to produce different solution behaviours. Although further validation is required, this research demonstrates the feasibility of analysing mechanic interactions as a distinct aspect of puzzle structure.

The findings also highlight an important limitation of the current formulation. Because MID is calculated from generation-time puzzle structures, the metric does not account for dynamic state changes that emerge through object movement and interaction during solving. This limitation provides a direction for future extensions of interaction-based puzzle analysis.

Together, these contributions provide a foundation or further research into interaction-aware puzzle generation and analysis.

The study also contributes an empirical comparison between interaction-based and structural approaches to puzzle analysis. By comparing MID with object-count-based measures, the research provides evidence regarding the strengths and limitations of different puzzle complexity descriptors. This comparison offers useful insights into how future puzzle-evaluation methods might combine structural and interaction-based information. Beyond the specific findings of this dissertation, the comparison study contributes methodological insight into how puzzle analysis approaches can be evaluated. Rather than assuming that a proposed metric should replace existing measures, the study examines how interaction-based and structural descriptors relate to one another. This approach provides a useful framework for future studies that seek to evaluate new puzzle-analysis metrics against established baselines.

6.3 Conclusion

This dissertation investigated two research questions concerning procedural generation and difficulty analysis in grid-based light puzzles.

For RQ1, the results demonstrated that CGRPG can generate solvable light-based puzzles. The framework successfully generated 459 solved puzzle instances from a dataset of 950 puzzles, showing that reverse-path construction can support solvability within the generation process. However, the results also indicated that constructing a solution path does not guarantee successful solver completion in every case, highlighting the influence of puzzle structure, search complexity, and solver limitations.

For RQ2, the findings provided partial support for the proposed MID metric. MID showed weak positive relationships with both Solve Time and Search Nodes. The comparison study produced mixed results: MID showed the strongest relationship with Solve Time, whereas object-count measures remained more strongly associated with Search Nodes. In addition, incremental contribution analysis suggested that MID captured information that was not fully represented by structural measures alone.

However, the observed relationships remained relatively weak, and the weighting configuration has not been independently validated. Consequently, the findings do not support MID as a validated difficulty metric. Instead, MID should be interpreted as an exploratory interaction-based descriptor that may complement existing structural approaches.

Overall, the findings demonstrate that puzzle generation and puzzle analysis can benefit from considering mechanic interactions alongside traditional structural information. Although the proposed MID metric requires further validation, the study provides initial evidence that interaction-related characteristics can be quantified and analysed within procedurally generated light-based puzzles. These findings contribute to ongoing research on puzzle generation, solver-based evaluation, and puzzle-complexity modelling.

Beyond the specific findings reported in this dissertation, the study demonstrates the value of integrating puzzle generation, solver-based evaluation, and interaction-based analysis within a unified framework. By combining these perspectives, the research provides a structured approach for investigating relationships between generated puzzle characteristics and solver behaviour. This integration may support future efforts to develop more interpretable and analytically informed procedural puzzle-generation systems.

6.4 Future Work

Several opportunities exist for extending this research.

- Human and Empirical Validation of MID

The current evaluation relied primarily on a custom solver agent and solver-effort measures. Future studies should investigate whether MID is associated with independent indicators of puzzle difficulty, including human-player assessments, expert evaluations, and alternative computational measures. Such validation would provide stronger evidence regarding the usefulness and generalisability of interaction-based descriptors across different evaluation settings. Such studies would also help determine whether interaction-based descriptors capture aspects of puzzle difficulty that are meaningful to players. Establishing stronger connections between solver-based measurements and player-based evaluations would improve confidence in the use of MID within puzzle-analysis research. Future validation studies may also investigate whether different player populations interpret puzzle complexity in similar ways. For example, experienced puzzle players and novice players may respond differently to mechanic interactions, providing additional insight into how interaction-based descriptors relate to perceived puzzle difficulty.

- Data-driven MID weighting.

  The weighting coefficients used in MID were informed by calibration analysis but remain exploratory. Future work could derive component weights through machine-learning methods, optimisation procedures, or regression analysis using larger datasets and human difficulty ratings. Such approaches may also allow weighting configurations to adapt to different puzzle domains rather than relying on a single predefined weighting hierarchy. This would improve the flexibility of MID and support comparative studies across multiple puzzle environments. Data-driven weighting approaches may also enable the relative contribution of individual MID components to be examined more systematically. Such analyses could help determine whether Component Density, Rule Heterogeneity, or Component Complexity consistently contribute to puzzle evaluation across different datasets and puzzle environments.

- Dynamic interaction modelling.

  The current MID formulation describes puzzle characteristics at generation time and does not account for dynamic state changes that emerge through object movement and mechanic interactions during solving. As discussed in Chapter 5, movable mirrors and refractors can create large numbers of additional puzzle states that are not represented within the current metric formulation. Consequently, solver effort may be influenced by dynamic interaction opportunities that remain outside the scope of MID.

  Future research could address this limitation by incorporating information about object mobility, state transitions, reachable configurations, or interaction probabilities observed during solving. Such extensions may provide a richer representation of puzzle complexity and improve the ability of interaction-based descriptors to explain both solver and player behavior.

- Expanded puzzle mechanics.

  This study focused on reflection and refraction mechanics within a grid-based light-puzzle environment. Future work could investigate additional mechanics and more complex puzzle domains to evaluate the generalisability of both CGRPG and MID. Additional mechanics may also produce new forms of interaction structures that are not present within the current puzzle environment. Evaluating MID in such contexts would provide insight into the extent to which interaction-based descriptors remain effective across different puzzle-generation domains. The introduction of additional mechanics may also enable the investigation of more complex interaction structures than those observed in the current environment. Studying such systems would provide insight into whether the principles underlying MID remain applicable when larger numbers of mechanics and interaction pathways are present.

- Hybrid difficulty models.

  The findings suggest that structural and interaction-based features may provide complementary information. Future studies could explore hybrid difficulty models that combine object-count measures, solution characteristics, and mechanic-interaction descriptors to improve the prediction of puzzle complexity. Hybrid approaches may be particularly valuable because different measures appeared to capture different aspects of puzzle-solving behaviour in the present study. Structural descriptors showed stronger associations with Search Nodes, whereas MID demonstrated stronger relationships with Solve Time. Future work could investigate whether combining these features leads to more comprehensive models of puzzle complexity than either approach alone. Such models could also be evaluated using human-player datasets to determine whether combinations of structural and interaction-based information provide better explanations of perceived puzzle difficulty. This would help establish whether hybrid approaches offer practical benefits beyond those observed using solver-based evaluation alone.

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Appendices

A. Calibration Dataset (50 Puzzles)

Appendix A presents the calibration dataset used during the development of the MID metric. The dataset contains 50 generated puzzles that were analysed during the weighting-selection process. The calibration stage was used to examine different weighting configurations and evaluate their relationships with solver-performance measures before selecting the final MID weighting scheme adopted in this study.

Puzzle Generation Variables

Column	Description
Run	Unique identifier for each generated puzzle.
Level	Level parameter used during puzzle generation.
Steps	Number of backward-chaining steps used to construct the solution path.
SolutionObjects	Number of objects belonging to the constructed solution path.
Mirrors	Number of mirrors located on the solution path.
Refractors	Number of refractors located on the solution path.
RuleTransitions	Number of mechanic transitions along the solution path.
Decoys	Number of non-essential objects added to increase visual complexity.
TotalObjects	Total number of placed puzzle objects.
GridWidth	Puzzle grid width.
GridHeight	Puzzle grid height.

Table 11 Puzzle Generation Variables

MID Variables

Column	Description
Cl	Component Density value.
Ch	Rule Heterogeneity value.
Ci	Component Complexity value.
MID	Weighted interaction-based metric score.

Table 12 MID Variables

Solver Outcome Variables

Column	Description
Solved	Solver outcome (1 = solved, 0 = unsolved).
SolvePhase	Phase in which the solution was identified.

Table 13 Solver Outcome Variables

Solver Performance Variables

Column	Description
SearchNodes	Number of states explored during logical search.
SearchTimeMs	Time spent performing logical search.
TotalPlacements	Number of object placements performed by the solver.
InPlaceRotations	Number of object rotations performed without relocation.
Relocations	Number of object movements performed during search.
ExecutionTimeMs	Execution time used for testing candidate solutions.
SweepIterations	Number of correction-sweep iterations performed.
SweepRelocations	Number of relocations performed during sweep operations.
SolveTimeMs	Total solver execution time.
GenerationTimeMs	Time required to generate the puzzle.

Table 14 Solver Performance Variables

Table 15 Calibration Dataset (50 Puzzles)

Run	Level	Steps	SolutionObjects	Mirrors	Refractors	RuleTransitions	Decoys	TotalObjects	GridWidth	GridHeight	Cl	Ch	Ci	MID	Solved	SolvePhase	SearchNodes	SearchTimeMs	TotalPlacements	InPlaceRotations	Relocations	ExecutionTimeMs	SweepIterations	SweepRelocations	SolveTimeMs	GenerationTimeMs
1	1	3	2	2	0	0	0	2	16	16	0.0078	0	0	0.0039	1	Sweep-S1	21	53.88	2	2	0	10351.97	12	0	10710.19	27.71
2	2	4	3	1	2	1	0	3	16	16	0.0117	0.5	0.6667	0.2892	0	Sweep	303	25.5	2	2	0	22820.41	101	0	23155.09	55.06
3	2	5	3	1	2	1	0	3	16	16	0.0117	0.5	0.6667	0.2892	1	1B	1	27.87	1	0	1	4370.32	0	0	5019.97	15.16
4	3	6	5	4	1	2	0	5	16	16	0.0195	0.5	0.2	0.1998	0	Sweep	794	41.45	3	3	0	22245.87	88	0	22591.05	11.73
5	3	7	4	2	2	3	0	4	16	16	0.0156	1	0.5	0.4078	0	Sweep	1089	59.3	3	3	0	18878.13	64	0	19239.49	10.86
6	3	8	5	2	3	2	0	5	16	16	0.0195	0.5	0.6	0.2798	1	Sweep-S1	5	9.9	1	0	1	6869.45	4	0	7184.75	10.77
7	4	9	7	5	2	4	0	7	16	16	0.0273	0.6667	0.2857	0.2708	1	Sweep-S1	77	18.62	5	5	0	28367.74	12	0	28689.99	11.1
8	5	3	2	2	0	0	1	3	16	16	0.0117	0	0	0.0059	1	Sweep-S1	6	9.22	1	1	0	5304.28	4	0	5614.94	10.42
9	6	4	3	2	1	2	1	4	16	16	0.0156	1	0.3333	0.3745	0	Sweep	734	6527.85	1	0	1	6056.09	61	1	12886.9	10.71
10	6	5	3	1	2	2	1	4	16	16	0.0156	1	0.6667	0.4411	1	Sweep-S1	6	10.3	1	0	1	7426.24	5	0	7737.73	11.18
11	7	6	5	4	1	1	1	6	16	16	0.0234	0.25	0.2	0.1267	0	Sweep	190	18.01	5	5	0	34199.89	116	0	34528.02	11.23
12	7	7	4	2	2	3	1	5	16	16	0.0195	1	0.5	0.4098	1	1B	1	9.7	1	0	1	4860.61	0	0	5474.64	10.31
13	8	8	7	3	4	5	1	8	16	16	0.0313	0.8333	0.5714	0.3799	0	Sweep	37007	2250.5	4	4	0	28073.84	132	0	30630.6	11.06
14	8	9	7	5	2	1	1	8	16	16	0.0313	0.1667	0.2857	0.1228	0	Sweep	798	19900.21	1	0	1	12727.78	93	1	32933.84	11.11
15	8	3	2	2	0	0	2	4	16	16	0.0156	0	0	0.0078	1	Trivial	0	0	0	0	0	0	0	0	301.3	10.44
16	9	4	3	2	1	1	2	5	16	16	0.0195	0.5	0.3333	0.2264	1	Sweep-S1	527	28.35	4	4	0	20729.06	4	0	21058.07	10.13
17	10	5	4	2	2	3	2	6	16	16	0.0234	1	0.5	0.4117	0	Sweep	32933	1797.18	5	5	0	31601.78	90	0	33703.98	10.13
18	10	6	4	4	0	0	2	6	16	16	0.0234	0	0	0.0117	0	Sweep	262302	8001.28	0	0	0	9062.56	93	0	17367.25	10.38
19	10	7	5	2	3	3	2	7	16	16	0.0273	0.75	0.6	0.3587	0	Sweep	33389	1792.6	5	5	0	31658.45	98	0	33756.13	10.07
20	10	8	7	3	4	5	2	9	16	16	0.0352	0.8333	0.5714	0.3819	0	Sweep	1334	56.88	7	7	0	53053.62	156	0	53410.89	10.74
21	10	9	4	2	2	2	2	6	16	16	0.0234	0.6667	0.5	0.3117	0	Sweep	4882	262.39	5	5	0	41617.49	200	0	42187.77	11.9
22	10	3	2	2	0	0	3	5	16	16	0.0195	0	0	0.0098	1	Sweep-S1	21	15.75	2	2	0	10326.23	12	0	10643.43	10.01
23	11	4	3	3	0	0	3	6	16	16	0.0234	0	0	0.0117	1	Trivial	0	0	0	0	0	0	0	0	304.2	9.46
24	12	5	4	3	1	1	3	7	16	16	0.0273	0.3333	0.25	0.1637	0	Sweep	108	7.69	3	3	0	25631.96	105	0	25946.78	10.19
25	12	6	4	2	2	3	3	7	16	16	0.0273	1	0.5	0.4137	0	Sweep	811	17461.79	1	0	1	12942.14	106	1	30709.96	9.95
26	12	7	4	3	1	2	3	7	16	16	0.0273	0.6667	0.25	0.2637	1	1B	1	8.85	1	0	1	7286.62	0	0	7902.22	10.19
27	13	8	6	5	1	1	3	9	16	16	0.0352	0.2	0.1667	0.1109	1	1B	1	8.3	1	0	1	8239.87	0	0	8853.64	10.07
28	14	9	8	4	4	5	3	11	16	16	0.043	0.7143	0.5	0.3358	0	Sweep	863	12922.42	0	0	0	23905.76	150	0	37133.73	10.38
29	14	3	2	2	0	0	4	6	16	16	0.0234	0	0	0.0117	1	Sweep-S1	19	8.06	1	1	0	5732.85	18	0	6044.92	9.77
30	15	4	3	2	1	1	4	7	16	16	0.0273	0.5	0.3333	0.2303	0	Sweep	782	11230.22	1	0	1	8240.54	85	1	19776.98	9.89
31	15	5	4	1	3	1	4	8	16	16	0.0313	0.3333	0.75	0.2656	1	Sweep-S3	69791	3787.72	4	4	0	28920.96	77	0	33009.7	10.38
32	16	6	4	3	1	1	4	8	16	16	0.0313	0.3333	0.25	0.1656	1	Sweep-S1	5	9.09	1	0	1	9009.09	4	0	9321.78	9.95
33	17	7	4	4	0	0	4	8	16	16	0.0313	0	0	0.0156	1	1B	1	9.22	1	0	1	6769.29	0	0	7387.88	10.07
34	18	8	6	3	3	4	4	10	16	16	0.0391	0.8	0.5	0.3595	0	Sweep	38090	2187.13	6	6	0	45344.12	129	0	47832.03	10.44
35	18	9	8	1	7	2	4	12	16	16	0.0469	0.2857	0.875	0.2842	0	Sweep	246	318.18	3	0	3	40112.98	172	0	40734.93	10.62
36	18	3	2	2	0	0	0	2	16	16	0.0078	0	0	0.0039	1	Trivial	0	0	0	0	0	0	0	0	304.93	11.29
37	19	4	3	2	1	1	0	3	16	16	0.0117	0.5	0.3333	0.2225	0	Sweep	754	5724.92	1	0	1	4618.41	49	1	10650.39	10.13
38	19	5	4	2	2	2	0	4	16	16	0.0156	0.6667	0.5	0.3078	1	1B	1	8.48	1	0	1	7658.81	0	0	8272.83	10.13
39	20	6	5	5	0	0	0	5	16	16	0.0195	0	0	0.0098	0	Sweep	681	27.47	3	3	0	25774.11	96	0	26108.83	10.31
40	20	7	6	4	2	3	0	6	16	16	0.0234	0.6	0.3333	0.2584	0	Sweep	37083	2030.58	4	4	0	32013.55	130	0	34349.18	10.62
41	20	8	6	2	4	4	0	6	16	16	0.0234	0.8	0.6667	0.3851	0	Sweep	70907	3698.43	5	5	0	32946.9	113	0	36948.91	9.46
42	20	9	4	3	1	1	0	4	16	16	0.0156	0.3333	0.25	0.1578	1	Sweep-S1	18	8.54	1	0	1	10074.28	17	0	10390.08	9.28
43	21	3	2	2	0	0	1	3	16	16	0.0117	0	0	0.0059	0	Sweep	758	8443.97	1	0	1	5177.06	53	1	13924.74	9.77
44	21	4	3	3	0	0	1	4	16	16	0.0156	0	0	0.0078	1	Sweep-S1	29	16.78	3	3	0	17208.13	20	0	17526.61	9.83
45	22	5	4	1	3	2	1	5	16	16	0.0195	0.6667	0.75	0.3598	1	1A	1115	139.77	4	4	0	19910.64	0	0	20655.76	10.13
46	23	6	4	2	2	3	1	5	16	16	0.0195	1	0.5	0.4098	0	Sweep	267	17.09	2	2	0	16598.75	73	0	16923.46	10.5
47	23	7	5	2	3	3	1	6	16	16	0.0234	0.75	0.6	0.3567	0	Sweep	95	15.38	3	3	0	21276.98	84	0	21593.38	9.64
48	23	8	7	5	2	1	1	8	16	16	0.0313	0.1667	0.2857	0.1228	1	Sweep-S1	205400	8000.85	0	0	0	593.38	6	0	8900.15	10.01
49	24	9	7	4	3	3	1	8	16	16	0.0313	0.5	0.4286	0.2513	0	Sweep	429400	8002.93	0	0	0	11489.38	117	0	19796.88	10.01
50	24	3	2	2	0	0	2	4	16	16	0.0156	0	0	0.0078	1	Sweep-S1	21	15.38	2	2	0	10332.03	12	0	10648.32	12.08

B. Evaluation Dataset

Appendix B presents the dataset used for the main evaluation procedures described in Chapter 4. The dataset contains 950 puzzles generated using the proposed Constraint-Guided Reverse Path Generation (CGRPG) framework. Measurements recorded for each puzzle were subsequently used in the correlation analyses, structural comparisons, and incremental contribution analyses reported in this dissertation.

Table 16 Main Evaluation Dataset (950 Puzzles)

Run	Level	Steps	SolutionObjects	Mirrors	Refractors	RuleTransitions	Decoys	TotalObjects	GridWidth	GridHeight	Cl	Ch	Ci	MID	Solved	SolvePhase	SearchNodes	SearchTimeMs	TotalPlacements	InPlaceRotations	Relocations	ExecutionTimeMs	SweepIterations	SweepRelocations	Solve TimeMs	GenerationTimeMs
51	25	4	3	2	1	1	2	5	16	16	0.0195	0.5	0.3333	0.2264	0	Sweep	746	941.16	1	0	1	6162.23	65	1	7409.55	10.74
52	25	5	3	2	1	2	2	5	16	16	0.0195	1	0.3333	0.3764	0	Sweep	33538	9642.33	0	0	0	10910.4	57	0	20859.62	10.13
53	25	6	5	3	2	2	2	7	16	16	0.0273	0.5	0.4	0.2437	0	1A	263352	8005.98	0	0	0	9945.19	102	0	18254.39	9.89
54	25	7	6	5	1	1	2	8	16	16	0.0313	0.2	0.1667	0.109	0	Sweep	798	19680.42	1	0	1	12567.26	93	1	32554.57	10.38
55	25	8	6	4	2	3	2	8	16	16	0.0313	0.6	0.3333	0.2623	0	Sweep	305200	8012.21	0	0	0	8793.58	89	0	17108.52	12.57
56	25	9	4	2	2	2	2	6	16	16	0.0234	0.6667	0.5	0.3117	1	1B	1	10.01	1	0	1	8780.88	0	0	9399.9	11.23
57	26	3	2	2	0	0	3	5	16	16	0.0195	0	0	0.0098	1	1B	1	9.64	1	0	1	8064.45	0	0	8683.84	10.62
58	27	4	3	1	2	2	3	6	16	16	0.0234	1	0.6667	0.4451	0	Sweep	33371	2039.8	3	3	0	20802.13	80	0	23145.51	11.11
59	27	5	4	4	0	0	3	7	16	16	0.0273	0	0	0.0137	1	Sweep-S3	246600	8007.32	0	0	0	5992.68	59	0	14299.68	10.99
60	28	6	5	2	3	4	3	8	16	16	0.0313	1	0.6	0.4356	0	Sweep	322000	8016.72	0	0	0	9555.18	95	0	17873.29	10.86
61	28	7	5	3	2	3	3	8	16	16	0.0313	0.75	0.4	0.3206	1	Trivial	0	0	0	0	0	0	0	0	303.71	13.06
62	29	8	7	6	1	2	3	10	16	16	0.0391	0.3333	0.1429	0.1481	0	Sweep	814	17913.33	1	0	1	24852.17	109	1	43070.07	10.99
63	29	9	5	3	2	4	3	8	16	16	0.0313	1	0.4	0.3956	0	Sweep	188	9.52	1	1	0	17056.52	114	0	17374.63	10.74
64	29	3	2	2	0	0	4	6	16	16	0.0234	0	0	0.0117	1	Sweep-S1	6	9.28	1	1	0	5862.43	4	0	6175.05	10.25
65	30	4	3	2	1	2	4	7	16	16	0.0273	1	0.3333	0.3803	0	Sweep	786	16444.34	1	0	1	10928.47	81	1	27675.54	10.86
66	30	5	4	1	3	2	4	8	16	16	0.0313	0.6667	0.75	0.3656	0	Sweep	307800	8005.86	0	0	0	8768.56	87	0	17082.28	11.47
67	30	6	4	2	2	3	4	8	16	16	0.0313	1	0.5	0.4156	1	Sweep-S1	6	10.25	1	0	1	7370.36	5	0	7683.11	10.5
68	31	7	6	3	3	2	4	10	16	16	0.0391	0.4	0.5	0.2395	1	Sweep-S1	34	17.09	7	7	0	36612.91	25	0	36936.89	10.99
69	32	8	3	3	0	0	4	7	16	16	0.0273	0	0	0.0137	0	Sweep	296200	8005.62	0	0	0	8133.79	82	0	16444.95	10.99
70	32	9	6	4	2	2	4	10	16	16	0.0391	0.4	0.3333	0.2062	1	1B	1	9.89	1	0	1	7567.63	0	0	8192.38	11.35
71	33	3	2	2	0	0	0	2	16	16	0.0078	0	0	0.0039	1	Sweep-S1	13	7.69	1	1	0	6526.37	4	0	6839.72	9.77
72	34	4	3	2	1	1	0	3	16	16	0.0117	0.5	0.3333	0.2225	0	Sweep	211	15.87	2	2	0	16596.07	74	0	16916.14	9.52
73	34	5	4	3	1	2	0	4	16	16	0.0156	0.6667	0.25	0.2578	1	1B	1	8.54	1	0	1	7013.43	0	0	7626.71	10.01
74	35	6	3	1	2	1	0	3	16	16	0.0117	0.5	0.6667	0.2892	1	Sweep-S1	25	8.18	1	0	1	10770.75	24	0	11079.83	9.64
75	36	7	6	3	3	3	0	6	16	16	0.0234	0.6	0.5	0.2917	0	Sweep	137	15.01	4	4	0	26957.28	126	0	27277.22	9.89
76	36	8	5	5	0	0	0	5	16	16	0.0195	0	0	0.0098	0	Sweep	770	13146.48	1	0	1	6224.61	65	1	19671.51	10.01
77	36	9	7	6	1	1	0	7	16	16	0.0273	0.1667	0.1429	0.0922	1	1B	1	8.54	1	0	1	9182.5	0	0	9801.39	10.13
78	37	3	2	2	0	0	1	3	16	16	0.0117	0	0	0.0059	0	Sweep	758	5633.42	1	0	1	5258.42	53	1	11196.17	9.4
79	37	4	3	1	2	2	1	4	16	16	0.0156	1	0.6667	0.4411	0	Sweep	766	5691.04	1	0	1	6017.58	61	1	12012.57	10.25
80	37	5	3	0	3	0	1	4	16	16	0.0156	0	1	0.2078	1	Sweep-S1	4129	212.77	1	0	1	7961.3	24	0	8475.71	10.25
81	38	6	5	2	3	3	1	6	16	16	0.0234	0.75	0.6	0.3567	0	Sweep	37099	1853.15	5	5	0	38001.1	154	0	40155.03	10.01
82	38	7	6	3	3	2	1	7	16	16	0.0273	0.4	0.5	0.2337	0	Sweep	74516	3860.6	5	5	0	44125.61	195	0	48290.04	9.77
83	38	8	4	3	1	2	1	5	16	16	0.0195	0.6667	0.25	0.2598	0	Sweep	774	13931.27	1	0	1	6634.16	69	1	20869.26	10.01
84	38	9	4	2	2	3	1	5	16	16	0.0195	1	0.5	0.4098	1	Sweep-S1	33	8.3	1	0	1	9690.31	32	0	10004.03	9.77
85	39	3	2	2	0	0	2	4	16	16	0.0156	0	0	0.0078	1	Sweep-S1	19	7.93	1	1	0	6438.11	18	0	6752.69	9.28
86	40	4	3	2	1	2	2	5	16	16	0.0195	1	0.3333	0.3764	0	Sweep	187	15.63	2	2	0	19163.33	98	0	19486.45	9.28
87	40	5	3	3	0	0	2	5	16	16	0.0195	0	0	0.0098	1	Sweep-S1	32781	1673.1	1	0	1	5419.8	4	0	7401.25	10.13
88	41	6	4	1	3	1	2	6	16	16	0.0234	0.3333	0.75	0.2617	0	Sweep	778	14728.03	1	0	1	9381.35	73	1	24420.41	10.01
89	41	7	5	3	2	1	2	7	16	16	0.0273	0.25	0.4	0.1687	0	Sweep	637	25.39	3	3	0	25347.78	114	0	25675.78	10.01
90	41	8	7	6	1	2	2	9	16	16	0.0352	0.3333	0.1429	0.1461	1	Sweep-S1	526	25.88	7	7	0	32921.51	12	0	33252.93	10.5
91	42	9	8	6	2	4	2	10	16	16	0.0391	0.5714	0.25	0.241	0	Sweep	465800	8003.17	0	0	0	14333.13	146	0	22643.8	10.38
92	42	3	2	2	0	0	3	5	16	16	0.0195	0	0	0.0098	1	Sweep-S1	6	9.28	1	0	1	7128.3	5	0	7444.82	10.01
93	43	4	3	1	2	1	3	6	16	16	0.0234	0.5	0.6667	0.2951	1	Sweep-S1	8299	456.54	3	3	0	22008.54	33	0	22772.22	10.25
94	44	5	3	2	1	1	3	6	16	16	0.0234	0.5	0.3333	0.2284	1	1B	1	8.54	1	0	1	6823	0	0	7432.37	10.74
95	45	6	5	2	3	3	3	8	16	16	0.0313	0.75	0.6	0.3606	0	Sweep	203	459.96	3	0	3	35354.49	137	0	36118.9	10.01
96	45	7	6	5	1	2	3	9	16	16	0.0352	0.4	0.1667	0.1709	1	1B	1	9.03	1	0	1	5829.1	0	0	6452.88	10.5
97	46	8	5	2	3	3	3	8	16	16	0.0313	0.75	0.6	0.3606	1	1B	1	8.79	1	0	1	7276.12	0	0	7895.51	10.25
98	47	9	7	4	3	3	3	10	16	16	0.0391	0.5	0.4286	0.2552	1	Sweep-S1	8	8.79	1	0	1	8262.7	7	0	8574.22	47.36
99	48	3	2	2	0	0	4	6	16	16	0.0234	0	0	0.0117	0	Sweep	262219	8002.69	0	0	0	6337.16	66	0	14642.33	10.25
100	48	4	3	1	2	2	4	7	16	16	0.0273	1	0.6667	0.447	1	1A	3	8.3	1	1	0	5762.7	0	0	6381.35	10.25

Run	Level	Steps	SolutionObjects	Mirrors	Refractors	RuleTransitions	Decoys	TotalObjects	GridWidth	GridHeight	Cl	Ch	Ci	MID	Solved	SolvePhase	SearchNodes	SearchTimeMs	TotalPlacements	InPlaceRotations	Relocations	ExecutionTimeMs	SweepIterations	SweepRelocations	Solve TimeMS	GenerationTimeMs
101	49	5	4	2	2	3	4	8	16	16	0.0313	1	0.5	0.4156	1	Sweep-S3	641	26.37	6	6	0	33586.67	54	0	33914.06	9.52
102	50	6	4	2	2	3	4	8	16	16	0.0313	1	0.5	0.4156	0	Sweep	794	12601.81	1	0	1	10763.43	89	1	23670.41	10.01
103	50	7	6	6	0	0	4	10	16	16	0.0391	0	0	0.0195	0	Sweep	381600	8002.93	0	0	0	9461.43	98	0	17771.73	10.5
104	50	8	6	4	2	2	4	10	16	16	0.0391	0.4	0.3333	0.2062	0	Sweep	398800	8005.86	0	0	0	11078.61	113	0	19385.99	10.74
105	50	9	6	4	2	2	4	10	16	16	0.0391	0.4	0.3333	0.2062	1	Sweep-S1	5	9.03	1	0	1	6848.88	4	0	7162.84	10.74
106	51	3	2	2	0	0	0	2	16	16	0.0078	0	0	0.0039	1	Sweep-S1	6	7.81	1	1	0	5772.22	4	0	6086.91	10.01
107	52	4	3	2	1	2	0	3	16	16	0.0117	1	0.3333	0.3725	0	Sweep	718	1005.37	1	0	1	5121.83	53	1	6429.44	10.01
108	52	5	4	4	0	0	0	4	16	16	0.0156	0	0	0.0078	1	Sweep-S1	5	8.3	1	0	1	6356.93	4	0	6671.88	10.01
109	53	6	5	2	3	1	0	5	16	16	0.0195	0.25	0.6	0.2048	1	Trivial	0	0	0	0	0	0	0	0	300.05	9.77
110	54	7	6	5	1	2	0	6	16	16	0.0234	0.4	0.1667	0.1651	0	Sweep	4860	252.69	3	3	0	23633.79	122	0	24191.89	9.77
111	54	8	6	5	1	2	0	6	16	16	0.0234	0.4	0.1667	0.1651	0	Sweep	262227	8008.06	0	0	0	7169.19	74	0	15482.42	10.25
112	54	9	6	4	2	2	0	6	16	16	0.0234	0.4	0.3333	0.1984	1	Sweep-S3	639	39.55	3	3	0	28664.55	118	0	29010.01	10.25
113	55	3	2	2	0	0	1	3	16	16	0.0117	0	0	0.0059	1	Sweep-S1	19	7.57	1	1	0	6434.57	18	0	6742.92	9.52
114	56	4	3	2	1	1	1	4	16	16	0.0156	0.5	0.3333	0.2245	0	Sweep	168	9.28	1	1	0	14530.76	94	0	14843.02	10.25
115	56	5	4	2	2	2	1	5	16	16	0.0195	0.6667	0.5	0.3098	0	Sweep	5330	236.82	2	2	0	22593.75	136	0	23137.45	9.52
116	56	6	5	4	1	1	1	6	16	16	0.0234	0.25	0.2	0.1267	0	Sweep	171	15.63	4	4	0	29806.15	96	0	30126.46	9.52
117	56	7	4	4	0	0	1	5	16	16	0.0195	0	0	0.0098	1	Sweep-S1	32781	1669.92	1	0	1	8150.64	4	0	10122.8	10.5
118	57	8	7	5	2	3	1	8	16	16	0.0313	0.5	0.2857	0.2228	0	1A	271035	8006.35	0	0	0	11130.13	113	0	19440.92	10.25
119	57	9	7	5	2	2	1	8	16	16	0.0313	0.3333	0.2857	0.1728	0	Sweep	798	13570.31	1	0	1	8985.84	93	1	22860.6	10.25
120	57	3	2	2	0	0	2	4	16	16	0.0156	0	0	0.0078	0	Sweep	4859	11142.58	0	0	0	4844.48	50	0	16289.06	10.01
121	57	4	3	1	2	1	2	5	16	16	0.0195	0.5	0.6667	0.2931	0	Sweep	4194	205.81	4	4	0	28804.44	88	0	29312.01	10.25
122	57	5	4	1	3	1	2	6	16	16	0.0234	0.3333	0.75	0.2617	0	Sweep	34004	1829.83	6	6	0	45580.57	139	0	47714.36	10.25
123	57	6	5	2	3	2	2	7	16	16	0.0273	0.5	0.6	0.2837	1	Trivial	0	0	0	0	0	0	0	0	300.05	10.5
124	58	7	4	1	3	1	2	6	16	16	0.0234	0.3333	0.75	0.2617	0	Sweep	41187	2044.92	5	5	0	33896.97	88	0	36247.31	10.01
125	58	8	7	4	3	5	2	9	16	16	0.0352	0.8333	0.4286	0.3533	1	Sweep-S1	228400	8005.13	0	0	0	1017.33	10	0	9322.75	10.25
126	59	9	6	4	2	3	2	8	16	16	0.0313	0.6	0.3333	0.2623	1	1B	1	9.03	1	0	1	6820.31	0	0	7437.01	12.21
127	60	3	2	2	0	0	3	5	16	16	0.0195	0	0	0.0098	1	Sweep-S1	21	15.63	5	5	0	24263.67	12	0	24581.79	12.94
128	61	4	3	2	1	1	3	6	16	16	0.0234	0.5	0.3333	0.2284	0	Sweep	184	10.01	1	1	0	17205.08	110	0	17522.46	14.16
129	61	5	4	3	1	2	3	7	16	16	0.0273	0.6667	0.25	0.2637	1	Sweep-S1	5	9.52	1	0	1	5989.99	4	0	6300.54	10.01
130	62	6	4	1	3	1	3	7	16	16	0.0273	0.3333	0.75	0.2637	0	Sweep	811	13900.39	1	0	1	10137.94	106	1	24338.62	9.52
131	62	7	6	1	5	2	3	9	16	16	0.0352	0.4	0.8333	0.3042	1	Sweep-S3	259400	8004.64	0	0	0	3620.85	36	0	11931.15	10.25
132	63	8	7	3	4	4	3	10	16	16	0.0391	0.6667	0.5714	0.3338	0	Sweep	447800	8008.06	0	0	0	13417.48	138	0	21726.56	10.25
133	63	9	6	4	2	2	3	9	16	16	0.0352	0.4	0.3333	0.2042	1	Sweep-S1	40977	2202.88	7	7	0	34796.14	6	0	37302.98	10.25
134	64	3	2	2	0	0	4	6	16	16	0.0234	0	0	0.0117	0	Sweep	762	10535.64	1	0	1	6940.43	73	1	17778.81	10.25
135	64	4	3	2	1	1	4	7	16	16	0.0273	0.5	0.3333	0.2303	0	Sweep	818	16315.19	1	0	1	14031.25	113	1	30652.34	10.25
136	64	5	4	2	2	2	4	8	16	16	0.0313	0.6667	0.5	0.3156	0	Sweep	794	12106.45	1	0	1	10701.66	89	1	23109.86	10.74
137	64	6	5	2	3	3	4	9	16	16	0.0352	0.75	0.6	0.3626	0	Sweep	806	22522.46	1	0	1	13753.17	101	1	36579.59	10.5
138	64	7	6	5	1	1	4	10	16	16	0.0391	0.2	0.1667	0.1129	0	Sweep	757	26.86	5	5	0	39262.21	170	0	39596.44	10.5
139	64	8	7	2	5	1	4	11	16	16	0.043	0.1667	0.7143	0.2143	1	1B	65	24.17	1	0	1	8658.2	0	0	9291.5	10.5
140	65	9	6	2	4	1	4	10	16	16	0.0391	0.2	0.6667	0.2129	1	1B	194	771	3	0	3	20057.86	0	0	21434.57	9.77
141	66	3	2	2	0	0	0	2	16	16	0.0078	0	0	0.0039	1	Sweep-S1	5	7.81	1	0	1	7915.53	4	0	8226.32	9.77
142	67	4	3	1	2	2	0	3	16	16	0.0117	1	0.6667	0.4392	1	Sweep-S1	6	7.57	1	0	1	8622.8	5	0	8935.79	9.77
143	68	5	4	2	2	3	0	4	16	16	0.0156	1	0.5	0.4078	0	Sweep	681	28.81	4	4	0	26709.47	96	0	27042.24	9.52
144	68	6	5	4	1	2	0	5	16	16	0.0195	0.5	0.2	0.1998	0	Sweep	777	27.34	4	4	0	38391.84	192	0	38722.41	66.41
145	68	7	6	2	4	2	0	6	16	16	0.0234	0.4	0.6667	0.2651	0	Sweep	262739	8006.59	0	0	0	8043.95	82	0	16353.52	12.7
146	68	8	7	3	4	5	0	7	16	16	0.0273	0.8333	0.5714	0.378	0	Sweep	41692	2244.14	7	7	0	46972.9	147	0	49524.66	12.94
147	68	9	6	4	2	2	0	6	16	16	0.0234	0.4	0.3333	0.1984	1	1B	1	8.06	1	0	1	6605.96	0	0	7217.53	10.01
148	69	3	2	2	0	0	1	3	16	16	0.0117	0	0	0.0059	0	Sweep	754	3948	1	0	1	4722.9	49	1	8977.78	9.77
149	69	4	3	2	1	2	1	4	16	16	0.0156	1	0.3333	0.3745	0	Sweep	235	18.31	2	2	0	21452.15	90	0	21771.97	11.23
150	69	5	4	1	3	2	1	5	16	16	0.0195	0.6667	0.75	0.3598	0	Sweep	1261	55.42	5	5	0	33735.11	99	0	34094.24	10.25

Run	Level	Steps	SolutionObjects	Mirrors	Refractors	RuleTransitio	Decoys	TotalObjects	GridWidth	GridHeight	Cl	Ch	Ci	MID	Solved	SolvePhase	SearchNodes	SearchTimeMs	TotalPlacements	InPlaceRotations	Relocations	ExecutionTimeMs	SweepIterations	SweepRelocations	Solve TimeMs	GenerationTimeMs
151	69	6	4	3	1	2	1	5	16	16	0.0195	0.6667	0.25	0.2598	1	Sweep-S1	37	7.57	1	1	0	9750.24	36	0	10059.57	9.77
152	70	7	6	3	3	1	1	7	16	16	0.0273	0.2	0.5	0.1737	0	Sweep	357000	8002.2	0	0	0	7156.74	74	0	15464.6	9.52
153	70	8	6	4	2	3	1	7	16	16	0.0273	0.6	0.3333	0.2603	0	Sweep	384000	8004.64	0	0	0	8716.06	89	0	17022.71	10.25
154	70	9	5	5	0	0	1	6	16	16	0.0234	0	0	0.0117	0	Sweep	782	14537.6	1	0	1	10164.06	77	1	25006.84	10.25
155	70	3	2	2	0	0	2	4	16	16	0.0156	0	0	0.0078	0	Sweep	4170	213.87	2	0	2	19443.6	64	0	19961.67	9.77
156	70	4	3	3	0	0	2	5	16	16	0.0195	0	0	0.0098	1	Trivial	0	0	0	0	0	0	0	0	303.22	10.5
157	71	5	4	2	2	1	2	6	16	16	0.0234	0.3333	0.5	0.2117	1	1A	8258	436.04	1	1	0	4886.96	0	0	5931.15	10.99
158	72	6	5	3	2	3	2	7	16	16	0.0273	0.75	0.4	0.3187	0	Sweep	790	15706.54	1	0	1	11030.76	85	1	27040.28	9.77
159	72	7	3	2	1	1	2	5	16	16	0.0195	0.5	0.3333	0.2264	1	1B	32777	1635.25	1	0	1	4590.58	0	0	6829.83	10.25
160	73	8	6	1	5	2	2	8	16	16	0.0313	0.4	0.8333	0.3023	0	Sweep	794	11604.74	1	0	1	8491.7	89	1	20398.68	10.01
161	73	9	8	4	4	6	2	10	16	16	0.0391	0.8571	0.5	0.3767	1	Sweep-S2	739	20472.41	1	0	1	6481.2	34	1	27256.84	10.25
162	74	3	2	2	0	0	3	5	16	16	0.0195	0	0	0.0098	1	Sweep-S1	21	15.14	4	4	0	18465.82	12	0	18785.89	10.01
163	75	4	3	0	3	0	3	6	16	16	0.0234	0	1	0.2117	1	Sweep-S1	5134	270.26	3	3	0	15507.32	3	0	16078.13	10.25
164	76	5	4	3	1	2	3	7	16	16	0.0273	0.6667	0.25	0.2637	0	1A	299124	8008.55	0	0	0	10386.96	107	0	18698.49	10.25
165	76	6	5	5	0	0	3	8	16	16	0.0313	0	0	0.0156	0	Sweep	665600	8006.84	0	0	0	24218.02	243	0	32530.27	10.25
166	76	7	6	3	3	3	3	9	16	16	0.0352	0.6	0.5	0.2976	0	Sweep	410600	8013.43	0	0	0	15519.29	155	0	23837.16	10.5
167	76	8	7	5	2	3	3	10	16	16	0.0391	0.5	0.2857	0.2267	1	Sweep-S1	8	9.03	1	0	1	8293.7	7	0	8605.47	11.23
168	77	9	5	3	2	1	3	8	16	16	0.0313	0.25	0.4	0.1706	1	Sweep-S1	6	10.25	1	0	1	7105.71	5	0	7421.88	10.74
169	78	3	2	2	0	0	4	6	16	16	0.0234	0	0	0.0117	1	Trivial	0	0	0	0	0	0	0	0	300.29	10.74
170	79	4	3	2	1	1	4	7	16	16	0.0273	0.5	0.3333	0.2303	1	Sweep-S1	5	8.79	1	0	1	7024.66	4	0	7339.6	10.5
171	80	5	4	3	1	2	4	8	16	16	0.0313	0.6667	0.25	0.2656	0	1A	262771	8006.59	0	0	0	9506.35	97	0	17816.65	12.21
172	80	6	4	0	4	0	4	8	16	16	0.0313	0	1	0.2156	0	Sweep	8909	485.84	4	4	0	29308.11	132	0	30099.85	11.23
173	80	7	6	3	3	3	4	10	16	16	0.0391	0.6	0.5	0.2995	1	1B	1	9.28	1	0	1	4732.67	0	0	5345.95	10.99
174	81	8	5	3	2	2	4	9	16	16	0.0352	0.5	0.4	0.2476	1	Sweep-S1	8	10.01	1	0	1	9891.11	7	0	10210.69	11.23
175	82	9	5	4	1	1	4	9	16	16	0.0352	0.25	0.2	0.1326	0	Sweep	359800	8012.45	0	0	0	12039.55	121	0	20359.62	10.74
176	82	3	2	2	0	0	0	2	16	16	0.0078	0	0	0.0039	1	Sweep-S1	5	9.03	1	0	1	7315.43	4	0	7626.95	10.5
177	83	4	3	2	1	1	0	3	16	16	0.0117	0.5	0.3333	0.2225	0	Sweep	754	5543.7	1	0	1	4802	49	1	10654.3	11.96
178	83	5	4	2	2	3	0	4	16	16	0.0156	1	0.5	0.4078	0	Sweep	742	30.76	3	3	0	27627.69	156	0	27959.72	10.99
179	83	6	4	2	2	1	0	4	16	16	0.0156	0.3333	0.5	0.2078	0	Sweep	4928	8199.46	0	0	0	9334.47	95	0	17838.87	10.74
180	83	7	6	3	3	4	0	6	16	16	0.0234	0.8	0.5	0.3517	1	Sweep-S1	13	8.3	1	1	0	5973.63	3	0	6286.87	10.5
181	84	8	6	4	2	2	0	6	16	16	0.0234	0.4	0.3333	0.1984	1	Sweep-S1	6	9.77	1	0	1	7371.34	5	0	7682.13	10.74
182	85	9	8	4	4	6	0	8	16	16	0.0313	0.8571	0.5	0.3728	0	Sweep	8443	554.69	6	6	0	44394.53	185	0	45257.08	10.5
183	85	3	2	2	0	0	1	3	16	16	0.0117	0	0	0.0059	1	Sweep-S1	5	10.25	1	0	1	7013.18	4	0	7328.37	10.74
184	86	4	3	3	0	0	1	4	16	16	0.0156	0	0	0.0078	1	Sweep-S1	37	9.52	1	1	0	9146.73	36	0	9456.79	10.25
185	87	5	3	2	1	2	1	4	16	16	0.0156	1	0.3333	0.3745	1	1B	1	9.03	1	0	1	7074.71	0	0	7693.36	14.16
186	88	6	5	5	0	0	1	6	16	16	0.0234	0	0	0.0117	1	1B	1	14.89	1	0	1	8340.09	0	0	8975.1	11.96
187	89	7	5	4	1	2	1	6	16	16	0.0234	0.5	0.2	0.2017	0	Sweep	650	36.87	3	3	0	23792.24	121	0	24136.96	93.99
188	89	8	7	6	1	1	1	8	16	16	0.0313	0.1667	0.1429	0.0942	0	Sweep	4239	220.46	6	6	0	36742.19	134	0	37270.51	11.47
189	89	9	8	5	3	5	1	9	16	16	0.0352	0.7143	0.375	0.3069	0	1A	295092	8007.81	0	0	0	11309.08	115	0	19623.54	11.23
190	89	3	2	2	0	0	2	4	16	16	0.0156	0	0	0.0078	1	Sweep-S1	21	18.07	4	4	0	27751.46	12	0	28077.39	10.99
191	90	4	3	2	1	2	2	5	16	16	0.0195	1	0.3333	0.3764	1	1B	1	10.74	1	0	1	7582.52	0	0	8198.98	10.99
192	91	5	4	2	2	2	2	6	16	16	0.0234	0.6667	0.5	0.3117	0	Sweep	5244	298.34	4	4	0	31855.96	114	0	32455.08	10.74
193	91	6	3	1	2	1	2	5	16	16	0.0195	0.5	0.6667	0.2931	1	Sweep-S1	5	9.28	1	1	0	3904.79	4	0	4215.33	11.23
194	92	7	5	3	2	3	2	7	16	16	0.0273	0.75	0.4	0.3187	1	Sweep-S1	5	9.28	1	0	1	8773.44	4	0	9088.38	10.25
195	93	8	7	3	4	4	2	9	16	16	0.0352	0.6667	0.5714	0.3319	0	Sweep	418200	8022.95	0	0	0	14138.18	143	0	22467.29	11.72
196	93	9	6	6	0	0	2	8	16	16	0.0313	0	0	0.0156	0	Sweep	798	19705.08	1	0	1	12822.75	93	1	32829.59	10.74
197	93	3	2	2	0	0	3	5	16	16	0.0195	0	0	0.0098	0	Sweep	746	1763.18	1	0	1	6230.47	65	1	8301.27	10.74
198	93	4	3	3	0	0	3	6	16	16	0.0234	0	0	0.0117	1	Sweep-S1	29	19.53	4	4	0	24095.7	20	0	24419.92	10.74
199	94	5	4	1	3	1	3	7	16	16	0.0273	0.3333	0.75	0.2637	0	Sweep	786	17347.66	1	0	1	10964.84	81	1	28615.72	11.23
200	94	6	3	1	2	1	3	6	16	16	0.0234	0.5	0.6667	0.2951	1	Sweep-S1	16	10.25	1	1	0	6128.91	7	0	6445.31	11.23

Run	Level	Steps	SolutionObjects	Mirrors	Refractors	RuleTransitions	Decoys	TotalObjects	GridWidth	GridHeight	Cl	Ch	Ci	MID	Solved	SolvePhase	SearchNodes	SearchTimeMs	TotalPlacements	InPlaceRotations	Relocations	ExecutionTimeMs	SweepIterations	SweepRelocations	Solve TimeMs	GenerationTimeMs
201	95	7	6	4	2	3	3	9	16	16	0.0352	0.6	0.3333	0.2642	1	Trivial	0	0	0	0	0	0	0	0	308.59	11.23
202	96	8	6	3	3	2	3	9	16	16	0.0352	0.4	0.5	0.2376	0	Sweep	373800	8004.88	0	0	0	11543.95	118	0	19855.96	11.23
203	96	9	8	3	5	3	3	11	16	16	0.043	0.4286	0.625	0.2751	0	Sweep	900	15918.95	0	0	0	30751.46	131	0	46976.07	11.23
204	96	3	2	2	0	0	4	6	16	16	0.0234	0	0	0.0117	1	Sweep-S1	13	8.79	1	1	0	6029.3	4	0	6344.73	12.21
205	97	4	3	2	1	2	4	7	16	16	0.0273	1	0.3333	0.3803	0	Sweep	917	13778.32	1	0	1	21158.2	212	1	35241.21	10.74
206	97	5	4	3	1	2	4	8	16	16	0.0313	0.6667	0.25	0.2656	0	Sweep	318200	8007.81	0	0	0	8665.04	89	0	16973.14	11.23
207	97	6	4	3	1	1	4	8	16	16	0.0313	0.3333	0.25	0.1656	1	Sweep-S2	739	12106.93	1	0	1	5932.13	34	1	18345.7	10.25
208	98	7	6	4	2	4	4	10	16	16	0.0391	0.8	0.3333	0.3262	1	Sweep-S1	13	9.77	1	1	0	7709.96	3	0	8021	10.74
209	99	8	3	1	2	2	4	7	16	16	0.0273	1	0.6667	0.447	1	Sweep-S1	6	9.77	1	0	1	5474.12	5	0	5788.09	10.25
210	100	9	7	2	5	1	4	11	16	16	0.043	0.1667	0.7143	0.2143	0	Sweep	818	8820.8	0	0	0	17072.75	105	0	26198.73	10.74
211	100	3	2	2	0	0	0	2	16	16	0.0078	0	0	0.0039	1	Sweep-S1	5	8.3	1	0	1	6811.52	4	0	7129.4	10.25
212	101	4	3	1	2	2	0	3	16	16	0.0117	1	0.6667	0.4392	1	Trivial	0	0	0	0	0	0	0	0	306.15	10.74
213	102	5	3	3	0	0	0	3	16	16	0.0117	0	0	0.0059	1	Sweep-S1	5	8.79	1	0	1	7579.1	4	0	7890.63	10.74
214	103	6	4	2	2	1	0	4	16	16	0.0156	0.3333	0.5	0.2078	1	Sweep-S1	33	23.44	3	3	0	15187.01	23	0	15514.65	10.25
215	104	7	5	3	2	3	0	5	16	16	0.0195	0.75	0.4	0.3148	0	Sweep	774	12337.89	1	0	1	9082.52	69	1	21728.52	11.23
216	104	8	5	3	2	3	0	5	16	16	0.0195	0.75	0.4	0.3148	1	Sweep-S1	86	22.95	5	5	0	22657.23	11	0	22983.89	11.23
217	105	9	6	2	4	2	0	6	16	16	0.0234	0.4	0.6667	0.2651	1	1B	1	12.21	1	0	1	8958.98	0	0	9581.54	10.74
218	106	3	2	2	0	0	1	3	16	16	0.0117	0	0	0.0059	1	Sweep-S1	21	17.58	2	2	0	10172.85	12	0	10495.61	10.74
219	107	4	3	2	1	1	1	4	16	16	0.0156	0.5	0.3333	0.2245	0	Sweep	259	21.97	3	3	0	19457.52	65	0	19789.55	14.65
220	107	5	3	2	1	2	1	4	16	16	0.0156	1	0.3333	0.3745	1	Sweep-S1	8	8.79	1	0	1	8265.14	7	0	8577.64	10.74
221	108	6	4	3	1	1	1	5	16	16	0.0195	0.3333	0.25	0.1598	1	1A	2	8.79	1	1	0	3156.74	0	0	3768.56	11.23
222	109	7	6	5	1	2	1	7	16	16	0.0273	0.4	0.1667	0.167	0	Sweep	790	9318.36	1	0	1	8371.58	85	1	17992.68	11.23
223	109	8	4	1	3	1	1	5	16	16	0.0195	0.3333	0.75	0.2598	0	Sweep	746	9322.75	1	0	1	6248.05	65	1	15875	10.74
224	109	9	8	7	1	2	1	9	16	16	0.0352	0.2857	0.125	0.1283	0	Sweep	349200	8006.35	0	0	0	11259.77	113	0	19566.89	10.74
225	109	3	2	2	0	0	2	4	16	16	0.0156	0	0	0.0078	1	Sweep-S1	21	18.07	2	2	0	10167.48	12	0	10490.23	11.23
226	110	4	3	1	2	1	2	5	16	16	0.0195	0.5	0.6667	0.2931	1	Sweep-S1	599	31.25	3	3	0	15266.6	5	0	15602.05	11.23
227	111	5	4	3	1	2	2	6	16	16	0.0234	0.6667	0.25	0.2617	0	Sweep	37058	2181.15	3	3	0	24061.52	121	0	26550.29	11.23
228	111	6	5	0	5	0	2	7	16	16	0.0273	0	1	0.2137	0	Sweep	41229	2406.25	6	6	0	38297.36	115	0	41012.7	12.7
229	111	7	6	4	2	2	2	8	16	16	0.0313	0.4	0.3333	0.2023	0	1A	262763	8004.88	0	0	0	9542.97	97	0	17848.63	11.23
230	111	8	6	4	2	2	2	8	16	16	0.0313	0.4	0.3333	0.2023	0	Sweep	396600	8024.41	0	0	0	13011.72	133	0	21340.82	11.72
231	111	9	8	4	4	4	2	10	16	16	0.0391	0.5714	0.5	0.291	0	Sweep	810	24076.66	1	0	1	14279.79	105	1	38661.13	11.72
232	111	3	2	2	0	0	3	5	16	16	0.0195	0	0	0.0098	1	Sweep-S1	32837	1849.61	1	0	1	7069.82	4	0	9225.59	10.74
233	112	4	3	2	1	2	3	6	16	16	0.0234	1	0.3333	0.3784	0	Sweep	766	10822.75	1	0	1	7613.28	77	1	18742.68	11.72
234	112	5	4	0	4	0	3	7	16	16	0.0273	0	1	0.2137	0	Sweep	820	13356.93	1	0	1	11138.18	115	1	24800.29	11.72
235	112	6	4	2	2	2	3	7	16	16	0.0273	0.6667	0.5	0.3137	0	Sweep	811	16488.28	1	0	1	13465.82	106	1	30262.7	10.74
236	112	7	6	3	3	2	3	9	16	16	0.0352	0.4	0.5	0.2376	1	1B	1	9.77	1	0	1	6770.02	0	0	7390.63	10.74
237	113	8	6	6	0	0	3	9	16	16	0.0352	0	0	0.0176	0	Sweep	369400	8007.81	0	0	0	12065.43	122	0	20375.49	11.23
238	113	9	7	6	1	2	3	10	16	16	0.0391	0.3333	0.1429	0.1481	0	1A	295655	8024.9	0	0	0	15610.35	158	0	23942.87	11.23
239	113	3	2	2	0	0	4	6	16	16	0.0234	0	0	0.0117	0	Sweep	778	13774.41	1	0	1	9760.74	73	1	23844.73	11.72
240	113	4	3	2	1	1	4	7	16	16	0.0273	0.5	0.3333	0.2303	1	Sweep-S1	6	9.28	1	0	1	6872.56	5	0	7190.92	11.23
241	114	5	3	1	2	1	4	7	16	16	0.0273	0.5	0.6667	0.297	1	Sweep-S1	6	10.25	1	0	1	6099.61	5	0	6415.04	10.74
242	115	6	5	4	1	1	4	9	16	16	0.0352	0.25	0.2	0.1326	1	Sweep-S1	13	8.3	1	1	0	5195.31	3	0	5508.3	12.21
243	116	7	6	2	4	3	4	10	16	16	0.0391	0.6	0.6667	0.3329	0	Sweep	372000	8000.98	0	0	0	11343.26	115	0	19646.97	11.23
244	116	8	6	6	0	0	4	10	16	16	0.0391	0	0	0.0195	0	Sweep	810	13112.79	1	0	1	10294.43	105	1	23712.4	11.72
245	116	9	8	5	3	3	4	12	16	16	0.0469	0.4286	0.375	0.227	0	Sweep	125	28.81	1	0	1	19399.9	116	0	19730.47	10.74
246	116	3	2	2	0	0	0	2	16	16	0.0078	0	0	0.0039	1	Sweep-S1	21	18.07	2	2	0	11099.61	12	0	11424.32	11.23
247	117	4	3	2	1	2	0	3	16	16	0.0117	1	0.3333	0.3725	0	Sweep	145	17.09	3	3	0	20562.99	72	0	20884.28	10.74
248	117	5	4	3	1	1	0	4	16	16	0.0156	0.3333	0.25	0.1578	0	Sweep	766	9636.72	1	0	1	6110.35	61	1	16051.76	10.74
249	117	6	5	3	2	3	0	5	16	16	0.0195	0.75	0.4	0.3148	0	Sweep	4259	241.21	4	4	0	26668.95	88	0	27213.38	11.23
250	117	7	5	5	0	0	0	5	16	16	0.0195	0	0	0.0098	0	Sweep	33607	10093.75	0	0	0	17844.24	126	0	28241.21	10.74

Run	Level	Steps	SolutionObjects	Mirrors	Refractors	RuleTransitions	Decoys	TotalObjects	GridWidth	GridHeight	Cl	Ch	Ci	MID	Solved	SolvePhase	SearchNodes	SearchTimeMs	TotalPlaceme	InPlaceRotations	Relocations	ExecutionTimeMs	SweepIterations	SweepRelocations	Solve TimeMs	GenerationTimeMs
251	117	8	4	3	1	1	0	4	16	16	0.0156	0.3333	0.25	0.1578	1	Sweep-S1	5	8.79	1	0	1	5899.9	4	0	6210.45	9.77
252	118	9	6	3	3	3	0	6	16	16	0.0234	0.6	0.5	0.2917	1	1B	1	9.28	1	0	1	7088.38	0	0	7709.47	13.18
253	119	3	2	2	0	0	1	3	16	16	0.0117	0	0	0.0059	1	Sweep-S1	19	8.3	1	1	0	5962.89	18	0	6278.32	11.72
254	120	4	3	2	1	1	1	4	16	16	0.0156	0.5	0.3333	0.2245	1	Sweep-S1	6	9.28	1	0	1	6876.95	5	0	7187.5	12.21
255	121	5	4	3	1	2	1	5	16	16	0.0195	0.6667	0.25	0.2598	0	Sweep	774	8462.4	1	0	1	6687.5	69	1	15456.05	12.21
256	121	6	4	1	3	1	1	5	16	16	0.0195	0.3333	0.75	0.2598	0	Sweep	1279	65.43	3	3	0	32643.55	182	0	33016.6	13.18
257	121	7	6	5	1	2	1	7	16	16	0.0273	0.4	0.1667	0.167	1	Sweep-S1	717	39.55	5	5	0	26636.72	68	0	26981.45	12.21
258	122	8	6	4	2	3	1	7	16	16	0.0273	0.6	0.3333	0.2603	0	Sweep	37053	1829.59	3	3	0	21420.41	106	0	23550.29	9.77
259	122	9	7	4	3	4	1	8	16	16	0.0313	0.6667	0.4286	0.3013	1	Sweep-S1	25	8.79	1	0	1	8687.01	24	0	9002.44	9.77
260	123	3	2	2	0	0	2	4	16	16	0.0156	0	0	0.0078	0	Sweep	730	7022.46	1	0	1	5480.96	57	1	12809.57	10.25
261	123	4	3	2	1	1	2	5	16	16	0.0195	0.5	0.3333	0.2264	1	Sweep-S1	5	7.81	1	0	1	7014.16	4	0	7329.1	10.25
262	124	5	4	2	2	1	2	6	16	16	0.0234	0.3333	0.5	0.2117	0	Sweep	782	7453.61	1	0	1	7420.41	77	1	15176.76	10.25
263	124	6	5	4	1	1	2	7	16	16	0.0273	0.25	0.2	0.1287	1	Sweep-S1	4227	256.35	6	6	0	36819.82	58	0	37382.32	10.25
264	125	7	6	4	2	3	2	8	16	16	0.0313	0.6	0.3333	0.2623	1	Sweep-S1	249600	8002.93	0	0	0	3125.98	30	0	11430.18	13.18
265	126	8	6	5	1	2	2	8	16	16	0.0313	0.4	0.1667	0.169	1	1B	1	8.3	1	0	1	7816.9	0	0	8434.08	10.74
266	127	9	8	6	2	1	2	10	16	16	0.0391	0.1429	0.25	0.1124	1	Sweep-S1	6	9.28	1	0	1	6607.42	5	0	6922.36	10.74
267	128	3	2	2	0	0	3	5	16	16	0.0195	0	0	0.0098	1	Trivial	0	0	0	0	0	0	0	0	302.25	10.25
268	129	4	3	2	1	2	3	6	16	16	0.0234	1	0.3333	0.3784	1	Sweep-S1	65566	3437.99	4	4	0	27584.47	21	0	31327.64	11.23
269	130	5	3	3	0	0	3	6	16	16	0.0234	0	0	0.0117	1	1B	1	8.3	1	0	1	5453.13	0	0	6068.36	10.74
270	131	6	5	4	1	1	3	8	16	16	0.0313	0.25	0.2	0.1306	1	1B	1	8.3	1	0	1	6819.34	0	0	7436.04	9.77
271	132	7	3	2	1	1	3	6	16	16	0.0234	0.5	0.3333	0.2284	1	1B	1	8.79	1	0	1	6577.15	0	0	7191.41	10.74
272	133	8	7	5	2	2	3	10	16	16	0.0391	0.3333	0.2857	0.1767	0	Sweep	814	23316.41	1	0	1	14738.77	109	1	38360.84	10.74
273	133	9	8	7	1	2	3	11	16	16	0.043	0.2857	0.125	0.1322	0	Sweep	212	83.98	2	0	2	29801.76	146	0	30193.36	9.77
274	133	3	2	2	0	0	4	6	16	16	0.0234	0	0	0.0117	0	Sweep	262219	8000.49	0	0	0	6385.25	66	0	14688.96	10.25
275	133	4	3	2	1	2	4	7	16	16	0.0273	1	0.3333	0.3803	1	Sweep-S1	86	15.14	4	4	0	18902.34	4	0	19223.63	10.74
276	134	5	4	2	2	3	4	8	16	16	0.0313	1	0.5	0.4156	1	Sweep-S1	8	9.28	1	0	1	8349.12	7	0	8664.06	9.77
277	135	6	4	3	1	2	4	8	16	16	0.0313	0.6667	0.25	0.2656	0	Sweep	408400	8000.49	0	0	0	10576.66	107	0	18883.3	10.25
278	135	7	6	2	4	3	4	10	16	16	0.0391	0.6	0.6667	0.3329	0	Sweep	393600	8002.44	0	0	0	10157.71	105	0	18460.45	10.74
279	135	8	5	1	4	2	4	9	16	16	0.0352	0.5	0.8	0.3276	0	Sweep	168	7.32	1	1	0	20932.13	150	0	21243.16	10.25
280	135	9	8	5	3	5	4	12	16	16	0.0469	0.7143	0.375	0.3127	1	Sweep-S1	14	8.3	1	0	1	8133.3	13	0	8446.78	10.25
281	136	3	2	2	0	0	0	2	16	16	0.0078	0	0	0.0039	1	Sweep-S1	5	7.81	1	0	1	8048.83	4	0	8360.84	9.28
282	137	4	3	2	1	1	0	3	16	16	0.0117	0.5	0.3333	0.2225	1	Sweep-S1	8	10.25	1	0	1	7858.4	7	0	8173.34	10.74
283	138	5	4	1	3	1	0	4	16	16	0.0156	0.3333	0.75	0.2578	1	1B	1	8.3	1	0	1	7834.96	0	0	8451.66	10.74
284	139	6	5	4	1	2	0	5	16	16	0.0195	0.5	0.2	0.1998	1	Trivial	0	0	0	0	0	0	0	0	301.76	9.77
285	140	7	5	4	1	2	0	5	16	16	0.0195	0.5	0.2	0.1998	0	Sweep	1114	42.48	3	3	0	22562.01	80	0	22909.67	9.77
286	140	8	7	7	0	0	0	7	16	16	0.0273	0	0	0.0137	1	1B	1	8.3	1	0	1	7570.31	0	0	8183.59	9.77
287	141	9	7	3	4	2	0	7	16	16	0.0273	0.3333	0.5714	0.228	0	Sweep	352000	8005.37	0	0	0	7245.12	74	0	15552.25	10.25
288	141	3	2	2	0	0	1	3	16	16	0.0117	0	0	0.0059	1	Sweep-S1	6	7.81	1	1	0	6264.65	4	0	6576.66	10.25
289	142	4	3	1	2	2	1	4	16	16	0.0156	1	0.6667	0.4411	0	Sweep	154	14.65	4	4	0	25347.17	80	0	25664.55	10.25
290	142	5	4	4	0	0	1	5	16	16	0.0195	0	0	0.0098	0	Sweep	774	13459.47	1	0	1	6776.37	69	1	20539.55	10.25
291	142	6	5	5	0	0	1	6	16	16	0.0234	0	0	0.0117	1	Sweep-S1	5	10.25	1	0	1	7773.44	4	0	8086.91	10.25
292	143	7	6	3	3	3	1	7	16	16	0.0273	0.6	0.5	0.2937	0	Sweep	70890	3975.1	6	6	0	43128.42	161	0	47411.13	11.23
293	143	8	5	4	1	2	1	6	16	16	0.0234	0.5	0.2	0.2017	1	Sweep-S1	8	8.79	1	0	1	8245.12	7	0	8560.55	10.74
294	144	9	6	4	2	2	1	7	16	16	0.0273	0.4	0.3333	0.2003	1	Sweep-S1	6	8.3	1	0	1	10708.98	5	0	11018.55	10.25
295	145	3	2	2	0	0	2	4	16	16	0.0156	0	0	0.0078	0	Sweep	762	6903.32	1	0	1	5515.63	57	1	12719.73	10.25
296	145	4	3	2	1	2	2	5	16	16	0.0195	1	0.3333	0.3764	0	Sweep	747	52.73	3	3	0	24678.22	98	0	25034.67	10.25
297	145	5	4	2	2	3	2	6	16	16	0.0234	1	0.5	0.4117	0	Sweep	262336	8005.37	0	0	0	12514.65	127	0	20826.17	10.74
298	145	6	5	4	1	1	2	7	16	16	0.0273	0.25	0.2	0.1287	0	Sweep	66203	3105.96	4	4	0	32109.86	145	0	35520.02	10.25
299	145	7	5	5	0	0	2	7	16	16	0.0273	0	0	0.0137	0	Sweep	33457	1738.77	5	5	0	34115.23	112	0	36155.76	10.74
300	145	8	7	4	3	1	2	9	16	16	0.0352	0.1667	0.4286	0.1533	0	Sweep	806	20450.2	1	0	1	13499.51	101	1	34250.49	10.25

Run	Level	Steps	SolutionObjects	Mirrors	Refractors	RuleTransitions	Decoys	TotalObjects	GridWidth	GridHeight	Cl	Ch	Ci	MID	Solved	SolvePhase	SearchNodes	SearchTimeMs	TotalPlacements	InPlaceRotations	Relocations	ExecutionTimeMs	SweepIterations	SweepRelocations	Solve TimeMs	GenerationTimeMs
301	145	9	7	2	5	3	2	9	16	16	0.0352	0.5	0.7143	0.3104	1	Sweep-S1	65647	3277.83	4	4	0	21181.64	45	0	24760.25	10.74
302	146	3	2	2	0	0	3	5	16	16	0.0195	0	0	0.0098	1	1B	1	8.79	1	0	1	6805.18	0	0	7418.46	9.77
303	147	4	3	1	2	1	3	6	16	16	0.0234	0.5	0.6667	0.2951	1	Sweep-S1	8	8.3	1	0	1	7809.08	7	0	8120.61	10.25
304	148	5	4	2	2	1	3	7	16	16	0.0273	0.3333	0.5	0.2137	0	Sweep	811	16505.86	1	0	1	13101.07	106	1	29909.67	9.77
305	148	6	5	3	2	3	3	8	16	16	0.0313	0.75	0.4	0.3206	0	Sweep	722	36.13	3	3	0	25476.07	120	0	25815.92	10.25
306	148	7	6	2	4	4	3	9	16	16	0.0352	0.8	0.6667	0.3909	0	Sweep	423400	8005.86	0	0	0	12554.2	126	0	20866.21	9.77
307	148	8	6	4	2	4	3	9	16	16	0.0352	0.8	0.3333	0.3242	1	Sweep-S1	179000	8005.37	0	0	0	344.73	3	0	8654.79	12.7
308	149	9	7	3	4	1	3	10	16	16	0.0391	0.1667	0.5714	0.1838	0	Sweep	426200	8011.23	0	0	0	14192.87	145	0	22506.84	13.67
309	149	3	2	2	0	0	4	6	16	16	0.0234	0	0	0.0117	1	Sweep-S1	69	8.79	1	1	0	5081.06	4	0	5396.97	10.74
310	150	4	3	1	2	1	4	7	16	16	0.0273	0.5	0.6667	0.297	1	Sweep-S1	676	36.62	5	5	0	26551.76	25	0	26896	10.25
311	151	5	4	2	2	1	4	8	16	16	0.0313	0.3333	0.5	0.2156	0	Sweep	597400	8005.86	0	0	0	21380.86	217	0	29687.99	10.25
312	151	6	3	1	2	1	4	7	16	16	0.0273	0.5	0.6667	0.297	0	Sweep	812	16486.82	1	0	1	13310.06	107	1	30099.12	10.25
313	151	7	6	3	3	3	4	10	16	16	0.0391	0.6	0.5	0.2995	1	1B	1	8.3	1	0	1	7292.48	0	0	7909.18	10.74
314	152	8	7	6	1	2	4	11	16	16	0.043	0.3333	0.1429	0.1501	1	Trivial	0	0	0	0	0	0	0	0	306.64	10.25
315	153	9	6	6	0	0	4	10	16	16	0.0391	0	0	0.0195	0	Sweep	408000	8002.44	0	0	0	10057.13	102	0	18359.86	9.77
316	153	3	2	2	0	0	0	2	16	16	0.0078	0	0	0.0039	1	Sweep-S1	21	15.14	2	2	0	11106.45	12	0	11424.32	9.77
317	154	4	3	3	0	0	0	3	16	16	0.0117	0	0	0.0059	1	Sweep-S1	37	7.81	1	1	0	9019.04	36	0	9333.5	9.77
318	155	5	3	1	2	1	0	3	16	16	0.0117	0.5	0.6667	0.2892	0	Sweep	758	8844.73	1	0	1	5238.77	53	1	14391.11	12.7
319	155	6	5	3	2	2	0	5	16	16	0.0195	0.5	0.4	0.2398	0	Sweep	1209	64.45	4	4	0	29129.88	111	0	29499.51	12.7
320	155	7	5	4	1	2	0	5	16	16	0.0195	0.5	0.2	0.1998	1	1B	32777	1846.19	1	0	1	6391.11	0	0	8841.8	14.65
321	156	8	6	3	3	3	0	6	16	16	0.0234	0.6	0.5	0.2917	0	Sweep	778	14290.04	1	0	1	9365.72	73	1	23956.05	12.7
322	156	9	7	5	2	2	0	7	16	16	0.0273	0.3333	0.2857	0.1708	0	Sweep	388600	8005.86	0	0	0	9810.55	100	0	18116.21	11.23
323	156	3	2	2	0	0	1	3	16	16	0.0117	0	0	0.0059	1	Sweep-S1	525	33.2	1	0	1	9005.86	4	0	9346.19	10.25
324	157	4	3	2	1	1	1	4	16	16	0.0156	0.5	0.3333	0.2245	0	Sweep	762	6810.55	1	0	1	5396.97	57	1	12513.67	10.25
325	157	5	4	1	3	2	1	5	16	16	0.0195	0.6667	0.75	0.3598	0	Sweep	4774	234.86	4	4	0	24704.59	91	0	25242.19	10.25
326	157	6	5	5	0	0	1	6	16	16	0.0234	0	0	0.0117	0	Sweep	37044	2100.1	3	3	0	33430.18	114	0	35834.47	10.74
327	157	7	5	4	1	2	1	6	16	16	0.0234	0.5	0.2	0.2017	1	Sweep-S1	78	7.81	1	1	0	5518.56	4	0	5828.61	9.77
328	158	8	7	2	5	4	1	8	16	16	0.0313	0.6667	0.7143	0.3585	0	Sweep	453000	8004.88	0	0	0	12831.54	131	0	21142.58	10.74
329	158	9	5	4	1	2	1	6	16	16	0.0234	0.5	0.2	0.2017	0	Sweep	36994	2021.97	3	3	0	25384.77	121	0	27708.5	12.21
330	158	3	2	2	0	0	2	4	16	16	0.0156	0	0	0.0078	1	Sweep-S1	21	15.63	3	3	0	14258.79	12	0	14579.59	10.25
331	159	4	3	1	2	2	2	5	16	16	0.0195	1	0.6667	0.4431	1	1B	1	8.79	1	0	1	6850.1	0	0	7465.33	10.74
332	160	5	3	0	3	0	2	5	16	16	0.0195	0	1	0.2098	1	Sweep-S1	32784	1630.37	1	0	1	5979	7	0	7913.57	10.25
333	161	6	4	2	2	3	2	6	16	16	0.0234	1	0.5	0.4117	1	Sweep-S1	8	8.3	1	0	1	8261.72	7	0	8573.24	9.77
334	162	7	4	2	2	1	2	6	16	16	0.0234	0.3333	0.5	0.2117	1	Sweep-S1	6	7.81	1	0	1	6620.12	5	0	6932.62	10.74
335	163	8	7	4	3	3	2	9	16	16	0.0352	0.5	0.4286	0.2533	0	Sweep	806	13918.46	1	0	1	12474.61	101	1	26694.34	10.74
336	163	9	7	4	3	3	2	9	16	16	0.0352	0.5	0.4286	0.2533	0	1A	332464	8002.93	0	0	0	16878.91	175	0	25184.57	9.77
337	163	3	2	2	0	0	3	5	16	16	0.0195	0	0	0.0098	1	Sweep-S1	6	7.32	1	1	0	5776.86	4	0	6089.36	10.25
338	164	4	3	2	1	1	3	6	16	16	0.0234	0.5	0.3333	0.2284	0	Sweep	780	13682.62	1	0	1	9568.85	75	1	23552.25	9.28
339	164	5	4	3	1	1	3	7	16	16	0.0273	0.3333	0.25	0.1637	0	Sweep	4285	207.52	5	5	0	38775.88	122	0	39288.57	10.74
340	164	6	5	1	4	1	3	8	16	16	0.0313	0.25	0.8	0.2506	0	Sweep	435600	8005.86	0	0	0	13864.26	141	0	22178.22	12.7
341	164	7	4	2	2	3	3	7	16	16	0.0273	1	0.5	0.4137	0	Sweep	9385	538.09	6	6	0	42484.86	104	0	43324.71	12.7
342	164	8	7	4	3	4	3	10	16	16	0.0391	0.6667	0.4286	0.3052	0	Sweep	4827	213.38	3	3	0	28221.68	153	0	28735.35	10.74
343	164	9	5	3	2	2	3	8	16	16	0.0313	0.5	0.4	0.2456	1	1B	1	9.28	1	0	1	4358.4	0	0	4974.12	13.18
344	165	3	2	2	0	0	4	6	16	16	0.0234	0	0	0.0117	1	Sweep-S1	21	32.23	3	3	0	13796.39	12	0	14135.74	12.7
345	166	4	3	1	2	1	4	7	16	16	0.0273	0.5	0.6667	0.297	0	Sweep	1187	71.29	6	6	0	40571.78	88	0	40944.34	45.41
346	166	5	3	1	2	2	4	7	16	16	0.0273	1	0.6667	0.447	1	Sweep-S1	5	8.79	1	0	1	6593.75	4	0	6905.76	398.44
347	167	6	5	3	2	1	4	9	16	16	0.0352	0.25	0.4	0.1726	0	Sweep	73929	3901.86	5	5	0	33184.57	128	0	37389.16	9.77
348	167	7	5	3	2	3	4	9	16	16	0.0352	0.75	0.4	0.3226	0	Sweep	4885	245.12	4	4	0	45075.68	212	0	45621.09	10.25
349	167	8	7	3	4	5	4	11	16	16	0.043	0.8333	0.5714	0.3858	1	Sweep-S2	725	18275.88	1	0	1	16543.46	20	1	35123.05	10.74
350	168	9	8	7	1	2	4	12	16	16	0.0469	0.2857	0.125	0.1342	0	Sweep	32968	1688.97	2	0	2	35536.62	198	0	37526.37	10.74

Run	Level	Steps	SolutionObjects	Mirrors	Refractors	RuleTransitions	Decoys	TotalObjects	GridWidth	GridHeight	Cl	Ch	Ci	MID	Solved	SolvePhase	SearchNodes	SearchTimeMs	TotalPlacements	InPlaceRotations	Relocations	ExecutionTimeMs	SweepIterations	SweepRelocations	Solve TimeMs	GenerationTimeMs
351	168	3	2	2	0	0	0	2	16	16	0.0078	0	0	0.0039	1	Sweep-S1	5	8.3	1	0	1	6538.09	4	0	6847.66	9.77
352	169	4	3	1	2	1	0	3	16	16	0.0117	0.5	0.6667	0.2892	0	Sweep	90	18.07	3	3	0	19728.52	72	0	20049.8	9.77
353	169	5	4	2	2	3	0	4	16	16	0.0156	1	0.5	0.4078	0	Sweep	1220	53.71	3	3	0	26556.15	122	0	26917.48	10.25
354	169	6	4	1	3	2	0	4	16	16	0.0156	0.6667	0.75	0.3578	0	Sweep	606	28.32	2	2	0	17530.27	76	0	17863.28	9.77
355	169	7	5	3	2	2	0	5	16	16	0.0195	0.5	0.4	0.2398	0	Sweep	358	16.11	3	3	0	39270.02	213	0	39592.29	9.77
356	169	8	4	3	1	2	0	4	16	16	0.0156	0.6667	0.25	0.2578	0	Sweep	766	7887.7	1	0	1	5942.87	61	1	14131.84	9.77
357	169	9	5	4	1	1	0	5	16	16	0.0195	0.25	0.2	0.1248	0	Sweep	774	12214.84	1	0	1	9055.66	69	1	21579.59	11.23
358	169	3	2	2	0	0	1	3	16	16	0.0117	0	0	0.0059	1	Sweep-S1	13	7.81	1	1	0	5778.32	4	0	6086.91	9.77
359	170	4	3	3	0	0	1	4	16	16	0.0156	0	0	0.0078	1	Trivial	0	0	0	0	0	0	0	0	300.29	9.77
360	171	5	3	2	1	2	1	4	16	16	0.0156	1	0.3333	0.3745	1	Sweep-S1	5	9.28	1	0	1	7328.13	4	0	7641.6	9.28
361	172	6	4	2	2	1	1	5	16	16	0.0195	0.3333	0.5	0.2098	1	Sweep-S1	5	9.28	1	0	1	7834.96	4	0	8146.48	12.21
362	173	7	6	3	3	3	1	7	16	16	0.0273	0.6	0.5	0.2937	0	1A	299189	8007.32	0	0	0	11327.64	115	0	19638.18	12.21
363	173	8	5	3	2	3	1	6	16	16	0.0234	0.75	0.4	0.3167	1	1B	1	11.23	1	0	1	8075.2	0	0	8696.29	12.7
364	174	9	8	3	5	4	1	9	16	16	0.0352	0.5714	0.625	0.314	1	1A	9	7.32	1	1	0	5995.12	0	0	6609.86	12.21
365	175	3	2	2	0	0	2	4	16	16	0.0156	0	0	0.0078	0	Sweep	770	7734.38	1	0	1	6435.55	65	1	14476.07	9.77
366	175	4	3	2	1	1	2	5	16	16	0.0195	0.5	0.3333	0.2264	1	Sweep-S1	32	15.63	4	4	0	20586.43	21	0	20903.32	9.77
367	176	5	4	2	2	2	2	6	16	16	0.0234	0.6667	0.5	0.3117	0	Sweep	4853	243.16	3	3	0	28548.34	172	0	29099.12	10.25
368	176	6	5	2	3	3	2	7	16	16	0.0273	0.75	0.6	0.3587	0	Sweep	4698	243.16	4	4	0	24274.9	88	0	24822.75	14.16
369	176	7	3	3	0	0	2	5	16	16	0.0195	0	0	0.0098	1	1B	1	8.79	1	0	1	6013.18	0	0	6632.81	9.77
370	177	8	7	5	2	4	2	9	16	16	0.0352	0.6667	0.2857	0.2747	0	Sweep	514800	8003.42	0	0	0	16051.27	163	0	24356.93	9.77
371	177	9	7	5	2	4	2	9	16	16	0.0352	0.6667	0.2857	0.2747	0	Sweep	841	18520.51	1	0	1	16466.8	136	1	35288.09	10.25
372	177	3	2	2	0	0	3	5	16	16	0.0195	0	0	0.0098	0	Sweep	770	12908.2	1	0	1	6167.48	65	1	19378.91	10.74
373	177	4	3	3	0	0	3	6	16	16	0.0234	0	0	0.0117	1	Sweep-S2	737	11279.79	1	0	1	3316.41	32	1	14904.3	10.25
374	178	5	4	3	1	2	3	7	16	16	0.0273	0.6667	0.25	0.2637	0	1A	295035	8001.95	0	0	0	11155.76	114	0	19464.84	10.25
375	178	6	3	0	3	0	3	6	16	16	0.0234	0	1	0.2117	1	Sweep-S1	6	8.3	1	0	1	9352.05	5	0	9660.65	9.77
376	179	7	6	3	3	3	3	9	16	16	0.0352	0.6	0.5	0.2976	0	Sweep	483000	8002.93	0	0	0	14763.18	147	0	23073.73	9.77
377	179	8	6	4	2	4	3	9	16	16	0.0352	0.8	0.3333	0.3242	0	1A	267396	8006.84	0	0	0	11029.3	114	0	19337.89	10.74
378	179	9	7	4	3	5	3	10	16	16	0.0391	0.8333	0.4286	0.3552	0	Sweep	557800	8007.81	0	0	0	19122.56	194	0	27436.52	10.25
379	179	3	2	2	0	0	4	6	16	16	0.0234	0	0	0.0117	1	Sweep-S1	21	15.63	5	5	0	25470.7	12	0	25791.02	10.25
380	180	4	3	2	1	2	4	7	16	16	0.0273	1	0.3333	0.3803	1	Sweep-S1	4181	224.61	4	4	0	24212.89	20	0	24741.21	9.77
381	181	5	4	2	2	1	4	8	16	16	0.0313	0.3333	0.5	0.2156	0	Sweep	245	16.6	7	7	0	45106.93	107	0	45428.22	10.25
382	181	6	3	2	1	1	4	7	16	16	0.0273	0.5	0.3333	0.2303	0	Sweep	812	14612.3	1	0	1	13193.36	107	1	28106.93	11.72
383	181	7	6	5	1	1	4	10	16	16	0.0391	0.2	0.1667	0.1129	0	1A	299274	8002.44	0	0	0	19013.67	193	0	27319.82	11.72
384	181	8	6	2	4	2	4	10	16	16	0.0391	0.4	0.6667	0.2729	1	Sweep-S1	5	10.74	1	0	1	6622.56	4	0	6937.99	11.72
385	182	9	7	5	2	2	4	11	16	16	0.043	0.3333	0.2857	0.1786	1	Sweep-S1	5	10.74	1	0	1	5375	4	0	5687.5	10.74
386	183	3	2	2	0	0	0	2	16	16	0.0078	0	0	0.0039	1	Sweep-S1	19	12.7	1	1	0	6491.21	18	0	6805.66	10.74
387	184	4	3	1	2	1	0	3	16	16	0.0117	0.5	0.6667	0.2892	0	Sweep	181	17.58	3	3	0	21604.49	92	0	21928.71	10.74
388	184	5	4	3	1	1	0	4	16	16	0.0156	0.3333	0.25	0.1578	1	Sweep-S1	5	9.77	1	0	1	6623.05	4	0	6934.57	10.74
389	185	6	5	5	0	0	0	5	16	16	0.0195	0	0	0.0098	0	Sweep	735	31.25	3	3	0	32414.06	206	0	32749.02	10.74
390	185	7	6	6	0	0	0	6	16	16	0.0234	0	0	0.0117	0	Sweep	820	40.04	3	3	0	34279.3	178	0	34625.98	10.74
391	185	8	4	3	1	1	0	4	16	16	0.0156	0.3333	0.25	0.1578	1	Sweep-S1	8	8.79	1	0	1	9271.48	7	0	9590.82	10.74
392	186	9	6	4	2	4	0	6	16	16	0.0234	0.8	0.3333	0.3184	0	Sweep	262291	8005.86	0	0	0	7964.84	82	0	16270.51	11.72
393	186	3	2	2	0	0	1	3	16	16	0.0117	0	0	0.0059	1	1B	1	9.77	1	0	1	7285.16	0	0	7905.27	10.74
394	187	4	3	3	0	0	1	4	16	16	0.0156	0	0	0.0078	1	Sweep-S1	37	8.79	1	1	0	9569.34	36	0	9882.81	10.74
395	188	5	4	3	1	2	1	5	16	16	0.0195	0.6667	0.25	0.2598	0	Sweep	164	9.77	1	1	0	14986.33	90	0	15297.85	10.74
396	188	6	4	2	2	2	1	5	16	16	0.0195	0.6667	0.5	0.3098	1	Sweep-S1	8	9.77	1	0	1	7887.7	7	0	8205.08	10.74
397	189	7	6	3	3	2	1	7	16	16	0.0273	0.4	0.5	0.2337	1	Sweep-S1	25	8.79	1	0	1	12113.28	24	0	12428.71	11.72
398	190	8	5	3	2	1	1	6	16	16	0.0234	0.25	0.4	0.1667	1	Sweep-S1	6	8.79	1	0	1	6853.52	5	0	7168.95	9.77
399	191	9	8	5	3	5	1	9	16	16	0.0352	0.7143	0.375	0.3069	1	Sweep-S3	313600	8001.95	0	0	0	7085.94	72	0	15394.53	9.77
400	192	3	2	2	0	0	2	4	16	16	0.0156	0	0	0.0078	1	Sweep-S1	19	7.81	1	1	0	6468.75	18	0	6783.2	10.74

Run	Level	Steps	SolutionObjects	Mirrors	Refractors	RuleTransitions	Decoys	TotalObjects	GridWidth	GridHeight	Cl	Ch	Ci	MID	Solved	SolvePhase	SearchNodes	SearchTimeMs	TotalPlacements	InPlaceRotations	Relocations	ExecutionTimeMs	SweepIterations	SweepRelocations	Solve TimeMs	GenerationTimeMs
401	193	4	3	1	2	2	2	5	16	16	0.0195	1	0.6667	0.4431	1	Sweep-S1	32780	1662.11	1	0	1	5717.77	3	0	7687.5	9.77
402	194	5	4	3	1	2	2	6	16	16	0.0234	0.6667	0.25	0.2617	0	Sweep	262294	8008.79	0	0	0	13736.33	141	0	22045.9	10.74
403	194	6	5	1	4	1	2	7	16	16	0.0273	0.25	0.8	0.2487	0	1A	303263	8002.93	0	0	0	14506.84	149	0	22813.48	10.74
404	194	7	6	2	4	3	2	8	16	16	0.0313	0.6	0.6667	0.329	0	1A	303289	8003.91	0	0	0	10837.89	111	0	19141.6	9.77
405	194	8	5	4	1	1	2	7	16	16	0.0273	0.25	0.2	0.1287	0	Sweep	782	12141.6	1	0	1	8297.85	85	1	20743.16	9.77
406	194	9	6	4	2	3	2	8	16	16	0.0313	0.6	0.3333	0.2623	1	Sweep-S1	189600	8002.93	0	0	0	246.09	2	0	8549.81	10.74
407	195	3	2	2	0	0	3	5	16	16	0.0195	0	0	0.0098	0	Sweep	770	6784.18	1	0	1	6198.24	65	1	13283.2	10.74
408	195	4	3	1	2	2	3	6	16	16	0.0234	1	0.6667	0.4451	0	Sweep	782	8834.96	1	0	1	7433.59	77	1	16571.29	9.77
409	195	5	4	2	2	2	3	7	16	16	0.0273	0.6667	0.5	0.3137	0	Sweep	391800	8002.93	0	0	0	9926.76	101	0	18229.49	10.74
410	195	6	5	4	1	1	3	8	16	16	0.0313	0.25	0.2	0.1306	0	Sweep	794	13533.2	1	0	1	10974.61	89	1	24812.5	10.74
411	195	7	6	5	1	2	3	9	16	16	0.0352	0.4	0.1667	0.1709	1	1A	18	7.81	1	1	0	5000	0	0	5617.19	10.74
412	196	8	3	3	0	0	3	6	16	16	0.0234	0	0	0.0117	1	Sweep-S1	5	8.79	1	0	1	6084.96	4	0	6401.37	9.77
413	197	9	8	5	3	4	3	11	16	16	0.043	0.5714	0.375	0.2679	1	1B	9	24.41	1	0	1	6358.4	0	0	6992.19	10.74
414	198	3	2	2	0	0	4	6	16	16	0.0234	0	0	0.0117	0	Sweep	786	11855.47	1	0	1	7936.52	81	1	20095.7	9.77
415	198	4	3	1	2	1	4	7	16	16	0.0273	0.5	0.6667	0.297	0	Sweep	386800	8008.79	0	0	0	9354.49	94	0	17666.02	9.77
416	198	5	3	2	1	1	4	7	16	16	0.0273	0.5	0.3333	0.2303	1	Sweep-S1	5	8.79	1	0	1	7051.76	4	0	7367.19	10.74
417	199	6	3	2	1	1	4	7	16	16	0.0273	0.5	0.3333	0.2303	0	Sweep	353400	8005.86	0	0	0	7112.31	73	0	15422.85	10.74
418	199	7	5	4	1	2	4	9	16	16	0.0352	0.5	0.2	0.2076	1	Sweep-S1	85	19.53	5	5	0	22255.86	12	0	22578.13	10.74
419	200	8	6	4	2	2	4	10	16	16	0.0391	0.4	0.3333	0.2062	0	Sweep	387400	8004.88	0	0	0	9393.56	97	0	17699.22	10.74
420	200	9	8	3	5	2	4	12	16	16	0.0469	0.2857	0.625	0.2342	1	Sweep-S1	14	24.41	1	0	1	9927.73	5	0	10259.77	12.7
421	201	3	2	2	0	0	0	2	16	16	0.0078	0	0	0.0039	1	Sweep-S1	6	7.81	1	1	0	5342.77	4	0	5651.37	8.79
422	202	4	3	1	2	1	0	3	16	16	0.0117	0.5	0.6667	0.2892	1	Sweep-S1	524	35.16	1	0	1	6991.21	3	0	7331.06	9.77
423	203	5	4	1	3	1	0	4	16	16	0.0156	0.3333	0.75	0.2578	1	Sweep-S3	1231	72.27	3	3	0	21216.8	61	0	21589.84	9.77
424	204	6	4	3	1	1	0	4	16	16	0.0156	0.3333	0.25	0.1578	1	1A	75	7.81	1	1	0	5047.85	0	0	5664.06	10.74
425	205	7	5	4	1	2	0	5	16	16	0.0195	0.5	0.2	0.1998	1	Sweep-S1	93	16.6	3	3	0	15756.84	20	0	16079.1	9.77
426	206	8	4	3	1	2	0	4	16	16	0.0156	0.6667	0.25	0.2578	0	Sweep	809	10463.87	1	0	1	10072.27	104	1	20838.87	8.79
427	206	9	8	5	3	3	0	8	16	16	0.0313	0.4286	0.375	0.2192	1	Sweep-S1	153	17.58	4	4	0	28113.28	14	0	28435.55	9.77
428	207	3	2	2	0	0	1	3	16	16	0.0117	0	0	0.0059	1	Sweep-S1	13	8.79	1	1	0	5599.61	4	0	5909.18	9.77
429	208	4	3	2	1	1	1	4	16	16	0.0156	0.5	0.3333	0.2245	1	1B	1	7.81	1	0	1	7277.34	0	0	7895.51	9.77
430	209	5	4	2	2	3	1	5	16	16	0.0195	1	0.5	0.4098	0	Sweep	770	8889.65	1	0	1	6156.25	65	1	15346.68	9.77
431	209	6	4	1	3	2	1	5	16	16	0.0195	0.6667	0.75	0.3598	0	Sweep	746	5291.02	1	0	1	6170.9	65	1	11765.63	9.77
432	209	7	6	5	1	1	1	7	16	16	0.0273	0.2	0.1667	0.107	0	Sweep	478600	8008.79	0	0	0	14581.05	148	0	22892.58	10.74
433	209	8	6	5	1	2	1	7	16	16	0.0273	0.4	0.1667	0.167	0	Sweep	38095	1921.88	3	3	0	25697.27	141	0	27923.83	9.77
434	209	9	7	3	4	3	1	8	16	16	0.0313	0.5	0.5714	0.2799	1	Sweep-S2	731	18464.84	1	0	1	5880.86	26	1	24651.37	9.77
435	210	3	2	2	0	0	2	4	16	16	0.0156	0	0	0.0078	1	Sweep-S1	6	7.81	1	1	0	5328.13	4	0	5641.6	10.74
436	211	4	3	2	1	2	2	5	16	16	0.0195	1	0.3333	0.3764	0	Sweep	33572	9176.76	0	0	0	9815.43	91	0	19294.92	9.77
437	211	5	4	2	2	2	2	6	16	16	0.0234	0.6667	0.5	0.3117	0	Sweep	262316	8002.93	0	0	0	10299.8	107	0	18602.54	9.77
438	211	6	4	2	2	2	2	6	16	16	0.0234	0.6667	0.5	0.3117	0	Sweep	778	14815.43	1	0	1	9458.98	73	1	24576.17	9.77
439	211	7	4	2	2	2	2	6	16	16	0.0234	0.6667	0.5	0.3117	0	Sweep	262219	8001.95	0	0	0	6377.93	66	0	14680.66	9.77
440	211	8	6	1	5	2	2	8	16	16	0.0313	0.4	0.8333	0.3023	0	Sweep	74397	3823.24	6	6	0	45087.89	130	0	49215.82	10.74
441	211	9	7	2	5	2	2	9	16	16	0.0352	0.3333	0.7143	0.2604	0	Sweep	372000	8000	0	0	0	10526.37	107	0	18833.98	10.74
442	211	3	2	2	0	0	3	5	16	16	0.0195	0	0	0.0098	1	Sweep-S1	19	10.74	1	1	0	5676.76	18	0	5993.16	9.77
443	212	4	3	2	1	2	3	6	16	16	0.0234	1	0.3333	0.3784	0	Sweep	262302	8004.88	0	0	0	9041.02	93	0	17349.61	9.77
444	212	5	4	1	3	1	3	7	16	16	0.0273	0.3333	0.75	0.2637	0	Sweep	710	26.37	7	7	0	48400.39	123	0	48729.49	10.74
445	212	6	4	3	1	1	3	7	16	16	0.0273	0.3333	0.25	0.1637	0	Sweep	814	16628.91	1	0	1	13444.34	109	1	30374.02	13.67
446	212	7	6	4	2	3	3	9	16	16	0.0352	0.6	0.3333	0.2642	0	Sweep	432800	8007.81	0	0	0	10132.81	105	0	18444.34	10.74
447	212	8	6	5	1	2	3	9	16	16	0.0352	0.4	0.1667	0.1709	0	Sweep	806	21387.7	1	0	1	13716.8	101	1	35406.25	10.74
448	212	9	8	4	4	4	3	11	16	16	0.043	0.5714	0.5	0.2929	0	Sweep	819	8830.08	0	0	0	31037.11	106	0	40168.95	9.77
449	212	3	2	2	0	0	4	6	16	16	0.0234	0	0	0.0117	1	Trivial	0	0	0	0	0	0	0	0	308.59	9.77
450	213	4	3	1	2	1	4	7	16	16	0.0273	0.5	0.6667	0.297	1	Sweep-S1	252800	8006.84	0	0	0	2795.9	28	0	11107.42	10.74

Run	Level	Steps	SolutionObjects	Mirrors	Refractors	RuleTransitions	Decoys	TotalObjects	GridWidth	GridHeight	Cl	Ch	Ci	MID	Solved	SolvePhase	SearchNodes	SearchTimeMs	TotalPlacements	InPlaceRotations	Relocations	ExecutionTimeMs	SweepIterations	SweepRelocations	Solve TimeMs	GenerationTimeMs
451	214	5	4	3	1	2	4	8	16	16	0.0313	0.6667	0.25	0.2656	0	Sweep	180	8.79	1	1	0	21532.23	170	0	21848.63	10.74
452	214	6	4	3	1	2	4	8	16	16	0.0313	0.6667	0.25	0.2656	0	Sweep	798	11004.88	1	0	1	8976.56	93	1	20290.04	10.74
453	214	7	4	1	3	2	4	8	16	16	0.0313	0.6667	0.75	0.3656	1	Sweep-S1	6	8.79	1	0	1	5785.16	5	0	6101.56	10.74
454	215	8	7	3	4	3	4	11	16	16	0.043	0.5	0.5714	0.2858	0	Sweep	814	18036.13	1	0	1	13628.91	109	1	31971.68	10.74
455	215	9	7	5	2	4	4	11	16	16	0.043	0.6667	0.2857	0.2786	1	1B	9	37.11	1	0	1	6373.05	0	0	7015.63	10.74
456	216	3	2	2	0	0	0	2	16	16	0.0078	0	0	0.0039	1	Sweep-S1	6	6.84	1	1	0	5510.74	4	0	5822.27	9.77
457	217	4	3	2	1	1	0	3	16	16	0.0117	0.5	0.3333	0.2225	1	Sweep-S1	8	8.79	1	0	1	10215.82	7	0	10530.27	9.77
458	218	5	3	2	1	2	0	3	16	16	0.0117	1	0.3333	0.3725	0	Sweep	754	4133.79	1	0	1	4640.63	49	1	9077.15	9.77
459	218	6	4	1	3	2	0	4	16	16	0.0156	0.6667	0.75	0.3578	1	Sweep-S1	6	9.77	1	0	1	7336.91	5	0	7647.46	14.65
460	219	7	5	4	1	2	0	5	16	16	0.0195	0.5	0.2	0.1998	0	Sweep	4382	223.63	3	3	0	27171.88	156	0	27701.17	10.74
461	219	8	6	5	1	2	0	6	16	16	0.0234	0.4	0.1667	0.1651	0	Sweep	689	37.11	3	3	0	23231.45	104	0	23577.15	9.77
462	219	9	8	5	3	4	0	8	16	16	0.0313	0.5714	0.375	0.2621	0	Sweep	466600	8006.84	0	0	0	14180.66	144	0	22493.16	9.77
463	219	3	2	2	0	0	1	3	16	16	0.0117	0	0	0.0059	1	Sweep-S1	19	7.81	1	1	0	6189.45	18	0	6504.88	9.77
464	220	4	3	2	1	1	1	4	16	16	0.0156	0.5	0.3333	0.2245	0	Sweep	265	17.58	2	2	0	16205.08	64	0	16525.39	12.7
465	220	5	3	3	0	0	1	4	16	16	0.0156	0	0	0.0078	1	1B	1	8.79	1	0	1	7288.09	0	0	7901.37	9.77
466	221	6	4	2	2	1	1	5	16	16	0.0195	0.3333	0.5	0.2098	1	Sweep-S1	4142	308.59	4	4	0	19872.07	21	0	20481.45	9.77
467	222	7	4	2	2	2	1	5	16	16	0.0195	0.6667	0.5	0.3098	0	Sweep	120	15.63	5	5	0	34687.5	93	0	35010.74	10.74
468	222	8	6	2	4	2	1	7	16	16	0.0273	0.4	0.6667	0.267	0	Sweep	69933	3431.64	4	4	0	31714.84	148	0	35451.17	10.74
469	222	9	8	6	2	3	1	9	16	16	0.0352	0.4286	0.25	0.1961	0	Sweep	363000	8000.98	0	0	0	8568.36	89	0	16871.09	9.77
470	222	3	2	2	0	0	2	4	16	16	0.0156	0	0	0.0078	1	Sweep-S1	4109	194.34	1	0	1	5556.64	4	0	6051.76	10.74
471	223	4	3	2	1	1	2	5	16	16	0.0195	0.5	0.3333	0.2264	0	Sweep	774	13711.91	1	0	1	6661.13	69	1	20679.69	9.77
472	223	5	4	1	3	2	2	6	16	16	0.0234	0.6667	0.75	0.3617	1	1B	1	7.81	1	0	1	9951.17	0	0	10564.45	10.74
473	224	6	5	2	3	3	2	7	16	16	0.0273	0.75	0.6	0.3587	0	Sweep	336800	8002.93	0	0	0	7112.31	73	0	15417.97	10.74
474	224	7	5	2	3	3	2	7	16	16	0.0273	0.75	0.6	0.3587	1	Sweep-S1	36888	1914.06	6	6	0	25817.38	14	0	28034.18	11.72
475	225	8	7	5	2	4	2	9	16	16	0.0352	0.6667	0.2857	0.2747	0	Sweep	37626	1973.63	5	5	0	43358.4	161	0	45638.67	9.77
476	225	9	7	3	4	3	2	9	16	16	0.0352	0.5	0.5714	0.2819	0	Sweep	463400	8006.84	0	0	0	13938.48	142	0	22253.91	13.67
477	225	3	2	2	0	0	3	5	16	16	0.0195	0	0	0.0098	1	Sweep-S1	6	7.81	1	1	0	6021.48	4	0	6331.06	11.72
478	226	4	3	1	2	2	3	6	16	16	0.0234	1	0.6667	0.4451	0	Sweep	36980	1833.98	5	5	0	32294.92	106	0	34435.55	10.74
479	226	5	4	0	4	0	3	7	16	16	0.0273	0	1	0.2137	1	Sweep-S1	6	7.81	1	0	1	8080.08	5	0	8394.53	10.74
480	227	6	5	3	2	2	3	8	16	16	0.0313	0.5	0.4	0.2456	0	Sweep	794	19683.59	1	0	1	12039.06	89	1	32025.39	9.77
481	227	7	5	3	2	1	3	8	16	16	0.0313	0.25	0.4	0.1706	1	Sweep-S1	5	7.81	1	0	1	8008.79	4	0	8322.27	10.74
482	228	8	7	3	4	2	3	10	16	16	0.0391	0.3333	0.5714	0.2338	0	Sweep	425400	8003.91	0	0	0	11569.34	116	0	19876.95	9.77
483	228	9	5	2	3	3	3	8	16	16	0.0313	0.75	0.6	0.3606	1	Sweep-S1	6	9.77	1	0	1	6195.31	5	0	6508.79	11.72
484	229	3	2	2	0	0	4	6	16	16	0.0234	0	0	0.0117	1	Sweep-S1	5	8.79	1	0	1	9525.39	4	0	9838.87	9.77
485	230	4	3	2	1	2	4	7	16	16	0.0273	1	0.3333	0.3803	1	Sweep-S1	5	7.81	1	0	1	8787.11	4	0	9098.63	10.74
486	231	5	3	1	2	1	4	7	16	16	0.0273	0.5	0.6667	0.297	1	1B	1	7.81	1	0	1	9426.76	0	0	10038.09	9.77
487	232	6	4	3	1	1	4	8	16	16	0.0313	0.3333	0.25	0.1656	1	1A	138	9.77	1	1	0	5794.92	0	0	6413.09	12.7
488	233	7	6	4	2	3	4	10	16	16	0.0391	0.6	0.3333	0.2662	0	Sweep	427200	8005.86	0	0	0	15004.88	151	0	23315.43	12.7
489	233	8	5	2	3	1	4	9	16	16	0.0352	0.25	0.6	0.2126	1	Sweep-S1	33	16.6	2	2	0	8558.59	14	0	8882.81	11.72
490	234	9	4	2	2	2	4	8	16	16	0.0313	0.6667	0.5	0.3156	1	Sweep-S1	30	7.81	1	1	0	8039.06	28	0	8351.56	9.77
491	235	3	2	2	0	0	0	2	16	16	0.0078	0	0	0.0039	1	Sweep-S1	6	7.81	1	1	0	5535.16	4	0	5845.7	9.77
492	236	4	3	2	1	1	0	3	16	16	0.0117	0.5	0.3333	0.2225	1	Sweep-S1	100	16.6	3	3	0	13786.13	27	0	14106.45	10.74
493	237	5	4	1	3	2	0	4	16	16	0.0156	0.6667	0.75	0.3578	0	Sweep	883	11136.72	1	0	1	17623.05	178	1	29062.5	9.77
494	237	6	5	4	1	1	0	5	16	16	0.0195	0.25	0.2	0.1248	0	Sweep	630	29.3	3	3	0	22577.15	99	0	22913.09	10.74
495	237	7	5	3	2	3	0	5	16	16	0.0195	0.75	0.4	0.3148	0	Sweep	92	9.77	1	1	0	12272.46	90	0	12586.91	10.74
496	237	8	6	4	2	3	0	6	16	16	0.0234	0.6	0.3333	0.2584	0	Sweep	9333	668.95	6	6	0	42244.14	108	0	43215.82	10.74
497	237	9	7	3	4	4	0	7	16	16	0.0273	0.6667	0.5714	0.328	0	1A	271004	8015.63	0	0	0	7179.69	73	0	15502.93	10.74
498	237	3	2	2	0	0	1	3	16	16	0.0117	0	0	0.0059	0	Sweep	758	5508.79	1	0	1	5303.71	53	1	11120.12	10.74
499	237	4	3	2	1	2	1	4	16	16	0.0156	1	0.3333	0.3745	0	Sweep	107	8.79	1	1	0	15084.96	96	0	15399.41	10.74
500	237	5	3	2	1	1	1	4	16	16	0.0156	0.5	0.3333	0.2245	0	Sweep	4859	9815.43	0	0	0	4962.89	50	0	15081.05	10.74

Run	Level	Steps	SolutionObjects	Mirrors	Refractors	RuleTransitions	Decoys	TotalObjects	GridWidth	GridHeight	Cl	Ch	Ci	MID	Solved	SolvePhase	SearchNodes	SearchTimeMs	TotalPlacements	InPlaceRotati	Relocations	ExecutionTimeMs	SweepIterations	SweepRelocations	Solve TimeMs	GenerationTimeMs
501	237	6	3	3	0	0	1	4	16	16	0.0156	0	0	0.0078	1	Sweep-S1	5	8.79	1	0	1	8273.44	4	0	8591.8	9.77
502	238	7	5	1	4	2	1	6	16	16	0.0234	0.5	0.8	0.3217	1	1A	3	9.77	1	1	0	4630.86	0	0	5250	10.74
503	239	8	7	4	3	3	1	8	16	16	0.0313	0.5	0.4286	0.2513	0	Sweep	42259	2485.35	7	7	0	47529.3	138	0	50323.24	10.74
504	239	9	8	4	4	1	1	9	16	16	0.0352	0.1429	0.5	0.1604	0	1A	296192	8007.81	0	0	0	19149.41	191	0	27463.87	11.72
505	239	3	2	2	0	0	2	4	16	16	0.0156	0	0	0.0078	1	Sweep-S1	13	8.79	1	1	0	6298.83	4	0	6617.19	9.77
506	240	4	3	3	0	0	2	5	16	16	0.0195	0	0	0.0098	1	Sweep-S1	29	18.55	4	4	0	19782.23	20	0	20105.47	11.72
507	241	5	4	4	0	0	2	6	16	16	0.0234	0	0	0.0117	1	Sweep-S1	93	22.46	5	5	0	21317.38	20	0	21640.63	11.72
508	242	6	5	3	2	3	2	7	16	16	0.0273	0.75	0.4	0.3187	1	Sweep-S1	5	6.84	1	1	0	4870.12	4	0	5183.59	10.74
509	243	7	6	3	3	4	2	8	16	16	0.0313	0.8	0.5	0.3556	1	Sweep-S1	18	8.79	1	0	1	7293.95	17	0	7603.52	9.77
510	244	8	7	6	1	2	2	9	16	16	0.0352	0.3333	0.1429	0.1461	0	Sweep	274	16.6	6	6	0	43894.53	136	0	44218.75	11.72
511	244	9	7	4	3	2	2	9	16	16	0.0352	0.3333	0.4286	0.2033	0	Sweep	502800	8004.88	0	0	0	15177.73	154	0	23484.38	10.74
512	244	3	2	2	0	0	3	5	16	16	0.0195	0	0	0.0098	1	Sweep-S1	32840	1596.68	1	0	1	7820.31	7	0	9721.68	9.77
513	245	4	3	2	1	1	3	6	16	16	0.0234	0.5	0.3333	0.2284	1	Sweep-S1	87	15.63	3	3	0	19691.41	12	0	20013.67	10.74
514	246	5	4	3	1	1	3	7	16	16	0.0273	0.3333	0.25	0.1637	0	Sweep	790	9678.71	1	0	1	8267.58	85	1	18250.98	10.74
515	246	6	5	4	1	2	3	8	16	16	0.0313	0.5	0.2	0.2056	1	Sweep-S1	5	10.74	2	0	2	16712.89	3	0	17027.34	9.77
516	247	7	6	3	3	3	3	9	16	16	0.0352	0.6	0.5	0.2976	0	1A	267379	8008.79	0	0	0	10237.3	106	0	18549.8	10.74
517	247	8	5	5	0	0	3	8	16	16	0.0313	0	0	0.0156	1	Sweep-S1	228000	8005.86	0	0	0	1197.27	11	0	9503.91	10.74
518	248	9	8	7	1	2	3	11	16	16	0.043	0.2857	0.125	0.1322	0	Sweep	851	9163.09	0	0	0	38029.3	138	0	47493.16	10.74
519	248	3	2	2	0	0	4	6	16	16	0.0234	0	0	0.0117	1	Sweep-S1	13	7.81	1	1	0	5831.06	4	0	6145.51	10.74
520	249	4	3	1	2	2	4	7	16	16	0.0273	1	0.6667	0.447	1	Sweep-S1	8244	407.23	6	6	0	34951.17	42	0	35665.04	10.74
521	250	5	3	2	1	1	4	7	16	16	0.0273	0.5	0.3333	0.2303	1	Trivial	0	0	0	0	0	0	0	0	307.62	11.72
522	251	6	4	4	0	0	4	8	16	16	0.0313	0	0	0.0156	1	Sweep-S1	5	8.79	1	0	1	7022.46	4	0	7333.98	10.74
523	252	7	6	3	3	2	4	10	16	16	0.0391	0.4	0.5	0.2395	0	Sweep	500200	8010.74	0	0	0	16966.8	173	0	25281.25	14.65
524	252	8	7	5	2	4	4	11	16	16	0.043	0.6667	0.2857	0.2786	1	Sweep-S1	82	7.81	1	1	0	14343.75	81	0	14654.3	12.7
525	253	9	6	3	3	5	4	10	16	16	0.0391	1	0.5	0.4195	1	Sweep-S1	6	10.74	1	1	0	5500	4	0	5816.41	12.7
526	254	3	2	2	0	0	0	2	16	16	0.0078	0	0	0.0039	1	Sweep-S1	21	14.65	2	2	0	10330.08	12	0	10648.44	9.77
527	255	4	3	3	0	0	0	3	16	16	0.0117	0	0	0.0059	0	Sweep	758	8592.77	1	0	1	5158.2	53	1	14054.69	9.77
528	255	5	4	2	2	1	0	4	16	16	0.0156	0.3333	0.5	0.2078	0	Sweep	211	16.6	4	4	0	27814.45	72	0	28133.79	8.79
529	255	6	5	1	4	2	0	5	16	16	0.0195	0.5	0.8	0.3198	0	Sweep	806	36.13	5	5	0	36156.25	148	0	36493.16	9.77
530	255	7	5	4	1	2	0	5	16	16	0.0195	0.5	0.2	0.1998	1	1B	32833	1625	1	0	1	8786.13	0	0	11020.51	9.77
531	256	8	6	4	2	2	0	6	16	16	0.0234	0.4	0.3333	0.1984	0	Sweep	4764	252.93	5	5	0	35882.81	146	0	36437.5	9.77
532	256	9	8	5	3	4	0	8	16	16	0.0313	0.5714	0.375	0.2621	0	Sweep	442000	8007.81	0	0	0	12288.09	125	0	20603.52	9.77
533	256	3	2	2	0	0	1	3	16	16	0.0117	0	0	0.0059	1	Sweep-S1	5	7.81	1	0	1	7331.06	4	0	7642.58	10.74
534	257	4	3	3	0	0	1	4	16	16	0.0156	0	0	0.0078	1	Sweep-S1	37	8.79	1	1	0	9441.41	36	0	9752.93	9.77
535	258	5	3	1	2	1	1	4	16	16	0.0156	0.5	0.6667	0.2911	0	Sweep	766	7782.23	1	0	1	5925.78	61	1	14011.72	10.74
536	258	6	3	3	0	0	1	4	16	16	0.0156	0	0	0.0078	1	Sweep-S1	5	7.81	1	1	0	5875	4	0	6188.48	9.77
537	259	7	4	3	1	2	1	5	16	16	0.0195	0.6667	0.25	0.2598	1	Sweep-S2	708	12794.92	1	0	1	356.45	3	1	13456.05	10.74
538	260	8	7	4	3	2	1	8	16	16	0.0313	0.3333	0.4286	0.2013	0	Sweep	393800	8006.84	0	0	0	9428.71	97	0	17740.23	9.77
539	260	9	7	4	3	4	1	8	16	16	0.0313	0.6667	0.4286	0.3013	0	1A	296203	8016.6	0	0	0	18999.02	193	0	27318.36	9.77
540	260	3	2	2	0	0	2	4	16	16	0.0156	0	0	0.0078	1	Trivial	0	0	0	0	0	0	0	0	304.69	9.77
541	261	4	3	1	2	1	2	5	16	16	0.0195	0.5	0.6667	0.2931	0	Sweep	770	12256.84	1	0	1	8212.89	65	1	20770.51	10.74
542	261	5	4	2	2	2	2	6	16	16	0.0234	0.6667	0.5	0.3117	1	1B	1	7.81	1	0	1	7290.04	0	0	7905.27	10.74
543	262	6	5	4	1	1	2	7	16	16	0.0273	0.25	0.2	0.1287	1	1B	1	7.81	1	0	1	5518.56	0	0	6128.91	9.77
544	263	7	6	4	2	2	2	8	16	16	0.0313	0.4	0.3333	0.2023	1	Sweep-S1	5	8.79	1	0	1	5086.91	4	0	5395.51	9.77
545	264	8	7	6	1	2	2	9	16	16	0.0352	0.3333	0.1429	0.1461	0	Sweep	463000	8002.93	0	0	0	13542.97	137	0	21851.56	9.77
546	264	9	7	4	3	4	2	9	16	16	0.0352	0.6667	0.4286	0.3033	0	Sweep	412400	8014.65	0	0	0	11960.94	121	0	20275.39	10.74
547	264	3	2	2	0	0	3	5	16	16	0.0195	0	0	0.0098	1	Sweep-S1	13	10.74	1	1	0	6269.53	4	0	6581.06	9.77
548	265	4	3	1	2	1	3	6	16	16	0.0234	0.5	0.6667	0.2951	1	Sweep-S1	1124	52.73	3	3	0	17398.44	25	0	17755.86	10.74
549	266	5	4	4	0	0	3	7	16	16	0.0273	0	0	0.0137	0	1A	295035	8000.98	0	0	0	11063.48	114	0	19366.21	10.74
550	266	6	4	2	2	1	3	7	16	16	0.0273	0.3333	0.5	0.2137	1	1B	1	6.84	1	0	1	7257.81	0	0	7878.91	9.77

Run	Level	Steps	SolutionObjects	Mirrors	Refractors	RuleTransitions	Decoys	TotalObjects	GridWidth	GridHeight	Cl	Ch	Ci	MID	Solved	SolvePhase	SearchNodes	SearchTimeMs	TotalPlacements	InPlaceRotations	Relocations	ExecutionTimeMs	SweepIterations	SweepRelocations	Solve TimeMs	GenerationTimeMs
551	267	7	5	1	4	2	3	8	16	16	0.0313	0.5	0.8	0.3256	0	Sweep	348400	8008.79	0	0	0	7893.56	82	0	16203.13	9.77
552	267	8	6	4	2	1	3	9	16	16	0.0352	0.2	0.3333	0.1442	0	Sweep	380600	8018.56	0	0	0	8715.82	90	0	17039.06	10.74
553	267	9	6	2	4	3	3	9	16	16	0.0352	0.6	0.6667	0.3309	1	Sweep-S1	663	39.06	4	4	0	18977.54	5	0	19323.24	10.74
554	268	3	2	2	0	0	4	6	16	16	0.0234	0	0	0.0117	1	Sweep-S1	13	7.81	1	1	0	5346.68	4	0	5660.16	10.74
555	269	4	3	3	0	0	4	7	16	16	0.0273	0	0	0.0137	0	Sweep	790	13149.41	1	0	1	8202.15	85	1	21654.3	9.77
556	269	5	3	2	1	1	4	7	16	16	0.0273	0.5	0.3333	0.2303	0	Sweep	819	16446.29	1	0	1	13733.4	114	1	30480.47	10.74
557	269	6	5	2	3	2	4	9	16	16	0.0352	0.5	0.6	0.2876	1	Sweep-S1	210400	8008.79	0	0	0	249.02	2	0	8563.48	9.77
558	270	7	5	2	3	3	4	9	16	16	0.0352	0.75	0.6	0.3626	1	Sweep-S1	4159	204.1	6	6	0	30875.98	52	0	31382.81	9.77
559	271	8	6	2	4	3	4	10	16	16	0.0391	0.6	0.6667	0.3329	0	Sweep	383000	8009.77	0	0	0	9428.71	97	0	17738.28	10.74
560	271	9	6	3	3	2	4	10	16	16	0.0391	0.4	0.5	0.2395	1	Sweep-S1	5	8.79	1	0	1	6525.39	4	0	6840.82	9.77
561	272	3	2	2	0	0	0	2	16	16	0.0078	0	0	0.0039	1	Sweep-S1	6	7.81	1	1	0	5286.13	4	0	5602.54	9.77
562	273	4	3	1	2	2	0	3	16	16	0.0117	1	0.6667	0.4392	0	Sweep	754	7411.13	1	0	1	4655.27	49	1	12368.16	9.77
563	273	5	4	2	2	2	0	4	16	16	0.0156	0.6667	0.5	0.3078	0	Sweep	790	35.16	2	2	0	23504.88	140	0	23843.75	9.77
564	273	6	5	3	2	3	0	5	16	16	0.0195	0.75	0.4	0.3148	1	Sweep-S2	724	12480.47	1	0	1	1971.68	19	1	14755.86	10.74
565	274	7	5	4	1	2	0	5	16	16	0.0195	0.5	0.2	0.1998	0	Sweep	750	1275.39	1	0	1	6649.41	69	1	8230.47	9.77
566	274	8	7	2	5	3	0	7	16	16	0.0273	0.5	0.7143	0.3065	1	Sweep-S1	6	8.79	1	0	1	6633.79	5	0	6945.31	10.74
567	275	9	3	2	1	2	0	3	16	16	0.0117	1	0.3333	0.3725	0	Sweep	754	5803.71	1	0	1	4726.56	49	1	10835.94	9.77
568	275	3	2	2	0	0	1	3	16	16	0.0117	0	0	0.0059	0	Sweep	758	5869.14	1	0	1	5248.05	53	1	11420.9	9.77
569	275	4	3	2	1	1	1	4	16	16	0.0156	0.5	0.3333	0.2245	1	1B	1	7.81	1	0	1	6814.45	0	0	7429.69	10.74
570	276	5	4	2	2	1	1	5	16	16	0.0195	0.3333	0.5	0.2098	0	Sweep	795	13262.7	1	0	1	8532.23	90	1	22103.52	9.77
571	276	6	4	3	1	2	1	5	16	16	0.0195	0.6667	0.25	0.2598	1	1B	1	8.79	1	0	1	7811.52	0	0	8428.71	9.77
572	277	7	6	5	1	2	1	7	16	16	0.0273	0.4	0.1667	0.167	0	Sweep	4787	240.23	6	6	0	48318.36	170	0	48859.38	9.77
573	277	8	5	2	3	1	1	6	16	16	0.0234	0.25	0.6	0.2067	0	Sweep	778	14955.08	1	0	1	9533.2	73	1	24791.99	9.77
574	277	9	8	5	3	5	1	9	16	16	0.0352	0.7143	0.375	0.3069	0	Sweep	366000	8006.84	0	0	0	9579.1	97	0	17892.58	9.77
575	277	3	2	2	0	0	2	4	16	16	0.0156	0	0	0.0078	1	Sweep-S1	19	7.81	1	1	0	6448.24	18	0	6760.74	9.77
576	278	4	3	2	1	2	2	5	16	16	0.0195	1	0.3333	0.3764	1	1B	1	7.81	1	0	1	8918.95	0	0	9530.27	9.77
577	279	5	3	2	1	1	2	5	16	16	0.0195	0.5	0.3333	0.2264	1	Sweep-S1	5	8.79	1	0	1	5349.61	4	0	5661.13	10.74
578	280	6	5	3	2	2	2	7	16	16	0.0273	0.5	0.4	0.2437	0	Sweep	65722	3424.81	4	4	0	27561.52	104	0	31291.02	9.77
579	280	7	6	3	3	2	2	8	16	16	0.0313	0.4	0.5	0.2356	0	Sweep	331200	8005.86	0	0	0	8000.98	81	0	16313.48	12.7
580	280	8	7	6	1	1	2	9	16	16	0.0352	0.1667	0.1429	0.0961	1	Sweep-S1	343800	8016.6	0	0	0	9457.03	92	0	17777.34	13.67
581	281	9	5	3	2	2	2	7	16	16	0.0273	0.5	0.4	0.2437	0	Sweep	357800	8007.81	0	0	0	7074.22	73	0	15386.72	10.74
582	281	3	2	2	0	0	3	5	16	16	0.0195	0	0	0.0098	1	Sweep-S1	6	8.79	2	2	0	10062.5	4	0	10372.07	10.74
583	282	4	3	2	1	1	3	6	16	16	0.0234	0.5	0.3333	0.2284	1	Sweep-S1	6	7.81	1	0	1	7099.61	5	0	7414.06	10.74
584	283	5	3	3	0	0	3	6	16	16	0.0234	0	0	0.0117	1	1B	1	7.81	1	0	1	7314.45	0	0	7933.59	10.74
585	284	6	4	2	2	3	3	7	16	16	0.0273	1	0.5	0.4137	0	Sweep	65644	3412.11	5	5	0	35848.63	97	0	39566.41	10.74
586	284	7	6	4	2	3	3	9	16	16	0.0352	0.6	0.3333	0.2642	0	Sweep	713200	8000.98	0	0	0	27799.8	279	0	36102.54	10.74
587	284	8	4	3	1	2	3	7	16	16	0.0273	0.6667	0.25	0.2637	1	Sweep-S1	37	7.81	1	1	0	9759.77	36	0	10073.24	10.74
588	285	9	6	4	2	4	3	9	16	16	0.0352	0.8	0.3333	0.3242	0	Sweep	557200	8006.84	0	0	0	18678.71	189	0	26991.21	10.74
589	285	3	2	2	0	0	4	6	16	16	0.0234	0	0	0.0117	0	Sweep	782	14007.81	1	0	1	10099.61	77	1	24410.16	11.72
590	285	4	3	3	0	0	4	7	16	16	0.0273	0	0	0.0137	0	Sweep	338600	8006.84	0	0	0	7186.52	74	0	15499.02	9.77
591	285	5	4	2	2	3	4	8	16	16	0.0313	1	0.5	0.4156	1	Sweep-S1	5	8.79	1	0	1	7106.45	4	0	7417.97	10.74
592	286	6	5	4	1	1	4	9	16	16	0.0352	0.25	0.2	0.1326	0	Sweep	505400	8009.77	0	0	0	14909.18	150	0	23220.7	10.74
593	286	7	5	4	1	2	4	9	16	16	0.0352	0.5	0.2	0.2076	1	Sweep-S1	6	6.84	3	3	0	14027.34	4	0	14340.82	10.74
594	287	8	5	1	4	2	4	9	16	16	0.0352	0.5	0.8	0.3276	0	Sweep	368000	8008.79	0	0	0	8652.34	90	0	16967.77	9.77
595	287	9	4	2	2	3	4	8	16	16	0.0313	1	0.5	0.4156	0	Sweep	794	18191.41	1	0	1	11751.95	89	1	30245.12	10.74
596	287	3	2	2	0	0	0	2	16	16	0.0078	0	0	0.0039	1	Sweep-S1	6	7.81	1	1	0	6011.72	4	0	6321.29	9.77
597	288	4	3	2	1	1	0	3	16	16	0.0117	0.5	0.3333	0.2225	0	Sweep	82	18.55	3	3	0	23071.29	72	0	23393.55	9.77
598	288	5	4	3	1	1	0	4	16	16	0.0156	0.3333	0.25	0.1578	1	1B	1	7.81	1	0	1	8009.77	0	0	8620.12	9.77
599	289	6	5	3	2	1	0	5	16	16	0.0195	0.25	0.4	0.1648	0	Sweep	227	28.32	5	5	0	28426.76	88	0	28761.72	9.77
600	289	7	4	2	2	1	0	4	16	16	0.0156	0.3333	0.5	0.2078	1	Sweep-S1	69	8.79	1	1	0	4669.92	4	0	4981.45	9.77

Run	Level	Steps	SolutionObjects	Mirrors	Refractors	RuleTransitions	Decoys	TotalObjects	GridWidth	GridHeight	Cl	Ch	Ci	MID	Solved	SolvePhase	SearchNodes	SearchTimeMs	TotalPlacements	InPlaceRotations	Relocations	ExecutionTimeMs	SweepIterations	SweepRelocations	Solve TimeMs	GenerationTimeMs
601	290	8	7	3	4	4	0	7	16	16	0.0273	0.6667	0.5714	0.328	1	Sweep-S1	8228	408.2	5	5	0	25139.65	19	0	25853.52	9.77
602	291	9	5	3	2	3	0	5	16	16	0.0195	0.75	0.4	0.3148	0	Sweep	748	34.18	2	2	0	20809.57	106	0	21146.48	9.77
603	291	3	2	2	0	0	1	3	16	16	0.0117	0	0	0.0059	1	1B	1	8.79	1	0	1	8294.92	0	0	8913.09	11.72
604	292	4	3	2	1	2	1	4	16	16	0.0156	1	0.3333	0.3745	1	Sweep-S1	604	54.69	3	3	0	15826.17	19	0	16190.43	11.72
605	293	5	4	4	0	0	1	5	16	16	0.0195	0	0	0.0098	1	Sweep-S1	58	7.81	1	1	0	12640.63	56	0	12953.13	9.77
606	294	6	5	2	3	3	1	6	16	16	0.0234	0.75	0.6	0.3567	1	Sweep-S1	70733	3672.85	3	3	0	12869.14	2	0	16843.75	10.74
607	295	7	6	4	2	1	1	7	16	16	0.0273	0.2	0.3333	0.1403	0	1A	262758	8009.77	0	0	0	8974.61	92	0	17284.18	9.77
608	295	8	7	4	3	4	1	8	16	16	0.0313	0.6667	0.4286	0.3013	0	Sweep	394000	8006.84	0	0	0	8586.91	89	0	16899.41	10.74
609	295	9	6	4	2	3	1	7	16	16	0.0273	0.6	0.3333	0.2603	0	Sweep	786	17208.98	1	0	1	10576.17	81	1	28091.8	10.74
610	295	3	2	2	0	0	2	4	16	16	0.0156	0	0	0.0078	1	1B	1	8.79	1	0	1	9318.36	0	0	9928.71	9.77
611	296	4	3	3	0	0	2	5	16	16	0.0195	0	0	0.0098	1	1B	1	7.81	1	0	1	7292.97	0	0	7910.16	9.77
612	297	5	4	3	1	2	2	6	16	16	0.0234	0.6667	0.25	0.2617	0	Sweep	8411	404.3	3	3	0	28420.9	138	0	29127.93	9.77
613	297	6	5	3	2	2	2	7	16	16	0.0273	0.5	0.4	0.2437	0	Sweep	4324	207.03	4	4	0	31982.42	154	0	32495.12	9.77
614	297	7	4	3	1	2	2	6	16	16	0.0234	0.6667	0.25	0.2617	1	Sweep-S1	5	8.79	1	0	1	5853.52	4	0	6167.97	9.77
615	298	8	6	5	1	1	2	8	16	16	0.0313	0.2	0.1667	0.109	1	1B	1	8.79	1	0	1	7318.36	0	0	7935.55	10.74
616	299	9	7	3	4	3	2	9	16	16	0.0352	0.5	0.5714	0.2819	1	1B	1	9.77	1	0	1	6575.2	0	0	7193.36	9.77
617	300	3	2	2	0	0	3	5	16	16	0.0195	0	0	0.0098	1	1B	1	8.79	1	0	1	7020.51	0	0	7639.65	8.79
618	301	4	3	2	1	1	3	6	16	16	0.0234	0.5	0.3333	0.2284	1	Trivial	0	0	0	0	0	0	0	0	299.8	10.74
619	302	5	4	3	1	1	3	7	16	16	0.0273	0.3333	0.25	0.1637	0	1A	262324	8002.93	0	0	0	10379.88	106	0	18682.62	9.77
620	302	6	5	3	2	3	3	8	16	16	0.0313	0.75	0.4	0.3206	0	Sweep	500000	8008.79	0	0	0	15518.55	158	0	23828.13	9.77
621	302	7	6	3	3	1	3	9	16	16	0.0352	0.2	0.5	0.1776	1	Sweep-S1	517	18.55	2	2	0	9456.06	4	0	9775.39	10.74
622	303	8	6	3	3	4	3	9	16	16	0.0352	0.8	0.5	0.3576	0	Sweep	607800	8001.95	0	0	0	21611.33	218	0	29916.02	10.74
623	303	9	8	3	5	3	3	11	16	16	0.043	0.4286	0.625	0.2751	1	1B	9	24.41	1	0	1	6605.47	0	0	7240.23	10.74
624	304	3	2	2	0	0	4	6	16	16	0.0234	0	0	0.0117	0	Sweep	778	11180.66	1	0	1	7000.98	73	1	18487.3	10.74
625	304	4	3	2	1	2	4	7	16	16	0.0273	1	0.3333	0.3803	0	Sweep	4209	227.54	7	7	0	44009.77	104	0	44539.06	9.77
626	304	5	4	1	3	2	4	8	16	16	0.0313	0.6667	0.75	0.3656	1	1B	1	7.81	1	0	1	8708.98	0	0	9321.29	9.77
627	305	6	4	3	1	2	4	8	16	16	0.0313	0.6667	0.25	0.2656	1	Sweep-S1	6	10.74	1	0	1	7869.14	5	0	8184.57	13.67
628	306	7	5	3	2	3	4	9	16	16	0.0352	0.75	0.4	0.3226	0	Sweep	4259	267.58	4	4	0	37284.18	146	0	37857.42	13.67
629	306	8	5	3	2	1	4	9	16	16	0.0352	0.25	0.4	0.1726	1	Sweep-S1	196	854.49	5	0	5	28221.68	1	0	29383.79	13.67
630	307	9	5	4	1	1	4	9	16	16	0.0352	0.25	0.2	0.1326	1	Sweep-S1	5	10.74	1	0	1	6862.31	4	0	7173.83	13.67
631	308	3	2	2	0	0	0	2	16	16	0.0078	0	0	0.0039	1	Trivial	0	0	0	0	0	0	0	0	305.66	11.72
632	309	4	3	1	2	1	0	3	16	16	0.0117	0.5	0.6667	0.2892	1	1A	146	16.6	1	1	0	3157.23	0	0	3784.18	12.7
633	310	5	4	3	1	2	0	4	16	16	0.0156	0.6667	0.25	0.2578	0	Sweep	267	16.6	3	3	0	26442.38	122	0	26767.58	9.77
634	310	6	5	3	2	2	0	5	16	16	0.0195	0.5	0.4	0.2398	1	Sweep-S1	9	8.79	1	1	0	4605.47	7	0	4919.92	9.77
635	311	7	6	3	3	4	0	6	16	16	0.0234	0.8	0.5	0.3517	0	Sweep	4351	228.52	5	5	0	33766.6	100	0	34296.88	10.74
636	311	8	7	3	4	2	0	7	16	16	0.0273	0.3333	0.5714	0.228	0	Sweep	343400	8030.27	0	0	0	9741.21	99	0	18075.2	13.67
637	311	9	8	4	4	4	0	8	16	16	0.0313	0.5714	0.5	0.2871	0	Sweep	796	13384.77	1	0	1	11294.92	91	1	24985.35	12.7
638	311	3	2	2	0	0	1	3	16	16	0.0117	0	0	0.0059	1	Sweep-S1	19	7.81	1	1	0	5732.42	18	0	6046.88	9.77
639	312	4	3	1	2	1	1	4	16	16	0.0156	0.5	0.6667	0.2911	0	Sweep	126	29.3	3	3	0	21808.59	100	0	22142.58	9.77
640	312	5	4	3	1	1	1	5	16	16	0.0195	0.3333	0.25	0.1598	1	Sweep-S1	4709	224.61	3	3	0	22049.8	91	0	22574.22	9.77
641	313	6	4	3	1	2	1	5	16	16	0.0195	0.6667	0.25	0.2598	1	Sweep-S1	68	7.81	1	1	0	9532.23	58	0	9842.77	9.77
642	314	7	6	3	3	1	1	7	16	16	0.0273	0.2	0.5	0.1737	0	Sweep	848	9984.38	1	0	1	14831.05	151	1	25119.14	9.77
643	314	8	7	5	2	4	1	8	16	16	0.0313	0.6667	0.2857	0.2728	0	Sweep	420200	8002.93	0	0	0	11020.51	114	0	19325.2	9.77
644	314	9	8	4	4	4	1	9	16	16	0.0352	0.5714	0.5	0.289	0	1A	267440	8004.88	0	0	0	9751.95	101	0	18061.52	10.74
645	314	3	2	2	0	0	2	4	16	16	0.0156	0	0	0.0078	1	1B	1	7.81	1	0	1	7287.11	0	0	7902.34	9.77
646	315	4	3	0	3	0	2	5	16	16	0.0195	0	1	0.2098	0	Sweep	4339	201.17	5	5	0	32088.87	104	0	32593.75	9.77
647	315	5	4	3	1	1	2	6	16	16	0.0234	0.3333	0.25	0.1617	0	Sweep	32963	2130.86	5	5	0	40337.89	122	0	42775.39	9.77
648	315	6	5	4	1	1	2	7	16	16	0.0273	0.25	0.2	0.1287	0	Sweep	4808	206.05	1	1	0	18739.26	125	0	19248.05	9.77
649	315	7	6	2	4	2	2	8	16	16	0.0313	0.4	0.6667	0.269	1	Sweep-S1	6	8.79	1	0	1	7927.73	5	0	8237.31	11.72
650	316	8	7	5	2	3	2	9	16	16	0.0352	0.5	0.2857	0.2247	0	Sweep	4730	206.05	4	4	0	27711.91	112	0	28219.73	10.74

Run	Level	Steps	SolutionObjects	Mirrors	Refractors	RuleTransitions	Decoys	TotalObjects	GridWidth	GridHeight	Cl	Ch	Ci	MID	Solved	SolvePhase	SearchNodes	SearchTimeMs	TotalPlacements	InPlaceRotations	Relocations	ExecutionTimeMs	SweepIterations	SweepRelocations	Solve TimeMs	GenerationTimeMs
651	316	9	7	5	2	3	2	9	16	16	0.0352	0.5	0.2857	0.2247	0	1A	529180	8006.84	0	0	0	21219.73	211	0	29531.25	18.55
652	316	3	2	2	0	0	3	5	16	16	0.0195	0	0	0.0098	1	Sweep-S1	21	16.6	3	3	0	16420.9	12	0	16743.16	10.74
653	317	4	3	2	1	2	3	6	16	16	0.0234	1	0.3333	0.3784	0	Sweep	5235	262.7	4	4	0	28746.09	106	0	29314.45	10.74
654	317	5	4	2	2	3	3	7	16	16	0.0273	1	0.5	0.4137	1	Sweep-S1	4714	232.42	5	5	0	28771.48	39	0	29305.66	11.72
655	318	6	5	5	0	0	3	8	16	16	0.0313	0	0	0.0156	0	Sweep	148	7.81	1	1	0	20632.81	138	0	20941.41	10.74
656	318	7	6	5	1	2	3	9	16	16	0.0352	0.4	0.1667	0.1709	0	Sweep	806	20597.66	1	0	1	13557.62	101	1	34458.01	10.74
657	318	8	4	3	1	1	3	7	16	16	0.0273	0.3333	0.25	0.1637	1	1B	1	8.79	1	0	1	7272.46	0	0	7887.7	9.77
658	319	9	6	4	2	3	3	9	16	16	0.0352	0.6	0.3333	0.2642	0	1A	266347	8011.72	0	0	0	9448.24	98	0	17764.65	10.74
659	319	3	2	2	0	0	4	6	16	16	0.0234	0	0	0.0117	1	Sweep-S1	6	7.81	1	1	0	6271.48	4	0	6583.01	10.74
660	320	4	3	2	1	2	4	7	16	16	0.0273	1	0.3333	0.3803	1	Sweep-S1	61	15.63	4	4	0	19862.3	36	0	20179.69	9.77
661	321	5	3	3	0	0	4	7	16	16	0.0273	0	0	0.0137	1	1B	1	8.79	1	0	1	4987.31	0	0	5604.49	10.74
662	322	6	5	2	3	2	4	9	16	16	0.0352	0.5	0.6	0.2876	0	1A	327948	8005.86	0	0	0	20009.77	202	0	28317.38	9.77
663	322	7	6	3	3	4	4	10	16	16	0.0391	0.8	0.5	0.3595	0	Sweep	596000	8006.84	0	0	0	20907.23	214	0	29214.84	9.77
664	322	8	7	3	4	3	4	11	16	16	0.043	0.5	0.5714	0.2858	1	Sweep-S1	517	34.18	1	0	1	7567.38	4	0	7903.32	10.74
665	323	9	7	4	3	6	4	11	16	16	0.043	1	0.4286	0.4072	0	Sweep	221	36.13	2	0	2	30346.68	155	0	30690.43	10.74
666	323	3	2	2	0	0	0	2	16	16	0.0078	0	0	0.0039	1	Sweep-S1	19	7.81	1	1	0	6231.45	18	0	6545.9	9.77
667	324	4	3	2	1	2	0	3	16	16	0.0117	1	0.3333	0.3725	0	Sweep	171	17.58	2	2	0	18166.99	82	0	18484.38	9.77
668	324	5	4	4	0	0	0	4	16	16	0.0156	0	0	0.0078	0	Sweep	809	11295.9	1	0	1	10156.25	104	1	21754.88	10.74
669	324	6	4	2	2	2	0	4	16	16	0.0156	0.6667	0.5	0.3078	0	Sweep	1229	54.69	2	2	0	22362.3	132	0	22722.66	10.74
670	324	7	6	4	2	2	0	6	16	16	0.0234	0.4	0.3333	0.1984	0	Sweep	262319	8018.56	0	0	0	10746.09	110	0	19068.36	9.77
671	324	8	7	4	3	3	0	7	16	16	0.0273	0.5	0.4286	0.2494	1	Sweep-S1	86	15.63	4	4	0	17803.71	4	0	18124.02	10.74
672	325	9	7	3	4	4	0	7	16	16	0.0273	0.6667	0.5714	0.328	1	Sweep-S1	237200	8000.98	0	0	0	1807.62	17	0	10109.38	9.77
673	326	3	2	2	0	0	1	3	16	16	0.0117	0	0	0.0059	1	1B	1	8.79	1	0	1	6805.66	0	0	7420.9	9.77
674	327	4	3	3	0	0	1	4	16	16	0.0156	0	0	0.0078	1	Trivial	0	0	0	0	0	0	0	0	303.71	11.72
675	328	5	3	2	1	2	1	4	16	16	0.0156	1	0.3333	0.3745	1	1B	1	9.77	1	0	1	7289.06	0	0	7908.2	13.67
676	329	6	5	4	1	2	1	6	16	16	0.0234	0.5	0.2	0.2017	0	Sweep	37067	2124.02	4	4	0	31667.97	130	0	34094.73	12.7
677	329	7	5	4	1	1	1	6	16	16	0.0234	0.25	0.2	0.1267	0	Sweep	41588	2137.7	4	4	0	33253.91	114	0	35696.29	9.77
678	329	8	6	4	2	3	1	7	16	16	0.0273	0.6	0.3333	0.2603	1	1B	1	7.81	1	0	1	6617.19	0	0	7229.49	10.74
679	330	9	7	6	1	2	1	8	16	16	0.0313	0.3333	0.1429	0.1442	0	1A	262868	8001.95	0	0	0	13486.33	139	0	21792.97	9.77
680	330	3	2	2	0	0	2	4	16	16	0.0156	0	0	0.0078	1	Sweep-S1	19	7.81	1	1	0	5973.63	18	0	6285.16	12.7
681	331	4	3	1	2	1	2	5	16	16	0.0195	0.5	0.6667	0.2931	1	Sweep-S1	108	16.6	4	4	0	17855.47	19	0	18178.71	10.74
682	332	5	4	3	1	1	2	6	16	16	0.0234	0.3333	0.25	0.1617	0	Sweep	164	15.63	5	5	0	33596.68	89	0	33918.95	9.77
683	332	6	5	3	2	2	2	7	16	16	0.0273	0.5	0.4	0.2437	1	Sweep-S1	8	7.81	1	0	1	8277.34	7	0	8587.89	9.77
684	333	7	6	1	5	2	2	8	16	16	0.0313	0.4	0.8333	0.3023	0	Sweep	827	20264.65	1	0	1	15355.47	122	1	35921.88	13.67
685	333	8	7	3	4	3	2	9	16	16	0.0352	0.5	0.5714	0.2819	1	Sweep-S1	199200	8001.95	0	0	0	764.65	7	0	9070.31	10.74
686	334	9	7	5	2	3	2	9	16	16	0.0352	0.5	0.2857	0.2247	0	Sweep	900	21497.07	1	0	1	23360.35	195	1	45166.02	11.72
687	334	3	2	2	0	0	3	5	16	16	0.0195	0	0	0.0098	1	Sweep-S1	6	8.79	1	1	0	6320.31	4	0	6636.72	10.74
688	335	4	3	2	1	1	3	6	16	16	0.0234	0.5	0.3333	0.2284	0	Sweep	262219	8001.95	0	0	0	6483.4	66	0	14790.04	11.72
689	335	5	3	1	2	2	3	6	16	16	0.0234	1	0.6667	0.4451	1	Sweep-S1	8	10.74	1	0	1	8279.3	7	0	8597.66	10.74
690	336	6	4	1	3	1	3	7	16	16	0.0273	0.3333	0.75	0.2637	0	Sweep	786	15821.29	1	0	1	10687.5	81	1	26809.57	10.74
691	336	7	5	4	1	2	3	8	16	16	0.0313	0.5	0.2	0.2056	0	Sweep	4730	255.86	4	4	0	31435.55	112	0	31991.21	11.72
692	336	8	6	4	2	2	3	9	16	16	0.0352	0.4	0.3333	0.2042	0	Sweep	802	12988.28	1	0	1	9382.81	97	1	22677.73	10.74
693	336	9	7	3	4	4	3	10	16	16	0.0391	0.6667	0.5714	0.3338	0	Sweep	332000	8001.95	0	0	0	9569.34	97	0	17879.88	10.74
694	336	3	2	2	0	0	4	6	16	16	0.0234	0	0	0.0117	1	Sweep-S1	5	9.77	1	0	1	6886.72	4	0	7197.27	9.77
695	337	4	3	2	1	1	4	7	16	16	0.0273	0.5	0.3333	0.2303	1	Sweep-S1	61	10.74	1	1	0	12078.13	59	0	12390.63	10.74
696	338	5	4	2	2	3	4	8	16	16	0.0313	1	0.5	0.4156	0	Sweep	794	11558.59	1	0	1	10767.58	89	1	22634.77	10.74
697	338	6	5	3	2	2	4	9	16	16	0.0352	0.5	0.4	0.2476	1	Sweep-S1	188600	8004.88	0	0	0	936.52	9	0	9247.07	11.72
698	339	7	3	1	2	1	4	7	16	16	0.0273	0.5	0.6667	0.297	1	Sweep-S1	33	8.79	1	0	1	10860.35	32	0	11174.8	14.65
699	340	8	7	3	4	3	4	11	16	16	0.043	0.5	0.5714	0.2858	1	Sweep-S1	11	42.97	1	0	1	11508.79	2	0	11855.47	12.7
700	341	9	7	4	3	3	4	11	16	16	0.043	0.5	0.4286	0.2572	0	Sweep	1452	11724.61	0	0	0	23462.89	235	0	35489.26	10.74

Run	Level	Steps	SolutionObjects	Mirrors	Refractors	RuleTransitions	Decoys	TotalObjects	GridWidth	GridHeight	Cl	Ch	Ci	MID	Solved	SolvePhase	SearchNodes	SearchTimeMs	TotalPlacements	InPlaceRotations	Relocations	ExecutionTimeMs	SweepIterations	SweepRelocations	Solve TimeMs	GenerationTimeMs
701	341	3	2	2	0	0	0	2	16	16	0.0078	0	0	0.0039	1	Sweep-S1	6	8.79	1	1	0	6291.02	4	0	6604.49	11.72
702	342	4	3	1	2	1	0	3	16	16	0.0117	0.5	0.6667	0.2892	1	Sweep-S1	245	17.58	2	2	0	13402.34	43	0	13729.49	11.72
703	343	5	4	3	1	2	0	4	16	16	0.0156	0.6667	0.25	0.2578	0	None	0	0	0	0	0	0	0	0	0.98	9.77
704	343	6	4	4	0	0	0	4	16	16	0.0156	0	0	0.0078	0	Sweep	762	12746.09	1	0	1	5503.91	57	1	18555.66	9.77
705	343	7	4	3	1	2	0	4	16	16	0.0156	0.6667	0.25	0.2578	1	1B	1	7.81	1	0	1	7273.44	0	0	7890.63	10.74
706	344	8	6	3	3	3	0	6	16	16	0.0234	0.6	0.5	0.2917	1	Sweep-S1	6	7.81	1	0	1	8054.69	5	0	8364.26	9.77
707	345	9	8	4	4	5	0	8	16	16	0.0313	0.7143	0.5	0.3299	0	Sweep	457000	8005.86	0	0	0	12918.95	132	0	21226.56	10.74
708	345	3	2	2	0	0	1	3	16	16	0.0117	0	0	0.0059	1	Trivial	0	0	0	0	0	0	0	0	302.73	9.77
709	346	4	3	3	0	0	1	4	16	16	0.0156	0	0	0.0078	1	Sweep-S1	29	30.27	3	3	0	13350.59	20	0	13683.59	10.74
710	347	5	3	2	1	1	1	4	16	16	0.0156	0.5	0.3333	0.2245	1	1B	1	8.79	1	0	1	7272.46	0	0	7888.67	9.77
711	348	6	4	2	2	1	1	5	16	16	0.0195	0.3333	0.5	0.2098	1	1A	3	7.81	1	1	0	5880.86	0	0	6497.07	10.74
712	349	7	6	4	2	2	1	7	16	16	0.0273	0.4	0.3333	0.2003	0	1A	299247	8012.7	0	0	0	16190.43	166	0	24507.81	10.74
713	349	8	7	4	3	6	1	8	16	16	0.0313	1	0.4286	0.4013	0	Sweep	41099	2173.83	5	5	0	35490.23	130	0	37966.8	10.74
714	349	9	7	5	2	3	1	8	16	16	0.0313	0.5	0.2857	0.2228	0	Sweep	414600	8005.86	0	0	0	11431.64	117	0	19744.14	10.74
715	349	3	2	2	0	0	2	4	16	16	0.0156	0	0	0.0078	0	Sweep	4827	4440.43	0	0	0	4817.38	50	0	9564.45	9.77
716	349	4	3	0	3	0	2	5	16	16	0.0195	0	1	0.2098	0	Sweep	770	7031.25	1	0	1	6231.45	65	1	13565.43	10.74
717	349	5	4	2	2	3	2	6	16	16	0.0234	1	0.5	0.4117	0	Sweep	262302	8005.86	0	0	0	9060.55	93	0	17368.16	9.77
718	349	6	3	3	0	0	2	5	16	16	0.0195	0	0	0.0098	1	Sweep-S2	695	2359.38	1	0	1	1468.75	14	1	4133.79	9.77
719	350	7	4	2	2	3	2	6	16	16	0.0234	1	0.5	0.4117	1	Sweep-S1	6	7.81	1	0	1	7842.77	5	0	8158.2	9.77
720	351	8	7	3	4	2	2	9	16	16	0.0352	0.3333	0.5714	0.2319	0	Sweep	430400	8005.86	0	0	0	11333.01	116	0	19642.58	9.77
721	351	9	8	5	3	3	2	10	16	16	0.0391	0.4286	0.375	0.2231	0	Sweep	433000	8001.95	0	0	0	12033.2	124	0	20342.77	12.7
722	351	3	2	2	0	0	3	5	16	16	0.0195	0	0	0.0098	1	Sweep-S1	19	7.81	1	1	0	6207.03	18	0	6521.48	10.74
723	352	4	3	2	1	1	3	6	16	16	0.0234	0.5	0.3333	0.2284	0	Sweep	262219	8003.91	0	0	0	6403.32	66	0	14711.91	10.74
724	352	5	4	2	2	3	3	7	16	16	0.0273	1	0.5	0.4137	1	1A	587	27.34	2	2	0	7873.05	0	0	8506.84	10.74
725	353	6	5	5	0	0	3	8	16	16	0.0313	0	0	0.0156	0	Sweep	798	18181.64	1	0	1	12426.76	93	1	30915.04	10.74
726	353	7	6	4	2	1	3	9	16	16	0.0352	0.2	0.3333	0.1442	0	Sweep	480600	8007.81	0	0	0	14536.13	149	0	22850.59	10.74
727	353	8	7	5	2	2	3	10	16	16	0.0391	0.3333	0.2857	0.1767	1	1B	1	8.79	1	0	1	8066.41	0	0	8686.52	10.74
728	354	9	5	1	4	1	3	8	16	16	0.0313	0.25	0.8	0.2506	1	Sweep-S1	8	8.79	1	0	1	8266.6	7	0	8577.15	9.77
729	355	3	2	2	0	0	4	6	16	16	0.0234	0	0	0.0117	0	Sweep	262218	8000	0	0	0	6322.27	65	0	14628.91	10.74
730	355	4	3	1	2	2	4	7	16	16	0.0273	1	0.6667	0.447	1	Sweep-S1	4624	249.02	6	6	0	32696.29	5	0	33246.09	10.74
731	356	5	4	3	1	1	4	8	16	16	0.0313	0.3333	0.25	0.1656	0	Sweep	867	15850.59	1	0	1	18441.41	162	1	34598.63	13.67
732	356	6	4	1	3	2	4	8	16	16	0.0313	0.6667	0.75	0.3656	0	Sweep	5272	248.05	8	8	0	57155.27	142	0	57711.91	9.77
733	356	7	6	3	3	3	4	10	16	16	0.0391	0.6	0.5	0.2995	0	Sweep	389200	8010.74	0	0	0	12275.39	124	0	20589.84	13.67
734	356	8	6	3	3	3	4	10	16	16	0.0391	0.6	0.5	0.2995	0	Sweep	406200	8012.7	0	0	0	12095.7	123	0	20410.16	10.74
735	356	9	7	4	3	4	4	11	16	16	0.043	0.6667	0.4286	0.3072	1	1B	65	41.02	1	0	1	6356.45	0	0	7007.81	11.72
736	357	3	2	2	0	0	0	2	16	16	0.0078	0	0	0.0039	1	Sweep-S1	13	7.81	1	1	0	6528.32	4	0	6840.82	10.74
737	358	4	3	1	2	2	0	3	16	16	0.0117	1	0.6667	0.4392	1	1B	1	7.81	1	0	1	7316.41	0	0	7930.66	9.77
738	359	5	3	0	3	0	0	3	16	16	0.0117	0	1	0.2059	0	Sweep	754	8174.81	1	0	1	4713.87	49	1	13190.43	9.77
739	359	6	5	4	1	2	0	5	16	16	0.0195	0.5	0.2	0.1998	0	Sweep	774	8741.21	1	0	1	6668.95	69	1	15714.84	9.77
740	359	7	6	2	4	1	0	6	16	16	0.0234	0.2	0.6667	0.2051	1	Sweep-S1	66115	3284.18	3	3	0	21477.54	66	0	25069.34	10.74
741	360	8	7	6	1	2	0	7	16	16	0.0273	0.3333	0.1429	0.1422	1	Sweep-S1	18	8.79	1	0	1	8029.3	17	0	8342.77	9.77
742	361	9	7	4	3	3	0	7	16	16	0.0273	0.5	0.4286	0.2494	0	Sweep	368800	8007.81	0	0	0	8864.26	90	0	17174.8	9.77
743	361	3	2	2	0	0	1	3	16	16	0.0117	0	0	0.0059	1	Sweep-S1	19	9.77	1	1	0	6508.79	18	0	6819.34	11.72
744	362	4	3	1	2	1	1	4	16	16	0.0156	0.5	0.6667	0.2911	1	Sweep-S1	536	23.44	3	3	0	13805.66	13	0	14131.84	11.72
745	363	5	4	3	1	1	1	5	16	16	0.0195	0.3333	0.25	0.1598	0	Sweep	677	26.37	2	2	0	16934.57	91	0	17264.65	9.77
746	363	6	5	4	1	1	1	6	16	16	0.0234	0.25	0.2	0.1267	1	1B	1	8.79	1	0	1	7249.02	0	0	7864.26	10.74
747	364	7	5	3	2	2	1	6	16	16	0.0234	0.5	0.4	0.2417	0	Sweep	262218	8003.91	0	0	0	6296.88	65	0	14602.54	10.74
748	364	8	5	3	2	2	1	6	16	16	0.0234	0.5	0.4	0.2417	1	Sweep-S1	79	7.81	1	1	0	13215.82	68	0	13530.27	13.67
749	365	9	7	4	3	2	1	8	16	16	0.0313	0.3333	0.4286	0.2013	0	Sweep	400800	8012.7	0	0	0	10339.84	107	0	18655.27	11.72
750	365	3	2	2	0	0	2	4	16	16	0.0156	0	0	0.0078	1	1B	1	7.81	1	0	1	8282.23	0	0	8896.48	9.77

Run	Level	Steps	SolutionObjects	Mirrors	Refractors	RuleTransitions	Decoys	TotalObjects	GridWidth	GridHeight	Cl	Ch	Ci	MID	Solved	SolvePhase	SearchNodes	SearchTimeMs	TotalPlacements	InPlaceRotations	Relocations	ExecutionTimeMs	SweepIterations	SweepRelocations	Solve TimeMs	GenerationTimeMs
751	366	4	3	2	1	1	2	5	16	16	0.0195	0.5	0.3333	0.2264	0	Sweep	796	12029.3	1	0	1	10698.24	91	1	23029.3	10.74
752	366	5	4	1	3	2	2	6	16	16	0.0234	0.6667	0.75	0.3617	1	Sweep-S1	4624	240.23	4	4	0	18499.02	6	0	19044.92	10.74
753	367	6	3	1	2	1	2	5	16	16	0.0195	0.5	0.6667	0.2931	1	Sweep-S1	6	9.77	1	0	1	8612.31	5	0	8926.76	9.77
754	368	7	5	3	2	1	2	7	16	16	0.0273	0.25	0.4	0.1687	1	Sweep-S1	116	28.32	3	3	0	18442.38	41	0	18777.34	10.74
755	369	8	7	5	2	2	2	9	16	16	0.0352	0.3333	0.2857	0.1747	1	Sweep-S1	5	8.79	1	0	1	9776.37	4	0	10087.89	13.67
756	370	9	8	5	3	5	2	10	16	16	0.0391	0.7143	0.375	0.3088	0	Sweep	367800	8004.88	0	0	0	9569.34	98	0	17879.88	12.7
757	370	3	2	2	0	0	3	5	16	16	0.0195	0	0	0.0098	1	Sweep-S1	13	8.79	1	1	0	5334.96	4	0	5649.41	11.72
758	371	4	3	2	1	1	3	6	16	16	0.0234	0.5	0.3333	0.2284	1	Sweep-S1	33	17.58	3	3	0	15600.59	23	0	15923.83	14.65
759	372	5	4	2	2	3	3	7	16	16	0.0273	1	0.5	0.4137	0	Sweep	196	16.6	4	4	0	29778.32	106	0	30096.68	9.77
760	372	6	5	3	2	4	3	8	16	16	0.0313	1	0.4	0.3956	1	Sweep-S1	598	41.02	5	5	0	26646.48	13	0	26995.12	9.77
761	373	7	6	2	4	3	3	9	16	16	0.0352	0.6	0.6667	0.3309	0	Sweep	281	619.14	4	0	4	41421.88	150	0	42341.8	10.74
762	373	8	7	4	3	3	3	10	16	16	0.0391	0.5	0.4286	0.2552	1	Sweep-S1	225800	8019.53	0	0	0	1690.43	16	0	10014.65	10.74
763	374	9	8	7	1	2	3	11	16	16	0.043	0.2857	0.125	0.1322	1	Sweep-S1	13	36.13	1	0	1	7528.32	4	0	7870.12	10.74
764	375	3	2	2	0	0	4	6	16	16	0.0234	0	0	0.0117	1	Sweep-S1	13	8.79	1	1	0	5346.68	4	0	5658.2	9.77
765	376	4	3	3	0	0	4	7	16	16	0.0273	0	0	0.0137	1	Sweep-S1	37	7.81	1	1	0	9724.61	36	0	10039.06	9.77
766	377	5	4	2	2	2	4	8	16	16	0.0313	0.6667	0.5	0.3156	0	Sweep	437400	8002.93	0	0	0	12565.43	128	0	20875.98	9.77
767	377	6	5	4	1	1	4	9	16	16	0.0352	0.25	0.2	0.1326	0	1A	295038	8020.51	0	0	0	11256.84	116	0	19581.05	9.77
768	377	7	6	5	1	2	4	10	16	16	0.0391	0.4	0.1667	0.1729	0	Sweep	473000	8005.86	0	0	0	15321.29	155	0	23627.93	10.74
769	377	8	7	4	3	5	4	11	16	16	0.043	0.8333	0.4286	0.3572	1	Sweep-S1	517	54.69	1	0	1	9914.06	4	0	10271.48	10.74
770	378	9	7	6	1	1	4	11	16	16	0.043	0.1667	0.1429	0.1001	1	1B	1	8.79	1	0	1	9008.79	0	0	9625	10.74
771	379	3	2	2	0	0	0	2	16	16	0.0078	0	0	0.0039	1	Sweep-S1	19	8.79	1	1	0	5715.82	18	0	6029.3	9.77
772	380	4	3	3	0	0	0	3	16	16	0.0117	0	0	0.0059	1	Sweep-S1	29	15.63	3	3	0	13893.55	20	0	14217.77	9.77
773	381	5	4	3	1	1	0	4	16	16	0.0156	0.3333	0.25	0.1578	0	Sweep	661	28.32	2	2	0	14832.03	74	0	15163.09	10.74
774	381	6	5	2	3	1	0	5	16	16	0.0195	0.25	0.6	0.2048	0	Sweep	774	12183.59	1	0	1	8649.41	69	1	21136.72	9.77
775	381	7	6	4	2	1	0	6	16	16	0.0234	0.2	0.3333	0.1384	0	Sweep	4890	231.45	4	4	0	34649.41	137	0	35183.59	10.74
776	381	8	7	4	3	3	0	7	16	16	0.0273	0.5	0.4286	0.2494	1	Trivial	0	0	0	0	0	0	0	0	302.73	9.77
777	382	9	4	4	0	0	0	4	16	16	0.0156	0	0	0.0078	1	Sweep-S1	21	17.58	4	4	0	17010.74	12	0	17332.03	9.77
778	383	3	2	2	0	0	1	3	16	16	0.0117	0	0	0.0059	0	Sweep	758	4923.83	1	0	1	5234.38	53	1	10463.87	9.77
779	383	4	3	2	1	2	1	4	16	16	0.0156	1	0.3333	0.3745	1	Sweep-S1	5	7.81	1	0	1	6748.05	4	0	7057.62	10.74
780	384	5	4	3	1	1	1	5	16	16	0.0195	0.3333	0.25	0.1598	0	Sweep	34078	3098.63	0	0	0	8967.77	93	0	12376.95	10.74
781	384	6	5	2	3	3	1	6	16	16	0.0234	0.75	0.6	0.3567	0	Sweep	69907	3875	4	4	0	30159.18	137	0	34337.89	16.6
782	384	7	4	3	1	1	1	5	16	16	0.0195	0.3333	0.25	0.1598	1	1B	1	7.81	1	0	1	6788.09	0	0	7408.2	10.74
783	385	8	5	3	2	3	1	6	16	16	0.0234	0.75	0.4	0.3167	1	Sweep-S1	196000	8002.93	0	0	0	253.91	2	0	8559.57	9.77
784	386	9	5	2	3	3	1	6	16	16	0.0234	0.75	0.6	0.3567	1	Sweep-S1	18	7.81	1	0	1	8950.2	17	0	9261.72	9.77
785	387	3	2	2	0	0	2	4	16	16	0.0156	0	0	0.0078	1	Sweep-S1	19	7.81	1	1	0	6260.74	18	0	6574.22	13.67
786	388	4	3	2	1	2	2	5	16	16	0.0195	1	0.3333	0.3764	1	Sweep-S1	5	7.81	1	0	1	8505.86	4	0	8821.29	9.77
787	389	5	3	2	1	1	2	5	16	16	0.0195	0.5	0.3333	0.2264	0	Sweep	774	12266.6	1	0	1	8729.49	69	1	21301.76	10.74
788	389	6	5	2	3	2	2	7	16	16	0.0273	0.5	0.6	0.2837	1	Sweep-S1	5	7.81	1	0	1	6759.77	4	0	7072.27	10.74
789	390	7	3	3	0	0	2	5	16	16	0.0195	0	0	0.0098	0	Sweep	34067	10279.3	0	0	0	7861.33	82	0	18442.38	9.77
790	390	8	7	5	2	3	2	9	16	16	0.0352	0.5	0.2857	0.2247	0	Sweep	379400	8003.91	0	0	0	8762.7	90	0	17067.38	10.74
791	390	9	7	5	2	2	2	9	16	16	0.0352	0.3333	0.2857	0.1747	1	Trivial	0	0	0	0	0	0	0	0	302.73	10.74
792	391	3	2	2	0	0	3	5	16	16	0.0195	0	0	0.0098	1	Trivial	0	0	0	0	0	0	0	0	301.76	9.77
793	392	4	3	1	2	1	3	6	16	16	0.0234	0.5	0.6667	0.2951	1	Sweep-S1	5	7.81	1	0	1	7254.88	4	0	7570.31	10.74
794	393	5	3	2	1	1	3	6	16	16	0.0234	0.5	0.3333	0.2284	1	1B	1	9.77	1	0	1	9220.7	0	0	9835.94	10.74
795	394	6	5	3	2	2	3	8	16	16	0.0313	0.5	0.4	0.2456	0	1A	296070	8020.51	0	0	0	12258.79	124	0	20583.98	10.74
796	394	7	4	3	1	2	3	7	16	16	0.0273	0.6667	0.25	0.2637	1	Trivial	0	0	0	0	0	0	0	0	304.69	14.65
797	395	8	6	4	2	3	3	9	16	16	0.0352	0.6	0.3333	0.2642	0	Sweep	444200	8006.84	0	0	0	15158.2	153	0	23471.68	12.7
798	395	9	6	3	3	2	3	9	16	16	0.0352	0.4	0.5	0.2376	0	Sweep	8340	532.23	6	6	0	49605.47	139	0	50441.41	12.7
799	395	3	2	2	0	0	4	6	16	16	0.0234	0	0	0.0117	1	Sweep-S1	6	9.77	1	1	0	5783.2	4	0	6095.7	9.77
800	396	4	3	2	1	2	4	7	16	16	0.0273	1	0.3333	0.3803	0	Sweep	180	15.63	4	4	0	26582.03	106	0	26904.3	11.72

Run	Level	Steps	SolutionObjects	Mirrors	Refractors	RuleTransitions	Decoys	TotalObjects	GridWidth	GridHeight	Cl	Ch	Ci	MID	Solved	SolvePhase	SearchNodes	SearchTimeMs	TotalPlacements	InPlaceRotations	Relocations	ExecutionTimeMs	SweepIterations	SweepRelocations	Solve TimeMs	GenerationTimeMs
801	396	5	4	2	2	2	4	8	16	16	0.0313	0.6667	0.5	0.3156	0	Sweep	340000	8005.86	0	0	0	7826.17	81	0	16132.81	9.77
802	396	6	5	5	0	0	4	9	16	16	0.0352	0	0	0.0176	0	Sweep	373000	8009.77	0	0	0	10388.67	106	0	18703.13	9.77
803	396	7	6	4	2	4	4	10	16	16	0.0391	0.8	0.3333	0.3262	1	Sweep-S1	198400	8003.91	0	0	0	179.69	1	0	8484.38	9.77
804	397	8	7	3	4	2	4	11	16	16	0.043	0.3333	0.5714	0.2358	0	Sweep	810	23644.53	1	0	1	14125	105	1	38072.27	11.72
805	397	9	6	3	3	3	4	10	16	16	0.0391	0.6	0.5	0.2995	1	Sweep-S1	198000	8005.86	0	0	0	1037.11	10	0	9351.56	9.77
806	398	3	2	2	0	0	0	2	16	16	0.0078	0	0	0.0039	1	Sweep-S1	5	7.81	1	0	1	7562.5	4	0	7873.05	9.77
807	399	4	3	2	1	2	0	3	16	16	0.0117	1	0.3333	0.3725	0	Sweep	754	3734.38	1	0	1	4650.39	49	1	8689.45	9.77
808	399	5	4	4	0	0	0	4	16	16	0.0156	0	0	0.0078	1	1B	1	7.81	1	0	1	7277.34	0	0	7900.39	9.77
809	400	6	4	3	1	2	0	4	16	16	0.0156	0.6667	0.25	0.2578	1	1B	1	7.81	1	0	1	6859.38	0	0	7474.61	9.77
810	401	7	6	4	2	3	0	6	16	16	0.0234	0.6	0.3333	0.2584	0	Sweep	4741	226.56	5	5	0	36044.92	122	0	36576.17	9.77
811	401	8	4	4	0	0	0	4	16	16	0.0156	0	0	0.0078	1	1B	2	7.81	2	0	2	15216.8	0	0	15837.89	9.77
812	402	9	5	4	1	1	0	5	16	16	0.0195	0.25	0.2	0.1248	1	1B	1	7.81	1	0	1	5978.52	0	0	6591.8	9.77
813	403	3	2	2	0	0	1	3	16	16	0.0117	0	0	0.0059	0	Sweep	758	6326.17	1	0	1	5181.64	53	1	11812.5	11.72
814	403	4	3	0	3	0	1	4	16	16	0.0156	0	1	0.2078	0	Sweep	762	10027.34	1	0	1	5476.56	57	1	15808.59	11.72
815	403	5	4	3	1	2	1	5	16	16	0.0195	0.6667	0.25	0.2598	1	Sweep-S1	166	17.58	5	5	0	24689.45	29	0	25007.81	11.72
816	404	6	4	2	2	2	1	5	16	16	0.0195	0.6667	0.5	0.3098	1	1B	1	7.81	1	0	1	7457.03	0	0	8070.31	13.67
817	405	7	4	0	4	0	1	5	16	16	0.0195	0	1	0.2098	0	Sweep	663	29.3	3	3	0	24822.27	84	0	25154.3	11.72
818	405	8	6	3	3	2	1	7	16	16	0.0273	0.4	0.5	0.2337	0	Sweep	404800	8005.86	0	0	0	10462.89	107	0	18769.53	9.77
819	405	9	8	6	2	1	1	9	16	16	0.0352	0.1429	0.25	0.1104	0	Sweep	4247	220.7	9	9	0	57470.7	140	0	57992.19	9.77
820	405	3	2	2	0	0	2	4	16	16	0.0156	0	0	0.0078	1	Sweep-S1	21	15.63	3	3	0	16255.86	12	0	16578.13	9.77
821	406	4	3	2	1	1	2	5	16	16	0.0195	0.5	0.3333	0.2264	1	Sweep-S1	23	13.67	5	5	0	22501.95	12	0	22824.22	9.77
822	407	5	4	2	2	3	2	6	16	16	0.0234	1	0.5	0.4117	1	Sweep-S1	6	9.77	1	0	1	8335.94	5	0	8650.39	9.77
823	408	6	5	2	3	2	2	7	16	16	0.0273	0.5	0.6	0.2837	1	Sweep-S1	33959	1642.58	6	6	0	33976.56	29	0	35925.78	11.72
824	409	7	6	5	1	2	2	8	16	16	0.0313	0.4	0.1667	0.169	0	1A	295138	8003.91	0	0	0	15115.23	153	0	23425.78	23.44
825	409	8	6	5	1	2	2	8	16	16	0.0313	0.4	0.1667	0.169	0	Sweep	362000	8007.81	0	0	0	7964.84	82	0	16279.3	11.72
826	409	9	8	5	3	4	2	10	16	16	0.0391	0.5714	0.375	0.266	0	Sweep	655400	8007.81	0	0	0	22806.64	232	0	31121.09	11.72
827	409	3	2	2	0	0	3	5	16	16	0.0195	0	0	0.0098	1	1B	1	9.77	1	0	1	9076.17	0	0	9691.41	9.77
828	410	4	3	1	2	2	3	6	16	16	0.0234	1	0.6667	0.4451	0	Sweep	5226	240.23	2	2	0	18107.42	96	0	18650.39	9.77
829	410	5	4	2	2	2	3	7	16	16	0.0273	0.6667	0.5	0.3137	0	Sweep	396800	8001.95	0	0	0	9839.84	101	0	18142.58	11.72
830	410	6	3	1	2	2	3	6	16	16	0.0234	1	0.6667	0.4451	0	Sweep	784	14105.47	1	0	1	10070.31	79	1	24482.42	9.77
831	410	7	5	5	0	0	3	8	16	16	0.0313	0	0	0.0156	0	Sweep	415800	8005.86	0	0	0	10302.73	106	0	18621.09	9.77
832	410	8	7	4	3	3	3	10	16	16	0.0391	0.5	0.4286	0.2552	0	Sweep	33486	1746.09	8	8	0	56443.36	195	0	58494.14	9.77
833	410	9	7	5	2	3	3	10	16	16	0.0391	0.5	0.2857	0.2267	0	Sweep	596200	8003.91	0	0	0	21195.31	211	0	29505.86	9.77
834	410	3	2	2	0	0	4	6	16	16	0.0234	0	0	0.0117	1	Trivial	0	0	0	0	0	0	0	0	306.64	9.77
835	411	4	3	2	1	2	4	7	16	16	0.0273	1	0.3333	0.3803	0	Sweep	348600	8009.77	0	0	0	7236.33	74	0	15548.83	11.72
836	411	5	4	1	3	2	4	8	16	16	0.0313	0.6667	0.75	0.3656	0	Sweep	5237	263.67	6	6	0	35552.73	107	0	36119.14	9.77
837	411	6	4	3	1	2	4	8	16	16	0.0313	0.6667	0.25	0.2656	1	Sweep-S1	37	5.86	1	1	0	9724.61	36	0	10037.11	9.77
838	412	7	6	5	1	1	4	10	16	16	0.0391	0.2	0.1667	0.1129	1	1B	1	9.77	1	0	1	7283.2	0	0	7898.44	9.77
839	413	8	7	7	0	0	4	11	16	16	0.043	0	0	0.0215	1	Trivial	0	0	0	0	0	0	0	0	304.69	9.77
840	414	9	6	4	2	1	4	10	16	16	0.0391	0.2	0.3333	0.1462	0	Sweep	496800	8005.86	0	0	0	13394.53	135	0	21701.17	9.77
841	414	3	2	2	0	0	0	2	16	16	0.0078	0	0	0.0039	1	Sweep-S1	6	9.77	1	1	0	6011.72	4	0	6324.22	9.77
842	415	4	3	1	2	1	0	3	16	16	0.0117	0.5	0.6667	0.2892	1	Sweep-S1	178	15.63	2	2	0	10984.38	32	0	11306.64	9.77
843	416	5	4	1	3	2	0	4	16	16	0.0156	0.6667	0.75	0.3578	1	Sweep-S1	182	7.81	1	1	0	11125	45	0	11439.45	11.72
844	417	6	3	2	1	2	0	3	16	16	0.0117	1	0.3333	0.3725	1	1B	1	7.81	1	0	1	7785.16	0	0	8398.44	9.77
845	418	7	6	3	3	4	0	6	16	16	0.0234	0.8	0.5	0.3517	0	Sweep	38033	2205.08	5	5	0	35785.16	128	0	38292.97	9.77
846	418	8	7	6	1	2	0	7	16	16	0.0273	0.3333	0.1429	0.1422	1	1B	1	9.77	1	0	1	6078.13	0	0	6693.36	11.72
847	419	9	8	5	3	5	0	8	16	16	0.0313	0.7143	0.375	0.3049	1	Sweep-S1	205400	8001.95	0	0	0	183.59	1	0	8488.28	11.72
848	420	3	2	2	0	0	1	3	16	16	0.0117	0	0	0.0059	1	Sweep-S1	19	7.81	1	1	0	6199.22	18	0	6511.72	9.77
849	421	4	3	2	1	1	1	4	16	16	0.0156	0.5	0.3333	0.2245	0	Sweep	83	9.77	1	1	0	11292.97	81	0	11603.52	9.77
850	421	5	4	1	3	2	1	5	16	16	0.0195	0.6667	0.75	0.3598	0	Sweep	1178	52.73	2	2	0	16300.78	72	0	16658.2	9.77

Run	Level	Steps	SolutionObjects	Mirrors	Refractors	RuleTransitions	Decoys	TotalObjects	GridWidth	GridHeight	Cl	Ch	Ci	MID	Solved	SolvePhase	SearchNodes	SearchTimeMs	TotalPlacements	InPlaceRotations	Relocations	ExecutionTimeMs	SweepIterations	SweepRelocations	Solve TimeMs	GenerationTimeMs
851	421	6	5	2	3	3	1	6	16	16	0.0234	0.75	0.6	0.3567	0	Sweep	90	9.77	1	0	1	15878.91	89	0	16189.45	11.72
852	421	7	6	4	2	4	1	7	16	16	0.0273	0.8	0.3333	0.3203	1	1B	1	7.81	1	0	1	5068.36	0	0	5681.64	9.77
853	422	8	7	6	1	2	1	8	16	16	0.0313	0.3333	0.1429	0.1442	0	Sweep	798	19824.22	1	0	1	12468.75	93	1	32597.66	9.77
854	422	9	6	2	4	3	1	7	16	16	0.0273	0.6	0.6667	0.327	0	Sweep	4900	244.14	6	6	0	41937.5	154	0	42482.42	9.77
855	422	3	2	2	0	0	2	4	16	16	0.0156	0	0	0.0078	1	Sweep-S1	6	7.81	1	1	0	6542.97	4	0	6857.42	9.77
856	423	4	3	2	1	1	2	5	16	16	0.0195	0.5	0.3333	0.2264	1	1A	75	9.77	1	1	0	6044.92	0	0	6662.11	9.77
857	424	5	4	4	0	0	2	6	16	16	0.0234	0	0	0.0117	0	Sweep	4268	218.75	3	3	0	25187.5	106	0	25710.94	9.77
858	424	6	4	3	1	2	2	6	16	16	0.0234	0.6667	0.25	0.2617	0	Sweep	806	13087.89	1	0	1	12308.59	101	1	25703.13	9.77
859	424	7	6	3	3	3	2	8	16	16	0.0313	0.6	0.5	0.2956	0	Sweep	410400	8007.81	0	0	0	10691.41	109	0	19001.95	9.77
860	424	8	5	2	3	2	2	7	16	16	0.0273	0.5	0.6	0.2837	1	Sweep-S1	5	7.81	1	0	1	9009.77	4	0	9324.22	11.72
861	425	9	7	5	2	3	2	9	16	16	0.0352	0.5	0.2857	0.2247	0	1A	262860	8001.95	0	0	0	12056.64	123	0	20363.28	9.77
862	425	3	2	2	0	0	3	5	16	16	0.0195	0	0	0.0098	1	Sweep-S1	21	15.63	2	2	0	10115.23	12	0	10439.45	9.77
863	426	4	3	2	1	1	3	6	16	16	0.0234	0.5	0.3333	0.2284	0	Sweep	262219	8005.86	0	0	0	6394.53	66	0	14707.03	9.77
864	426	5	3	2	1	1	3	6	16	16	0.0234	0.5	0.3333	0.2284	0	Sweep	782	9855.47	1	0	1	7453.13	77	1	17615.23	11.72
865	426	6	5	3	2	2	3	8	16	16	0.0313	0.5	0.4	0.2456	1	Sweep-S1	6	11.72	1	0	1	7595.7	5	0	7914.06	9.77
866	427	7	4	2	2	1	3	7	16	16	0.0273	0.3333	0.5	0.2137	0	Sweep	786	15800.78	1	0	1	10308.59	81	1	26416.02	11.72
867	427	8	5	5	0	0	3	8	16	16	0.0313	0	0	0.0156	0	Sweep	338600	8000	0	0	0	7767.58	80	0	16070.31	11.72
868	427	9	4	2	2	2	3	7	16	16	0.0273	0.6667	0.5	0.3137	1	Sweep-S1	6	7.81	1	0	1	8335.94	5	0	8650.39	11.72
869	428	3	2	2	0	0	4	6	16	16	0.0234	0	0	0.0117	1	Sweep-S1	6	7.81	1	0	1	7162.11	5	0	7476.56	9.77
870	429	4	3	1	2	2	4	7	16	16	0.0273	1	0.6667	0.447	1	1B	1	7.81	1	0	1	7296.88	0	0	7912.11	11.72
871	430	5	4	2	2	1	4	8	16	16	0.0313	0.3333	0.5	0.2156	0	Sweep	33082	1634.77	4	4	0	32142.58	176	0	34082.03	9.77
872	430	6	4	4	0	0	4	8	16	16	0.0313	0	0	0.0156	1	Sweep-S1	5	9.77	1	0	1	9011.72	4	0	9326.17	9.77
873	431	7	6	3	3	4	4	10	16	16	0.0391	0.8	0.5	0.3595	0	Sweep	415000	8011.72	0	0	0	11304.69	117	0	19615.23	9.77
874	431	8	7	4	3	4	4	11	16	16	0.043	0.6667	0.4286	0.3072	1	Sweep-S1	715	16521.48	0	0	0	14048.83	2	0	30875	9.77
875	432	9	6	3	3	4	4	10	16	16	0.0391	0.8	0.5	0.3595	1	Sweep-S1	4	9.77	1	0	1	6560.55	3	0	6876.95	11.72
876	433	3	2	2	0	0	0	2	16	16	0.0078	0	0	0.0039	0	Sweep	750	3410.16	1	0	1	4601.56	45	1	8320.31	11.72
877	433	4	3	0	3	0	0	3	16	16	0.0117	0	1	0.2059	0	Sweep	780	7642.58	1	0	1	7361.33	75	1	15308.59	11.72
878	433	5	3	2	1	1	0	3	16	16	0.0117	0.5	0.3333	0.2225	1	Sweep-S1	6	9.77	1	0	1	7099.61	5	0	7416.02	11.72
879	434	6	4	2	2	3	0	4	16	16	0.0156	1	0.5	0.4078	1	1A	75	9.77	1	1	0	5267.58	0	0	5894.53	11.72
880	435	7	6	2	4	2	0	6	16	16	0.0234	0.4	0.6667	0.2651	1	Sweep-S1	99	17.58	2	2	0	14712.89	34	0	15031.25	11.72
881	436	8	6	2	4	2	0	6	16	16	0.0234	0.4	0.6667	0.2651	0	Sweep	99	7.81	1	1	0	13666.02	97	0	13982.42	9.77
882	436	9	8	5	3	4	0	8	16	16	0.0313	0.5714	0.375	0.2621	1	1B	1	9.77	1	0	1	6845.7	0	0	7466.8	11.72
883	437	3	2	2	0	0	1	3	16	16	0.0117	0	0	0.0059	0	Sweep	1250	1457.03	0	0	0	3966.8	41	0	5730.47	9.77
884	437	4	3	2	1	2	1	4	16	16	0.0156	1	0.3333	0.3745	0	Sweep	97	19.53	4	4	0	24898.44	80	0	25226.56	11.72
885	437	5	4	4	0	0	1	5	16	16	0.0195	0	0	0.0098	0	Sweep	107	9.77	1	1	0	16841.8	98	0	17158.2	11.72
886	437	6	5	3	2	1	1	6	16	16	0.0234	0.25	0.4	0.1667	0	Sweep	8910	470.7	3	3	0	28271.48	124	0	29048.83	9.77
887	437	7	4	4	0	0	1	5	16	16	0.0195	0	0	0.0098	1	1B	1	9.77	1	0	1	8796.88	0	0	9417.97	9.77
888	438	8	7	3	4	3	1	8	16	16	0.0313	0.5	0.5714	0.2799	0	1A	332975	8007.81	0	0	0	10167.97	100	0	18478.52	11.72
889	438	9	6	5	1	2	1	7	16	16	0.0273	0.4	0.1667	0.167	1	1B	1	9.77	1	0	1	6871.09	0	0	7494.14	11.72
890	439	3	2	2	0	0	2	4	16	16	0.0156	0	0	0.0078	1	1B	1	7.81	1	0	1	7291.02	0	0	7908.2	11.72
891	440	4	3	2	1	1	2	5	16	16	0.0195	0.5	0.3333	0.2264	0	Sweep	33539	9302.73	0	0	0	10855.47	58	0	20460.94	11.72
892	440	5	4	3	1	2	2	6	16	16	0.0234	0.6667	0.25	0.2617	0	Sweep	782	13443.36	1	0	1	9679.69	77	1	23423.83	11.72
893	440	6	4	1	3	1	2	6	16	16	0.0234	0.3333	0.75	0.2617	0	Sweep	8809	478.52	3	3	0	23992.19	95	0	24773.44	11.72
894	440	7	6	3	3	3	2	8	16	16	0.0313	0.6	0.5	0.2956	1	Sweep-S1	6	7.81	1	0	1	7175.78	5	0	7490.23	11.72
895	441	8	4	2	2	1	2	6	16	16	0.0234	0.3333	0.5	0.2117	1	Sweep-S1	8820	535.16	5	5	0	26511.72	51	0	27351.56	9.77
896	442	9	8	6	2	2	2	10	16	16	0.0391	0.2857	0.25	0.1552	0	Sweep	394200	8001.95	0	0	0	10292.97	106	0	18595.7	11.72
897	442	3	2	2	0	0	3	5	16	16	0.0195	0	0	0.0098	1	Sweep-S1	8	9.77	1	0	1	8750	7	0	9060.55	11.72
898	443	4	3	2	1	1	3	6	16	16	0.0234	0.5	0.3333	0.2284	1	1B	1	9.77	1	0	1	5365.23	0	0	5984.38	11.72
899	444	5	4	4	0	0	3	7	16	16	0.0273	0	0	0.0137	1	Sweep-S1	4125	203.13	5	5	0	29724.61	20	0	30228.52	9.77
900	445	6	4	2	2	3	3	7	16	16	0.0273	1	0.5	0.4137	1	1B	1	7.81	1	0	1	6847.66	0	0	7470.7	9.77

Run	Level	Steps	SolutionObje	Mirrors	Refractors	RuleTransitions	Decoys	TotalObjects	GridWidth	GridHeight	Cl	Ch	Ci	MID	Solved	SolvePhase	SearchNodes	SearchTimeMs	TotalPlaceme	InPlaceRotations	Relocations	ExecutionTimeMs	SweepIterations	SweepRelocations	Solve TimeMs	GenerationTimeMs
901	446	7	3	2	1	1	3	6	16	16	0.0234	0.5	0.3333	0.2284	0	Sweep	811	14738.28	1	0	1	12779.3	106	1	27824.22	9.77
902	446	8	4	2	2	2	3	7	16	16	0.0273	0.6667	0.5	0.3137	0	Sweep	1226	56.64	4	4	0	33576.17	129	0	33935.55	9.77
903	446	9	6	2	4	3	3	9	16	16	0.0352	0.6	0.6667	0.3309	0	Sweep	618800	8007.81	0	0	0	21642.58	219	0	29957.03	11.72
904	446	3	2	2	0	0	4	6	16	16	0.0234	0	0	0.0117	0	Sweep	262219	8005.86	0	0	0	6410.16	66	0	14720.7	9.77
905	446	4	3	3	0	0	4	7	16	16	0.0273	0	0	0.0137	1	1B	1	9.77	1	0	1	6830.08	0	0	7447.27	9.77
906	447	5	4	2	2	3	4	8	16	16	0.0313	1	0.5	0.4156	1	Sweep-S1	8785	453.13	4	4	0	25482.42	6	0	26238.28	9.77
907	448	6	4	4	0	0	4	8	16	16	0.0313	0	0	0.0156	0	Sweep	99	7.81	1	1	0	16228.52	97	0	16542.97	11.72
908	448	7	6	2	4	4	4	10	16	16	0.0391	0.8	0.6667	0.3929	0	Sweep	567400	8005.86	0	0	0	17199.22	174	0	25509.77	11.72
909	448	8	6	3	3	3	4	10	16	16	0.0391	0.6	0.5	0.2995	0	Sweep	869	14724.61	1	0	1	18732.42	164	1	33763.67	9.77
910	448	9	7	3	4	5	4	11	16	16	0.043	0.8333	0.5714	0.3858	1	Sweep-S1	70	23.44	1	0	1	8115.23	5	0	8443.36	11.72
911	449	3	2	2	0	0	0	2	16	16	0.0078	0	0	0.0039	1	Sweep-S1	6	9.77	1	1	0	5537.11	4	0	5851.56	9.77
912	450	4	3	1	2	1	0	3	16	16	0.0117	0.5	0.6667	0.2892	0	Sweep	220	17.58	2	2	0	16722.66	74	0	17041.02	11.72
913	450	5	3	3	0	0	0	3	16	16	0.0117	0	0	0.0059	0	Sweep	1292	1740.23	0	0	0	6429.69	67	0	8470.7	9.77
914	450	6	5	3	2	3	0	5	16	16	0.0195	0.75	0.4	0.3148	0	Sweep	9400	443.36	2	2	0	19955.08	109	0	20707.03	9.77
915	450	7	3	2	1	1	0	3	16	16	0.0117	0.5	0.3333	0.2225	0	Sweep	786	5195.31	1	0	1	8021.48	81	1	13519.53	9.77
916	450	8	7	4	3	3	0	7	16	16	0.0273	0.5	0.4286	0.2494	1	Sweep-S1	8	7.81	1	0	1	7816.41	7	0	8130.86	9.77
917	451	9	7	3	4	4	0	7	16	16	0.0273	0.6667	0.5714	0.328	1	Sweep-S1	15	15.63	4	4	0	19916.02	5	0	20232.42	9.77
918	452	3	2	2	0	0	1	3	16	16	0.0117	0	0	0.0059	0	Sweep	1282	2861.33	0	0	0	5515.63	57	0	8681.64	13.67
919	452	4	3	2	1	2	1	4	16	16	0.0156	1	0.3333	0.3745	1	Sweep-S1	52	7.81	1	1	0	9412.11	42	0	9720.7	13.67
920	453	5	4	4	0	0	1	5	16	16	0.0195	0	0	0.0098	0	Sweep	774	12539.06	1	0	1	6658.2	69	1	19500	11.72
921	453	6	5	2	3	3	1	6	16	16	0.0234	0.75	0.6	0.3567	1	Trivial	0	0	0	0	0	0	0	0	298.83	9.77
922	454	7	5	4	1	1	1	6	16	16	0.0234	0.25	0.2	0.1267	0	Sweep	780	12642.58	1	0	1	9572.27	75	1	22521.48	9.77
923	454	8	6	3	3	3	1	7	16	16	0.0273	0.6	0.5	0.2937	1	1B	1	7.81	1	0	1	9785.16	0	0	10402.34	9.77
924	455	9	7	6	1	2	1	8	16	16	0.0313	0.3333	0.1429	0.1442	0	Sweep	640	23.44	2	2	0	22941.41	127	0	23271.48	9.77
925	455	3	2	2	0	0	2	4	16	16	0.0156	0	0	0.0078	1	Sweep-S1	21	15.63	3	3	0	14425.78	12	0	14742.19	9.77
926	456	4	3	1	2	1	2	5	16	16	0.0195	0.5	0.6667	0.2931	1	Sweep-S1	8	7.81	1	0	1	8357.42	7	0	8667.97	11.72
927	457	5	4	3	1	2	2	6	16	16	0.0234	0.6667	0.25	0.2617	0	Sweep	782	8060.55	1	0	1	7458.98	77	1	15820.31	11.72
928	457	6	4	3	1	2	2	6	16	16	0.0234	0.6667	0.25	0.2617	0	Sweep	98	7.81	1	1	0	15414.06	97	0	15724.61	9.77
929	457	7	4	4	0	0	2	6	16	16	0.0234	0	0	0.0117	1	1B	1	7.81	1	0	1	7279.3	0	0	7894.53	9.77
930	458	8	6	3	3	3	2	8	16	16	0.0313	0.6	0.5	0.2956	0	1A	327871	8005.86	0	0	0	16265.63	166	0	24578.13	11.72
931	458	9	3	3	0	0	2	5	16	16	0.0195	0	0	0.0098	0	Sweep	33538	10509.77	0	0	0	6667.97	57	0	17482.42	9.77
932	458	3	2	2	0	0	3	5	16	16	0.0195	0	0	0.0098	1	Sweep-S1	21	15.63	4	4	0	21960.94	12	0	22277.34	9.77
933	459	4	3	3	0	0	3	6	16	16	0.0234	0	0	0.0117	1	Sweep-S1	37	7.81	1	1	0	8962.89	36	0	9275.39	9.77
934	460	5	4	2	2	3	3	7	16	16	0.0273	1	0.5	0.4137	0	Sweep	794	11037.11	1	0	1	8673.83	89	1	20011.72	11.72
935	460	6	4	2	2	3	3	7	16	16	0.0273	1	0.5	0.4137	0	Sweep	337000	8005.86	0	0	0	7154.3	74	0	15464.84	9.77
936	460	7	6	4	2	3	3	9	16	16	0.0352	0.6	0.3333	0.2642	0	1A	295629	8005.86	0	0	0	12744.14	131	0	21054.69	9.77
937	460	8	6	4	2	4	3	9	16	16	0.0352	0.8	0.3333	0.3242	0	Sweep	372000	8000	0	0	0	8863.28	90	0	17164.06	13.67
938	460	9	7	3	4	1	3	10	16	16	0.0391	0.1667	0.5714	0.1838	0	Sweep	33418	1800.78	2	2	0	23900.39	128	0	26005.86	9.77
939	460	3	2	2	0	0	4	6	16	16	0.0234	0	0	0.0117	0	Sweep	778	11533.2	1	0	1	7005.86	73	1	18845.7	11.72
940	460	4	3	1	2	1	4	7	16	16	0.0273	0.5	0.6667	0.297	0	Sweep	1164	70.31	6	6	0	40158.2	123	0	40529.3	11.72
941	460	5	4	3	1	2	4	8	16	16	0.0313	0.6667	0.25	0.2656	0	Sweep	188	9.77	1	1	0	17121.09	114	0	17435.55	11.72
942	460	6	3	3	0	0	4	7	16	16	0.0273	0	0	0.0137	1	Sweep-S1	5	7.81	1	0	1	8515.63	4	0	8828.13	9.77
943	461	7	5	2	3	1	4	9	16	16	0.0352	0.25	0.6	0.2126	0	Sweep	33426	1984.38	7	7	0	48654.3	136	0	50947.27	11.72
944	461	8	4	1	3	2	4	8	16	16	0.0313	0.6667	0.75	0.3656	0	Sweep	831	17826.17	1	0	1	15460.94	126	1	33589.84	11.72
945	461	9	8	3	5	5	4	12	16	16	0.0469	0.7143	0.625	0.3627	0	Sweep	821	9642.58	0	0	0	37373.05	108	0	47324.22	9.77
946	461	3	2	2	0	0	0	2	16	16	0.0078	0	0	0.0039	1	Sweep-S1	21	13.67	2	2	0	10349.61	12	0	10667.97	11.72
947	462	4	3	3	0	0	0	3	16	16	0.0117	0	0	0.0059	1	Sweep-S1	29	15.63	3	3	0	13660.16	20	0	13978.52	9.77
948	463	5	3	1	2	1	0	3	16	16	0.0117	0.5	0.6667	0.2892	0	Sweep	758	5681.64	1	0	1	5169.92	53	1	11158.2	9.77
949	463	6	4	2	2	3	0	4	16	16	0.0156	1	0.5	0.4078	1	1B	1	7.81	1	0	1	6517.58	0	0	7138.67	9.77
950	464	7	6	5	1	1	0	6	16	16	0.0234	0.2	0.1667	0.1051	1	Sweep-S1	8	9.77	1	0	1	7873.05	7	0	8185.55	11.72

Run	Level	Steps	SolutionObjects	Mirrors	Refractors	RuleTransitio	Decoys	TotalObjects	GridWidth	GridHeight	Cl	Ch	Ci	MID	Solved	SolvePhase	SearchNodes	SearchTimeMs	TotalPlacements	InPlaceRotations	Relocations	ExecutionTimeMs	SweepIterations	SweepRelocations	SolveTimeMs	GenerationTimeMs
951	465	8	5	4	1	2	0	5	16	16	0.0195	0.5	0.2	0.1998	1	Sweep-S1	85	15.63	4	4	0	21390.63	20	0	21708.98	9.77
952	466	9	8	6	2	4	0	8	16	16	0.0313	0.5714	0.25	0.2371	1	Sweep-S1	227400	8003.91	0	0	0	1560.55	15	0	9867.19	11.72
953	467	3	2	2	0	0	1	3	16	16	0.0117	0	0	0.0059	1	Sweep-S1	6	9.77	1	1	0	6009.77	4	0	6326.17	9.77
954	468	4	3	2	1	2	1	4	16	16	0.0156	1	0.3333	0.3745	0	Sweep	766	7400.39	1	0	1	5880.86	61	1	13582.03	9.77
955	468	5	4	2	2	2	1	5	16	16	0.0195	0.6667	0.5	0.3098	0	Sweep	801	13218.75	1	0	1	9320.31	96	1	22845.7	9.77
956	468	6	5	4	1	2	1	6	16	16	0.0234	0.5	0.2	0.2017	1	Sweep-S1	15	7.81	1	1	0	7707.03	6	0	8015.63	9.77
957	469	7	4	3	1	2	1	5	16	16	0.0195	0.6667	0.25	0.2598	1	Sweep-S1	5	7.81	1	0	1	6634.77	4	0	6941.41	11.72
958	470	8	5	4	1	2	1	6	16	16	0.0234	0.5	0.2	0.2017	0	Sweep	4819	322.27	5	5	0	39996.09	146	0	40625	9.77
959	470	9	8	3	5	3	1	9	16	16	0.0352	0.4286	0.625	0.2711	0	Sweep	519400	8003.91	0	0	0	17095.7	174	0	25400.39	11.72
960	470	3	2	2	0	0	2	4	16	16	0.0156	0	0	0.0078	1	Sweep-S1	6	7.81	2	2	0	9828.13	4	0	10142.58	11.72
961	471	4	3	3	0	0	2	5	16	16	0.0195	0	0	0.0098	1	Sweep-S1	29	15.63	4	4	0	21863.28	20	0	22183.59	9.77
962	472	5	4	3	1	2	2	6	16	16	0.0234	0.6667	0.25	0.2617	0	Sweep	778	13244.14	1	0	1	9417.97	73	1	22966.8	9.77
963	472	6	4	4	0	0	2	6	16	16	0.0234	0	0	0.0117	1	Sweep-S1	8	7.81	1	0	1	8771.48	7	0	9083.98	9.77
964	473	7	6	3	3	3	2	8	16	16	0.0313	0.6	0.5	0.2956	0	Sweep	404800	8005.86	0	0	0	10933.59	113	0	19240.23	11.72
965	473	8	6	1	5	2	2	8	16	16	0.0313	0.4	0.8333	0.3023	1	1B	1	7.81	1	0	1	6041.02	0	0	6654.3	11.72
966	474	9	8	6	2	3	2	10	16	16	0.0391	0.4286	0.25	0.1981	0	Sweep	862	16560.55	1	0	1	18226.56	157	1	35089.84	11.72
967	474	3	2	2	0	0	3	5	16	16	0.0195	0	0	0.0098	0	Sweep	33539	10351.56	0	0	0	11445.31	58	0	22105.47	11.72
968	474	4	3	1	2	1	3	6	16	16	0.0234	0.5	0.6667	0.2951	1	Sweep-S2	721	8093.75	1	0	1	1750	16	1	10150.39	9.77
969	475	5	4	2	2	2	3	7	16	16	0.0273	0.6667	0.5	0.3137	0	Sweep	362400	8007.81	0	0	0	7048.83	73	0	15359.38	9.77
970	475	6	5	4	1	1	3	8	16	16	0.0313	0.25	0.2	0.1306	0	Sweep	1242	54.69	4	4	0	33642.58	145	0	34003.91	9.77
971	475	7	5	4	1	2	3	8	16	16	0.0313	0.5	0.2	0.2056	1	Sweep-S1	5	7.81	1	0	1	7343.75	4	0	7654.3	9.77
972	476	8	4	3	1	1	3	7	16	16	0.0273	0.3333	0.25	0.1637	1	Sweep-S1	8329	429.69	4	4	0	30179.69	64	0	30914.06	11.72
973	477	9	7	6	1	2	3	10	16	16	0.0391	0.3333	0.1429	0.1481	1	Sweep-S1	220200	8007.81	0	0	0	1384.77	13	0	9693.36	9.77
974	478	3	2	2	0	0	4	6	16	16	0.0234	0	0	0.0117	0	Sweep	262234	8005.86	0	0	0	7847.66	81	0	16160.16	11.72
975	478	4	3	3	0	0	4	7	16	16	0.0273	0	0	0.0137	1	Trivial	0	0	0	0	0	0	0	0	308.59	11.72
976	479	5	4	3	1	1	4	8	16	16	0.0313	0.3333	0.25	0.1656	0	Sweep	667	29.3	6	6	0	46636.72	138	0	46972.66	9.77
977	479	6	3	1	2	1	4	7	16	16	0.0273	0.5	0.6667	0.297	0	Sweep	348200	8003.91	0	0	0	7246.09	74	0	15550.78	9.77
978	479	7	6	3	3	2	4	10	16	16	0.0391	0.4	0.5	0.2395	0	Sweep	445400	8007.81	0	0	0	13441.41	135	0	21753.91	11.72
979	479	8	6	3	3	3	4	10	16	16	0.0391	0.6	0.5	0.2995	0	Sweep	814	14080.08	1	0	1	13384.77	109	1	27769.53	9.77
980	479	9	6	5	1	2	4	10	16	16	0.0391	0.4	0.1667	0.1729	0	Sweep	421800	8007.81	0	0	0	11712.89	121	0	20023.44	9.77
981	479	3	2	2	0	0	0	2	16	16	0.0078	0	0	0.0039	0	Sweep	750	3593.75	1	0	1	4472.66	45	1	8375	9.77
982	479	4	3	1	2	1	0	3	16	16	0.0117	0.5	0.6667	0.2892	0	Sweep	787	7884.77	1	0	1	7958.98	82	1	16144.53	9.77
983	479	5	3	3	0	0	0	3	16	16	0.0117	0	0	0.0059	1	1B	1	7.81	1	0	1	4603.52	0	0	5212.89	9.77
984	480	6	5	4	1	1	0	5	16	16	0.0195	0.25	0.2	0.1248	0	Sweep	33580	9601.56	0	0	0	14490.23	99	0	24396.48	11.72
985	480	7	6	4	2	3	0	6	16	16	0.0234	0.6	0.3333	0.2584	0	Sweep	222	13.67	6	6	0	43179.69	147	0	43496.09	9.77
986	480	8	6	4	2	2	0	6	16	16	0.0234	0.4	0.3333	0.1984	0	Sweep	304	15.63	3	3	0	30744.14	166	0	31066.41	11.72
987	480	9	4	2	2	3	0	4	16	16	0.0156	1	0.5	0.4078	1	Sweep-S1	8	11.72	1	0	1	7060.55	7	0	7378.91	11.72
988	481	3	2	2	0	0	1	3	16	16	0.0117	0	0	0.0059	1	Sweep-S1	21	15.63	2	2	0	10841.8	12	0	11164.06	9.77
989	482	4	3	1	2	1	1	4	16	16	0.0156	0.5	0.6667	0.2911	1	Sweep-S1	151	19.53	3	3	0	14585.94	5	0	14906.25	11.72
990	483	5	4	2	2	3	1	5	16	16	0.0195	1	0.5	0.4098	0	Sweep	770	12083.98	1	0	1	6181.64	65	1	18568.36	9.77
991	483	6	5	2	3	3	1	6	16	16	0.0234	0.75	0.6	0.3567	0	Sweep	8427	423.83	4	4	0	29017.58	96	0	29748.05	9.77
992	483	7	4	3	1	2	1	5	16	16	0.0195	0.6667	0.25	0.2598	0	Sweep	770	12296.88	1	0	1	8294.92	65	1	20896.48	9.77
993	483	8	7	4	3	5	1	8	16	16	0.0313	0.8333	0.4286	0.3513	0	Sweep	361600	8021.48	0	0	0	8699.22	90	0	17023.44	11.72
994	483	9	6	3	3	3	1	7	16	16	0.0273	0.6	0.5	0.2937	0	Sweep	790	15587.89	1	0	1	11171.88	85	1	27064.45	11.72
995	483	3	2	2	0	0	2	4	16	16	0.0156	0	0	0.0078	1	Sweep-S1	21	27.34	2	2	0	10349.61	12	0	10679.69	9.77
996	484	4	3	2	1	1	2	5	16	16	0.0195	0.5	0.3333	0.2264	0	Sweep	91	15.63	2	2	0	18849.61	72	0	19171.88	9.77
997	484	5	4	2	2	3	2	6	16	16	0.0234	1	0.5	0.4117	0	Sweep	32964	1599.61	5	5	0	37101.56	122	0	39001.95	9.77
998	484	6	5	2	3	3	2	7	16	16	0.0273	0.75	0.6	0.3587	0	Sweep	33426	2060.55	6	6	0	39230.47	136	0	41593.75	9.77
999	484	7	4	3	1	2	2	6	16	16	0.0234	0.6667	0.25	0.2617	0	Sweep	262218	8000	0	0	0	6283.2	65	0	14587.89	9.77
1000	484	8	5	2	3	1	2	7	16	16	0.0273	0.25	0.6	0.2087	0	Sweep	786	13408.2	1	0	1	7726.56	81	1	21439.45	9.77

C: Weight Calibration Results

Appendix C presents the calibration results used during the selection of the MID weighting configuration. Multiple weighting combinations were evaluated using the calibration dataset, and Pearson correlation coefficients were calculated against solver-performance measures.

The purpose of this analysis was not to identify a universally optimal weighting configuration, but rather to examine whether different MID formulations exhibited consistent relationships with solver behaviour. The results were subsequently used to inform the weighting configuration adopted in Chapter 3.

MID	Solve Time,Search Node	Search Node
MID(0.5/0.2/0.3)	0.494497027	-0.013568794
MID(0.5/0.1/0.4)	0.507320547	0.006319416
MID(0.6/0.1/0.3)	0.512034508	0.004728611
MID(0.7/0.1/0.2)	0.517604637	0.002175117
MID(0.1/0.1/0.8)	0.496155703	0.009229044
MID(0.1/0.2/0.7)	0.494178585	-0.002436364
MID(0.1/0.3/0.6)	0.48778414	-0.013553501
MID(0.1/0.4/0.5)	0.477817966	-0.023888395
MID0.1/0.5/0.4)	0.465222898	-0.033302474
MID(0.1/0.6/0.3)	0.450899511	-0.041743776
MID(0.1/0.7/0.2)	0.435619828	-0.049226739
MID(0.1/0.8/0.1)	0.419992688	-0.055809399
MID(0.2/0.1/0.7)	0.498144299	0.008769073
MID(0.2/0.2/0.6)	0.494937969	-0.004264741
MID(0.2/0.3/0.5)	0.486380156	-0.016536475
MID(0.2/0.4/0.4)	0.473733212	-0.027752768
MID(0.2/0.5/0.3)	0.458327934	-0.037768935
MID(0.2/0.6/0.2)	0.441352763	-0.046563735
MID(0.2/0.7/0.1)	0.423751239	-0.054199903
MID(0.3/0.1/0.6)	0.500566331	0.008180177
MID(0.3/0.2/0.5)	0.495502597	-0.006570877
MID(0.3/0.3/0.4)	0.483879098	-0.02022485
MID(0.3/0.3/0.3)	0.477199547	-0.026023643
MID(0.3/0.4/0.3)	0.467643793	-0.032418772
MID(0.3/0.5/0.2)	0.448698546	-0.043026908
MID(0.4/0.1/0.5)	0.503562933	0.007400064
MID(0.4/0.2/0.4)	0.495577208	-0.009560672
MID(0.4/0.3/0.3)	0.479488511	-0.02487385
MID(0.4/0.4/0.2)	0.458408163	-0.038112604
MID(0.4/0.5/0.1)	0.435058961	-0.049234152
MID(0.5/0.3/0.2)	0.568124665	0.024611551
MID(0.5/0.4/0.1)	0.444117995	-0.045113131
MID(0.6/0.2/0.2)	0.490668461	-0.019163178
MID(0.7/0.2/0.1)	0.479943628	-0.02733373
MID(0.8/0.1/0.1)	0.521534519	-0.002485719
MID(.33/.33/.33)	0.477199547	-0.026023643
MID(0.6/0.3/0.1)	0.456334842	-0.039332506
MID(0.3/0.6/0.1)	0.428592784	-0.052096965

Table 17 Weight Calibration Results